

GLOBAL CREDIT TRADER

The Hybrid Renaissance

- The momentum behind hybrid debt issuance—which began in early 2024 following a change to Moody’s rating methodology—has accelerated in 2026. While the elevated pace of year-to-date hybrid supply is especially notable in the EUR market, the USD market is also on track to set a new annual record.
- The amount of outstanding USD hybrid debt has doubled vs. early 2024. Even so, hybrid spreads have *outperformed* both BBBs and BBs since that time. The recent performance of EUR hybrids vs. BBBs has been more range-bound, owing to the more mature stage of the market (and significant outperformance in 2022).
- For IG-focused investors searching for incremental spread (vs. BBBs), hybrids can be an attractive solution, as they offer a spread premium to compensate for structural subordination—typically within an IG-rated parent company capital structure—and certain issuer-friendly terms. That said, determining whether this compensation is adequate requires a clear understanding of issuer-specific capital management priorities and sector-specific headwinds.
- USD hybrids offer (on average) 61bp of additional spread vs. BBBs. We view this as moderately attractive in the context of the tight spread environment and expect the spread premium to compress vs. BBBs as the hybrid market grows further.
- In the EUR market, the incremental spread ‘pickup’ of hybrids vs. BBBs is 53bp. We do not see much scope for further absolute spread compression, given the more challenging growth-inflation-monetary policy mix in the region. That said, the additional carry of hybrids vs. EUR BBBs can be attractive for a range of investor types. Given our more cautious positioning in the EUR market (relative to USD), we would focus on higher-quality hybrid issuers in more economically resilient sectors.

Amanda Lynam, CPA

+1(212)934-1895 |
amanda.lynam@gs.com
Goldman Sachs & Co. LLC

Spencer Rogers, CFA

+1(801)884-1104 |
spencer.rogers@gs.com
Goldman Sachs & Co. LLC

Sara Grut

+44(20)7774-8622 | sara.grut@gs.com
Goldman Sachs International

Shamshad Ali

+1(212)902-6712 |
shamshad.ali@gs.com
Goldman Sachs & Co. LLC

The Hybrid Renaissance

Hybrid supply: On pace to set new records, especially in the EUR market

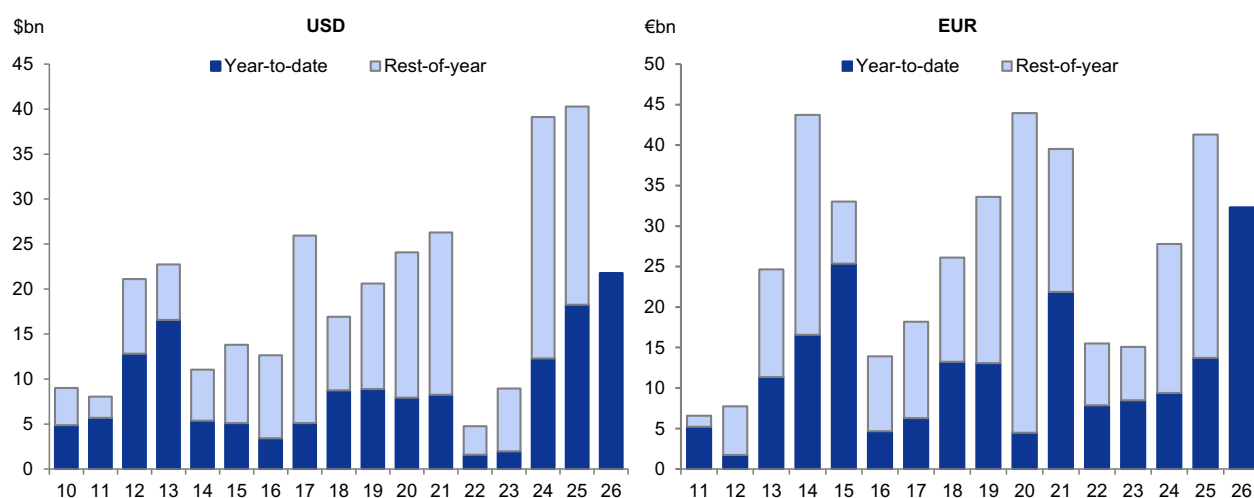
The issuance tailwind behind the hybrid debt market—which began in early 2024 following a change to the Moody’s rating methodology—has accelerated in 2026. As shown in Exhibit 1, both the USD and EUR markets are on pace to set new annual records for hybrid gross supply.

In the USD market, there are now more than a dozen hybrid bonds with \$1 billion or more of notional outstanding—and some of the largest hybrid new issues were launched in recent months. The elevated gross supply of EUR hybrids partly reflects a more mature market, which has had a high share of bonds being called this year. For comparison, while gross supply is up 132% year to date vs. the same period last year, the notional amount outstanding of the EUR hybrid market has increased by a more modest 9% since the end of 2025.

Utilities have been among the most active sectors in both markets, accounting for 50% and 40% of hybrid supply in the USD and EUR markets, respectively, since 2024 (Exhibits 2 and 3).

Exhibit 1: Hybrid supply is running at the fastest pace of the post financial crisis era, in both the USD and EUR markets

Gross issuance of USD (left panel) and EUR (right panel) non-bank hybrids

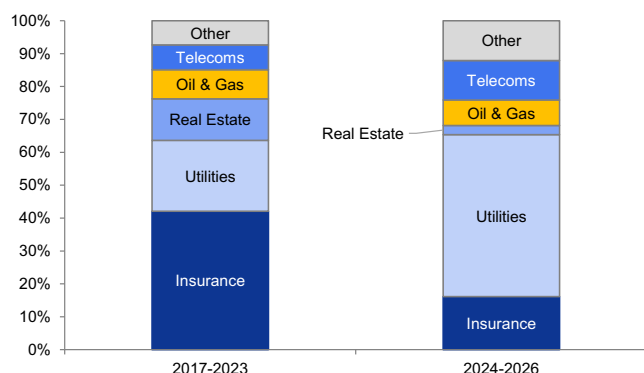


Note: As of June 1, 2026.

Source: Dealogic (ION Analytics), Goldman Sachs Global Investment Research

Exhibit 2: In the USD market, Utilities have generated a large share of hybrid supply in recent years

Share of total USD non-bank hybrid supply, by sector

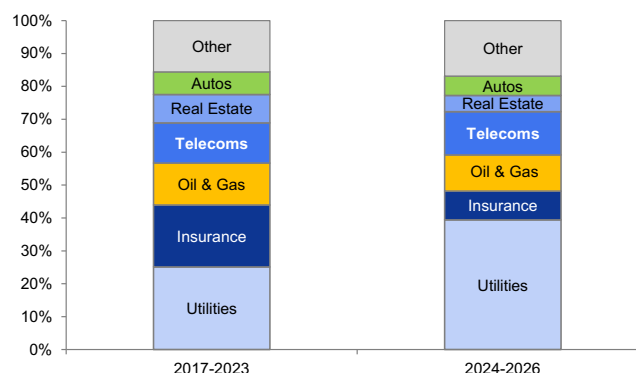


Note: As of June 1, 2026.

Source: Dealogic (ION Analytics), Goldman Sachs Global Investment Research

Exhibit 3: Utilities are also active hybrid issuers in the EUR market, although there is some additional sector diversification compared to the USD market

Share of total EUR non-bank hybrid supply, by sector



Note: Note: As of June 1, 2026.

Source: Dealogic (ION Analytics), Goldman Sachs Global Investment Research

The drivers of hybrid issuance and investment

With the broader debt capital markets open to a wide range of borrowers, many market participants have questioned why an issuer might opt for a subordinated (higher coupon) hybrid instrument vs. a more traditional financing option in senior unsecured form.

Issuers in capital intensive industries—which often operate with large debt capital structures—tend to favor hybrid debt for two key reasons. First is the partial equity credit it receives from the rating agencies, which reflects its ‘equity like’ characteristics, including subordination in the capital structure and the ability to defer coupon payments without triggering a default. For a capital structure with limited ‘headroom’ within a targeted leverage or rating band, this partial equity credit can be attractive—especially since it does not involve diluting shareholders.

Second is the favorable tax deductibility of hybrid interest (relative to preferred stock, for example). For a profitable corporate taxpayer, the tax shield can help optimize the weighted average cost of capital—even if the hybrid debt has a moderately higher coupon.

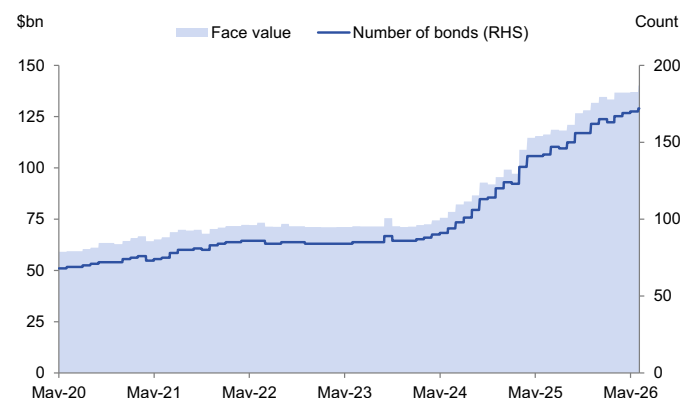
Placing the growth of the hybrid market (across regions) in context

Exhibits 4 and 5 show the evolution of the size of the hybrid asset class across regions. We start in the USD market and first turn to the ICE-BAML USD Developed Markets Corporate (ex-Banks) Hybrid Bond Index as a proxy for the USD hybrid universe.

As shown in Exhibit 4, the USD Hybrid Index had \$140 billion of notional outstanding as of June 3, 2026—up 96% vs. January 2024. The number of bonds in the index has also doubled since that time (to 172, currently). For context, the notional size of the ICE-BAML USD BBB Corporate Index grew 5% (to \$4.3 trillion) over that same time frame and the BB Corporate Index grew 41% (to \$880 billion). Of note, these BBB and BB indices include hybrids.

Exhibit 4: The size of the USD hybrid market has doubled since early 2024

Face value and number of bonds in the ICE-BAML USD Developed Markets Corporate (ex-Banks) Hybrid Bond Index



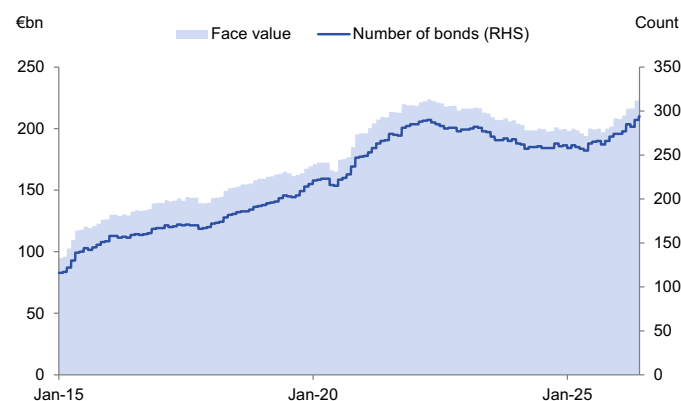
Note: As of June 2, 2026.

Source: ICE-BAML, Goldman Sachs Global Investment Research

In contrast to its USD peer, the EUR hybrid market is more mature. It experienced the most significant growth prior to mid-2022 (Exhibit 5). Here, we are using the ICE EUR Hybrid Corporate and High Yield Index as a proxy for the EUR hybrid market. As shown below, the June 2026 notional value (€226 billion) is 2% above the June 2022 peak. For context, the EUR BBB index has grown 6% since that time (to €1.56 trillion), while the EUR BB index has contracted 11% (to €282 billion). Here, too, the BBB and BB indices include hybrids.

Exhibit 5: The peak in sizing of the EUR Hybrid market was in mid-2022

Face value and number of bonds in the ICE EUR Hybrid Corporate & High Yield Index



Note: As of June 3, 2026.

Source: ICE, Goldman Sachs Global Investment Research

Assessing relative value—with some important caveats

It is important to acknowledge up front that the composition of these EUR and USD hybrid indices can make direct relative value comparisons to the broader corporate credit market challenging. For example, in both regions, the hybrid indices contain a *mix* of ratings across IG and HY (for example, blending BBBs and BBs together, in one index),

and fewer—yet more concentrated—industry exposures.

Exhibits 6 through 9 provide snapshot summaries of the USD and EUR hybrid indices by rating and by industry. Given the rating distributions shown in Exhibits 6 and 8, we view BBB rated debt (and, to a lesser extent, BBs) as the most relevant comparable to hybrids. Additionally, the concentrated hybrid sector weights shown in Exhibits 7 and 9 are important to keep in mind when interpreting the index-level relative value relationships that follow.

Exhibit 6: The EUR Hybrid Index is tilted primarily towards BBBs and BBs

Rating weights (by market value %)

Rating	EUR Hybrid	EUR BBB	EUR BB
A+	4.2	0.0	0.0
A	7.9	0.0	0.0
A-	6.7	0.0	0.0
BBB+	9.4	45.9	0.0
BBB	11.9	33.6	0.0
BBB-	28.8	20.4	0.0
BB+	13.7	0.0	44.9
BB	14.5	0.0	32.1
BB-	1.5	0.0	23.1
B+	0.5	0.0	0.0
B	0.8	0.0	0.0
CCC	0.1	0.0	0.0

Note: As of June 2, 2026.

Source: ICE and ICE-BAML, Goldman Sachs Global Investment Research

Exhibit 8: The USD Hybrid market also has a strong skew toward BBBs and BBs

Rating weights (by market value %)

Rating	USD Hybrid	USD BBB	USD BB
A+	1.6	0.0	0.0
A	1.4	0.0	0.0
A-	11.4	0.0	0.0
BBB+	5.6	37.0	0.0
BBB	19.6	40.6	0.0
BBB-	35.2	22.3	0.0
BB+	19.1	0.0	23.5
BB	6.0	0.0	35.3
BB-	0.0	0.0	41.1

Note: As of June 2, 2026.

Source: ICE and ICE-BAML, Goldman Sachs Global Investment Research

Exhibit 7: The EUR Hybrid market has large weights to Utilities, Insurance, Telecom, Energy and Autos

Industry weights (by market value %)

Sector	EUR Hybrid	EUR BBB	EUR BB
Automotive	7.4	5.4	13.8
Banking	0.0	17.8	4.6
Basic Industry	0.6	5.0	6.0
Capital Goods	0.9	3.1	4.0
Consumer Goods	4.1	5.8	3.5
Energy	10.9	2.9	2.6
Financial Services	0.9	3.8	5.5
Healthcare	3.1	4.0	5.4
Insurance	25.5	3.4	0.0
Leisure	0.5	0.6	4.3
Media	0.0	1.3	0.9
Real Estate	3.3	8.3	6.0
Retail	0.0	2.7	3.6
Services	0.8	1.6	5.4
Technology & Electronics	0.7	2.3	2.8
Telecommunications	10.8	8.2	15.6
Transportation	2.4	4.3	5.0
Utility	28.1	19.4	11.0

Note: As of June 2, 2026.

Source: ICE and ICE-BAML, Goldman Sachs Global Investment Research

Exhibit 9: The USD Hybrid market has more concentrated sector weights, relative to the broader BBB and BB markets

Industry weights (by market value %)

Sector	USD Hybrid	USD BBB	USD BB
Automotive	0.4	3.0	3.3
Banking	0.0	6.5	0.2
Basic Industry	0.0	6.5	9.8
Capital Goods	0.0	5.0	5.8
Consumer Goods	0.0	5.1	2.6
Energy	14.4	11.9	12.9
Financial Services	3.5	8.6	9.0
Healthcare	3.3	9.0	7.5
Insurance	31.6	3.8	1.1
Leisure	0.0	1.4	5.6
Media	0.0	1.9	9.1
Real Estate	0.0	4.7	3.7
Retail	0.0	3.3	5.8
Services	0.0	1.3	5.2
Technology & Electronics	0.0	7.4	5.6
Telecommunications	11.7	7.0	6.7
Transportation	0.0	2.0	1.6
Utility	35.2	11.7	4.4

Note: As of June 2, 2026.

Source: ICE and ICE-BAML, Goldman Sachs Global Investment Research

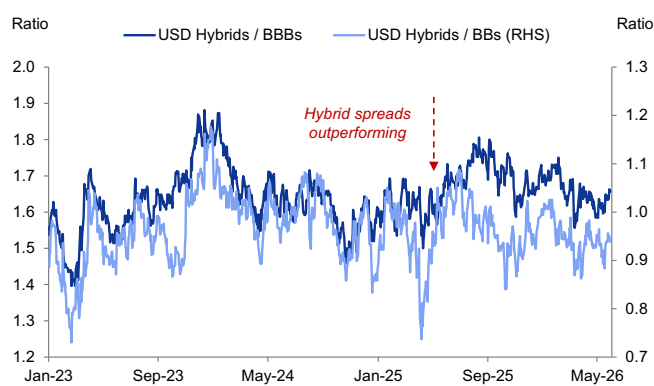
Performance: Solid (despite recent growth) for USD hybrids and range-bound for EUR

The swift growth in the USD hybrid market has not derailed performance. As Exhibit 10 illustrates, the spread ratio of the USD Hybrid Index vs. both BBBs and BBs has *compressed* since early 2024—indicating hybrid outperformance.

In the EUR market, the recent performance backdrop for hybrids has been largely range-bound—at least compared to BBBs (Exhibit 11). This likely reflects, in our view, the more mature stage of the market and the significant periods of outperformance in prior years (most recently in 2022, for example).

Exhibit 10: The performance of hybrids vs. the broader BBB and BB cohorts has been resilient over the past year, even as it has grown

Spread ratios: USD hybrid OAS vs. BBBs and BBs

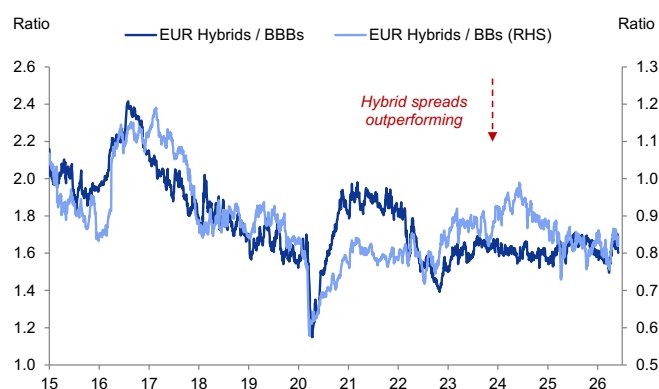


Note: As of June 2, 2026.

Source: ICE-BAML, Goldman Sachs Global Investment Research

Exhibit 11: After a period of EUR hybrid outperformance in 2022, the relative value relationship vs. EUR BBBs has been range-bound

Spread ratios: EUR hybrid OAS vs. BBBs and BBs



Note: As of June 2, 2026.

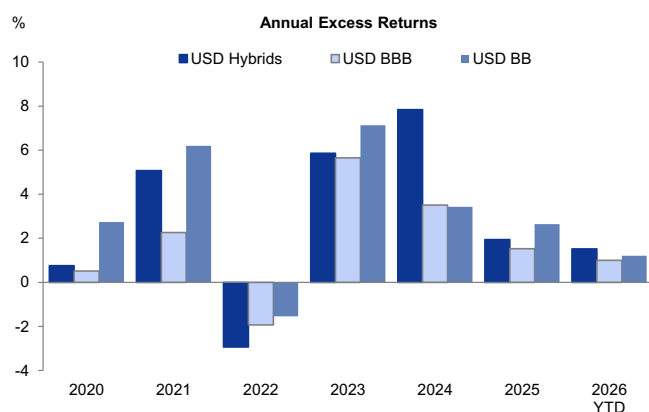
Source: ICE-BAML, Goldman Sachs Global Investment Research

The more mature state of the EUR hybrid market is also a key reason why we show a longer performance history in Exhibit 11 relative to the shorter performance history for USD hybrids in Exhibit 10. As highlighted earlier in Exhibit 4, the USD hybrid market did not cross \$90 billion in notional value until late 2024. Meanwhile, the EUR hybrid market stood at €95 billion in notional outstanding as early as 2015 (again, Exhibit 5).

Exhibits 12 and 13 provide a comparison of excess returns, which remove the performance influence from interest rates and focus instead on credit risk. The takeaway is clear, in our view: in periods where excess returns are positive, hybrids tend to outperform their BBB peers (albeit to varying degrees across markets and periods). However, in a year such as 2022 when excess returns were negative, hybrids underperformed.

Exhibit 12: Hybrids have outperformed BBBs on an excess return basis for the past few years, but have lagged during risk-off periods such as 2022

Annual and year-to-date 2026 excess returns for the ICE-BAML USD Developed Markets Corporate (ex-Banks) Hybrid Bond Index, and the USD BBB and BB Corporate indices

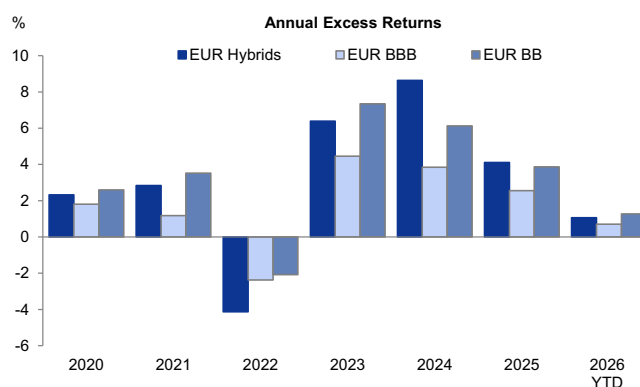


Note: As of June 2, 2026.

Source: ICE-BAML, Goldman Sachs Global Investment Research

Exhibit 13: Hybrids have also frequently outperformed BBBs in the EUR market, but have lagged in risk-off periods

Annual and year-to-date excess returns for the ICE EUR Hybrid Index, and the EUR BBB and BB Corporate indices



Note: As of June 2, 2026.

Source: ICE, Goldman Sachs Global Investment Research

Hybrids generally trade *structure risk* for *quality risk*

For IG-focused investors searching for ways to capture incremental spread without deviating too far down the quality spectrum, hybrids can be an attractive solution given the spread premium they offer vs. senior debt.

This additional spread premia serves as ‘compensation’ for hybrids’ subordination as well as some issuer-friendly structural features, such as the ability to defer interest payments in certain situations (without triggering a default).

That said, determining whether this compensation is adequate requires a clear understanding of the company-specific capital management priorities for each hybrid issuer. It also requires a thorough analysis of the potential long-term fundamental headwinds facing the issuers’ industries to evaluate the risk of financial distress (which may necessitate deferral of interest).

Very few investors expect hybrid issuers to *actually utilize* the provision to defer interest—especially if that issuer already has a large debt capital structure which requires frequent refinancing. The assumption is that deferring the hybrid’s coupon payments may make it more difficult—due to investor push back from overlapping buyer bases—to refinance the ‘regular’ (non-subordinated) debt when the time comes. In many instances, deferring a hybrid coupon may also prevent an issuer from paying its common equity dividend.

The value of this interest deferral provision to the issuer increases during times of economic weakness, which can often result in hybrids underperforming during risk-off moves (again, Exhibit 12 and 13).

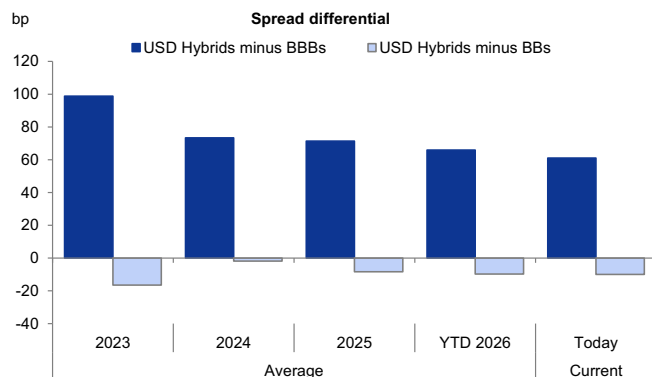
Quantifying the compensation for subordination

To quantify this compensation premium, we again use the USD and EUR hybrid indices mentioned earlier, comparing index-level average (and current) spreads relative to the

broader BBB and BB markets. Leaving the important rating and industry compositional difference caveats aside, we show the index-level absolute spread differentials across regions in Exhibit 14 and 15.

Exhibit 14: USD hybrids trade, on average, 61bp wide to USD BBBs and 10bp tight to USD BBs

Average index-level spread differentials

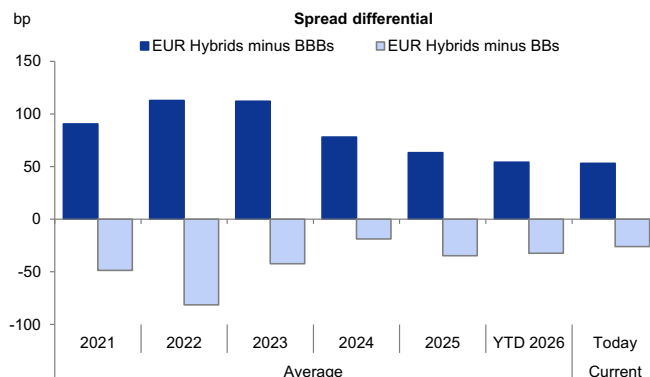


Note: As of June 2, 2026.

Source: ICE-BAML, Goldman Sachs Global Investment Research

Exhibit 15: EUR hybrids trade, on average, 53bp wide to EUR BBBs and 26bp tight to EUR BBs

Average index-level spread differentials



Note: As of June 2, 2026.

Source: ICE, Goldman Sachs Global Investment Research

For example, the absolute spread differential between USD hybrids and USD BBBs compressed from 99bp (on average) in 2023 to 61bp, currently (again, Exhibit 14). While this is narrower vs. a few years ago, it is consistent with the *directional* trend of spread tightening in the USD market. As a result, we view this index-level compensation as moderately attractive considering spreads for many markets are hovering at the tight end of the post-financial crisis history (Exhibits 37 and 38 show spread percentile ranks vs. history). We also expect the hybrid spread premium to compress as the market grows further.

In the EUR market, the hybrid vs. BBB spread differential is currently 53bp (again, Exhibit 15). This tighter spread differential likely reflects, in part, the more mature positioning of the EUR hybrid market, as investors and issuers have become very familiar with the structures over the past decade.

We do not see much scope for further absolute spread tightening (compression) in EUR hybrids, given the more challenging growth-inflation-monetary policy mix stemming from the ongoing energy shock. That said, the additional carry vs. EUR BBBs *can* be attractive for a range of investor types. Given our more cautious positioning in the EUR market (relative to USD), we would focus on higher-quality hybrid issuers in more economically resilient sectors.

The authors would like to thank Patricia Pacheco for her contribution to this report.

Forecast dashboard: Spreads, returns, issuance, defaults, and ratings migrations

Exhibit 16: USD and EUR spread forecasts

Index	Updated as of Jun. 03, 2026			
	Current	2026Q2	2026Q3	2026Q4
USD spreads				
USD IG	73	95	92	90
USD Fin	76	103	98	94
USD Non-Fin	70	92	89	86
USD HY	263	335	325	315
EUR spreads				
EUR IG	86	120	117	114
EUR Fin	92	127	122	118
EUR Non-Fin	83	116	113	111
EUR HY	261	375	368	360

Source: Bloomberg, ICE-BAML, iBoxx, Goldman Sachs Global Investment Research

Exhibit 18: Issuance forecasts

Market	2026 YTD (bn)	2026 full-year forecast (bn)
USD market		
Gross issuance		
IG	\$1,109	\$2,100
HY	\$159	\$400
Leveraged loans	\$182	--
Net issuance		
IG	\$478	\$850
HY	\$43	\$160
EUR market		
Gross issuance		
IG	€ 426	€ 850
HY	€ 47	€ 125
Net issuance		
IG	€ 98	€ 265
HY	€ 21	€ 60

Source: Dealogic, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 17: USD and EUR excess returns forecast

Index	Updated as of Jun. 03, 2026		
	Current duration	2026	2026 full-year forecast (%)
USD excess returns			
USD IG	6.8	0.8%	0.0%
USD HY	3.0	1.3%	1.3%
EUR excess returns			
EUR IG	4.7	0.8%	0.0%
EUR HY	2.7	1.1%	-0.4%

Source: Bloomberg, ICE-BAML, iBoxx, Goldman Sachs Global Investment Research

Exhibit 19: Ratings migration and default forecasts

Default rates are issuer-weighted and 12-month trailing

Market	Updated as of Jun. 03, 2026	
	2026 YTD	2026 full-year forecast
USD market		
Rising stars	\$43bn	\$75bn
Fallen angels	\$31bn	\$60bn
HY defaults (12-month trailing)	3.6%	3.0%
Leveraged loan defaults (12-month trailing)	5.1%	4.75%
EUR market		
Rising stars	€ 6bn	€20bn
Fallen angels	€ 4bn	€30bn
HY defaults (12-month trailing)	4.5%	4.7%
Leveraged loan defaults (12-month trailing)	3.3%	4.0%

Source: Bloomberg, Moody's, Goldman Sachs Global Investment Research

Forecast dashboard: Relative value views

Exhibit 20: Relative value views

Relative value views		
Corporate Credit		
Regional	Overweight USD vs. EUR	The growth, inflation and monetary policy mix is somewhat more favorable in the US, relative to Europe. As a result, and relative to the USD market, we forecast a wider peak (and a slower recovery in 2H2026) in EUR credit spreads.
	Overweight IG vs. HY	We have a slight preference for IG vs. HY in the USD market, owing to valuations and our economists' forecasts for below-potential growth. We view the BB/BBB spread ratio as fairly priced amid a somewhat more challenging growth and inflation mix, and have a neutral view between the two rating cohorts.
	Overweight BBBs vs. higher-rated cohorts	We prefer owning BBBs over the higher-rated segments of USD IG (As and AAs), as a way to capture additional spread. We also see scope for balance sheet deterioration at the highest end of the rating spectrum in IG.
USD market	Overweight BBs, Neutral CCCs, Underweight Bs	Current spread levels indicate that single-Bs offer the lowest credit risk premium relative to expected losses among all HY rating buckets.
	Overweight HY bonds vs. loans	HY bonds still provide a superior trade-off between carry and duration compared to leveraged loans. In addition, loans' substantial software sector exposure is likely to present a sentiment overhang, at least in the near-term.
	Overweight CLO mezz tranches vs. leveraged loans Neutral CLO vs. USD IG Neutral CLO vs. CMOs	For floating rate investors, we prefer CLO mezzanine tranches vs. leveraged loans. Historical data points to low impairment rates for BB and BBB CLO tranches, and current valuations offer attractive carry and a cushion against moderate to elevated default outcomes. Elsewhere, we are neutral between senior CLO tranches and USD IG; we would use spread widening to add exposure in CLOs. We are also neutral between CLO AAA and CMO floaters.
	Prefer Agency MBS vs. IG	Agency MBS offers an attractive diversification benefit vs. a portfolio of IG corporate bonds.
	Overweight Hybrids vs. BBBs	We view the additional spread compensation of hybrids vs. BBBs as moderately attractive in the context of the tight spread environment. We also expect the hybrid spread premium to compress vs. BBBs, as the hybrid market grows further.
	Overweight IG vs. HY	We prefer IG over HY given the more challenging growth, inflation, and monetary policy mix in the Euro Area.
EUR market	Overweight As vs. BBBs	In the EUR market, we prefer As over BBBs, as the weaker growth backdrop warrants more of an up-in-quality stance. That said, we would not move too far up the quality spectrum, as we expect supply technicals will be a headwind in the EUR AA market.
	Overweight HY bonds vs. loans	Similar to the US, we prefer HY bonds given the loan market's weighting toward software (and the potential negative sentiment overhang). The reaction function of the ECB may also increase borrowing costs for floating rate issuers, given the central bank's focus on inflationary pressures.
	Overweight BBs, Neutral Bs, Underweight CCCs	At current valuations and given our expectation for heightened dispersion, we prefer moving up in quality within EUR HY.
	Underweight AT1s vs. EUR HY	Given downside tail risks and a limited valuation cushion, we recommend an underweight on AT1s vs. EUR HY.
	Neutral Hybrids vs. BBBs	The additional carry vs. EUR BBBs is unlikely to compress further. Given our more cautious positioning in the EUR market (relative to USD), we would focus on higher-quality hybrid issuers in more economically resilient sectors.

Source: Goldman Sachs Global Investment Research

Performance dashboard: Cash and synthetic markets

Exhibit 21: Performance of USD and EUR markets

Market	Current*		Total return			Spread change (YTD,bp)	Yield change (YTD, %)
	Spread	Yield	MTD	QTD	YTD		
USD Corporate bonds							
IG	73	5.2	-0.26	0.95	0.41	-5	0.38
AA	48	5.0	-0.25	0.58	0.26	-1	0.38
A	62	5.1	-0.26	0.83	0.44	-3	0.35
BBB	93	5.4	-0.24	1.28	0.80	-8	0.32
HY	263	7.1	-0.19	2.00	1.49	-3	0.56
BB	165	6.3	-0.13	1.86	1.47	-4	0.32
B	299	7.8	-0.18	2.31	1.93	-2	0.41
CCC	947	13.8	-0.17	2.06	-0.19	62	0.91
USD Leveraged Loans							
Leveraged loan index	437	8.2	0.00	0.09	-0.46	42	0.4
BB	257	6.3	0.00	0.00	0.70	2	-0.1
B	448	8.3	0.00	0.00	-0.93	57	0.5
CCC	1877	23.9	0.00	0.00	-5.09	314	3.3
USD Securitized products							
Agency MBS	23	5.0	-0.28	0.08	0.48	1	0.4
ABS (Autos)	59	4.6	-0.04	0.54	1.21	-7	0.4
ABS (Credit cards)	31	4.3	-0.09	0.41	0.92	-2	0.4
ABS (Student loans)	101	4.9	0.08	0.75	1.84	128	-0.1
USD Macro indices							
CDX IG	51	--	-0.02	0.66	0.43	1	--
CDX HY	305	--	-0.17	3.15	1.28	-11	--
EUR Corporate bonds							
IG	88	3.7	-0.28	1.56	0.54	-2	0.3
AA	63	3.6	-0.33	1.36	0.41	2	0.3
A	81	3.7	-0.30	1.51	0.51	-3	0.2
BBB	101	3.8	-0.25	1.64	0.59	-1	0.3
HY	261	5.8	0.02	3.03	1.26	-9	0.3
AT1	275	5.5	-0.12	3.47	2.06	-12	0.2
EUR Leveraged loans							
Leveraged loan index	497	7.2	0.00	0.18	-0.87	47	0.6
BB	282	5.0	0.00	0.00	0.75	-70	-0.6
B	471	6.9	0.00	0.00	-1.30	56	0.7
EUR Macro indices							
iTraxx Main	54	--	-0.04	0.97	0.4	3	--
iTraxx Xover	263	--	-0.18	4.80	2.1	19	--
iTraxx SenFin	57	--	-0.06	1.15	0.6	2	--
iTraxx SubFin	93	--	-0.09	2.12	1.2	0	--

*Data as of previous day close.

For USD IG and HY we use the Bloomberg indices with history back to Jun-89 and Jan-94, respectively. Corp rating level, Agency MBS and ABS we use ICE-BAML indices with history back to Dec-96, EUR IG we use the iBoxx QW5A index with history back to Jan-99, EUR HY we use the ICE-BAML index with history back to Dec-97, and USD and EUR leveraged loans we use the Morningstar LSTA LLI and ELLI indices with data going back to Jan-97 and Jan-02, respectively.

Source: Bloomberg, iBoxx, ICE-BAML, Pitchbook LCD, Goldman Sachs Global Investment Research

Performance dashboard: Sectors

Exhibit 22: Sector performance: USD IG

Sector	Metrics							Spread change				Total return			
	Index weight	Mkt val (\$bn)	OAS	Percentile rank (2010-today)	YTW	Dur	Rating	1-week	MTD	QTD	YTD	1-week	MTD	QTD	YTD
Basic Industry	2%	179.5	76		5.3	7.5	BAA1/BAA2	0	0	-21	-14	-0.1	-0.2	1.4	1.0
Chemicals	1%	81.2	81		5.4	7.7	BAA1/BAA2	1	1	-20	-16	-0.1	-0.2	1.5	1.4
Metals & Mining	1%	72.0	65		5.2	7.7	A3/BAA1	-1	-1	-24	-13	0.0	-0.2	1.5	0.7
Paper	0%	26.3	86		5.3	6.0	BAA2/BAA3	0	0	-19	-10	-0.1	-0.2	1.2	0.4
Capital Goods	6%	412.2	57		5.0	6.8	A3/BAA1	1	1	-19	-11	-0.1	-0.2	0.8	0.5
Aerospace & Defense	2%	150.9	63		5.2	8.3	BAA1/BAA2	1	1	-19	-12	-0.1	-0.3	1.1	0.6
Building Materials	1%	40.4	73		5.2	6.7	BAA1/BAA2	5	5	-16	-5	-0.1	-0.2	1.0	0.4
Diversified Manufacturing	2%	118.0	56		5.0	6.6	A3/BAA1	0	0	-19	-8	-0.1	-0.2	0.8	0.5
Construction Machinery	1%	57.6	34		4.6	4.7	A1/A2	3	3	-24	-17	-0.1	-0.2	0.5	0.4
Packaging	0%	14.3	71		5.0	4.5	BAA2/BAA3	0	0	-21	-10	-0.1	-0.2	0.7	0.4
Environmental	0%	31.0	42		4.8	5.8	A3/A3	1	1	-20	-14	-0.1	-0.2	0.3	0.2
Communications	8%	562.2	89		5.5	8.7	A3/BAA1	2	2	-11	-6	-0.1	-0.3	0.8	0.1
Cable & Satellite	1%	107.7	132		6.0	9.4	BAA1/BAA2	2	2	-5	1	-0.1	-0.4	0.5	0.2
Media & Entertainment	2%	136.4	77		5.4	8.7	A1/A2	1	1	-6	-9	0.0	-0.2	0.4	-1.1
Wireless	2%	125.3	74		5.2	7.3	BAA1/BAA2	1	1	-18	-8	-0.1	-0.3	1.0	0.6
Wirelines	3%	192.9	84		5.5	9.3	BAA1/BAA2	3	3	-12	-6	-0.2	-0.4	0.9	0.5
Consumer Cyclical	7%	542.0	65		5.1	6.3	A2/A3	2	2	-21	-9	-0.1	-0.3	0.9	0.4
Automotive	2%	170.0	80		5.1	4.3	A3/BAA1	2	2	-24	-10	-0.1	-0.3	1.1	0.8
Leisure	0%	4.8	94		5.3	6.4	BAA2/BAA2	2	2	-21	-8	-0.2	-0.4	1.5	0.0
Gaming	0%	22.8	102		5.3	4.8	BAA2/BAA3	1	1	-22	-8	-0.2	-0.3	1.3	0.7
Home Construction	0%	9.7	81		5.1	5.1	BAA1/BAA2	0	0	-17	-3	0.0	-0.2	0.8	0.4
Lodging	0%	20.0	67		5.0	4.8	BAA2/BAA3	1	1	-24	-9	-0.1	-0.3	0.9	0.3
Retailers	3%	252.6	50		5.0	7.8	A1/A2	3	3	-18	-5	-0.1	-0.3	0.7	0.2
Restaurants	0%	33.4	56		5.1	7.6	BAA1/BAA2	0	0	-22	-12	-0.1	-0.2	0.9	0.5
Consumer Cyc Services	0%	28.8	72		5.1	6.2	A3/BAA1	1	1	-17	0	0.0	-0.2	0.8	0.1
Consumer Non-Cyclical	15%	1123.6	62		5.2	7.9	A2/A3	1	1	-18	-8	-0.1	-0.2	0.9	0.3
Consumer Products	1%	59.5	44		4.8	6.0	A2/A3	2	2	-20	-12	-0.2	-0.3	0.6	0.4
Food & Beverage	3%	232.1	67		5.2	7.7	A3/BAA1	2	2	-19	-3	-0.1	-0.3	1.0	0.1
Healthcare	4%	327.2	70		5.2	7.9	A3/BAA1	0	0	-17	-8	-0.1	-0.2	0.8	0.3
Pharmaceuticals	5%	405.9	53		5.1	8.5	A1/A2	2	2	-18	-10	-0.1	-0.3	0.8	0.3
Supermarkets	0%	13.5	81		5.5	9.6	BAA1/BAA2	0	0	-15	-6	0.0	-0.2	1.1	0.3
Tobacco	1%	85.5	69		5.1	6.6	A3/BAA1	1	1	-18	-11	-0.2	-0.3	0.9	0.4
Energy	7%	532.3	74		5.3	7.7	A3/BAA1	1	1	-19	-17	-0.1	-0.2	1.2	1.3
Independent Energy	1%	92.9	73		5.3	7.6	BAA1/BAA2	0	0	-22	-23	-0.1	-0.2	1.5	2.1
Integrated Energy	1%	103.3	49		5.1	8.3	AA3/A1	1	1	-19	-16	-0.1	-0.2	0.8	0.7
Oil Field Services	0%	24.6	75		5.3	7.7	A3/BAA1	3	3	-19	-23	-0.2	-0.3	1.2	1.5
Refining	0%	30.3	83		5.3	7.2	BAA1/BAA2	-1	-1	-19	-21	-0.1	-0.2	1.3	1.7
Midstream	4%	281.1	82		5.4	7.5	BAA1/BAA2	1	1	-18	-15	-0.2	-0.3	1.3	1.2
Technology	10%	759.6	75		5.2	7.3	A2/A3	2	2	-18	-2	0.0	-0.3	1.1	0.4
Transportation	2%	147.0	66		5.3	9.3	A3/BAA1	1	1	-19	-9	-0.1	-0.2	1.1	0.6
Airlines	0%	16.5	84		5.1	3.7	A2/A3	3	3	-18	-7	0.0	-0.1	0.8	0.8
Railroads	1%	88.8	61		5.4	11.0	A3/BAA1	2	2	-19	-10	-0.1	-0.2	1.2	0.5
Transportation Services	1%	41.6	69		5.2	7.9	A3/BAA1	0	0	-18	-8	-0.1	-0.2	1.0	0.7
Other Industrial	1%	38.6	58		5.3	10.4	AA3/A1	-1	-1	-17	-8	0.0	-0.1	0.8	0.5
Utility	10%	762.6	78		5.3	8.2	A3/BAA1	1	1	-16	-8	-0.2	-0.3	0.9	0.5
Electric	9%	687.8	77		5.3	8.2	A3/BAA1	1	1	-16	-8	-0.2	-0.3	0.9	0.4
Natural Gas	1%	58.0	87		5.4	7.9	A3/BAA1	2	2	-17	-9	-0.2	-0.3	1.1	0.6
Other Utility	0%	16.8	66		5.2	8.1	BAA1/BAA2	2	2	-17	-5	-0.2	-0.3	0.8	0.0
Financial Institutions	32%	2410.1	76		5.1	5.4	A2/A3	1	1	-19	-2	-0.1	-0.2	0.9	0.2
Banking	22%	1647.6	71		5.0	4.9	A2/A3	1	1	-17	0	-0.1	-0.3	0.8	0.2
Brokerage Asset Managers Exchanges	2%	119.3	75		5.2	6.2	A3/BAA1	1	1	-20	2	-0.2	-0.3	1.1	0.2
Finance Companies	1%	107.8	147		5.7	3.6	BAA2/BAA3	0	0	-19	4	0.0	-0.2	1.5	0.5
Insurance	4%	333.2	82		5.4	8.3	A3/BAA1	0	0	-23	-9	-0.1	-0.2	1.5	0.4
REITS	3%	194.7	70		5.0	5.4	A3/BAA1	0	0	-21	-9	-0.1	-0.2	0.8	0.4
Other Financial	0%	7.5	91		5.2	4.7	A3/BAA1	0	0	-19	-10	-0.1	-0.2	1.1	1.0
US Corporate IG		7469.6	73		5.2	7.0	A3/BAA1	1	1	-18	-6	-0.1	-0.3	0.9	0.4

Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 23: Sector performance: USD HY

Sector	Metrics							Spread change				Total return			
	Index weight	Mkt val (\$bn)	OAS	Percentile rank (2010-today)	YTW	Dur	Rating	1-week	MTD	QTD	YTD	1-week	MTD	QTD	YTD
Basic Industry	6%	93.1	246		7.0	3.3	BA3/B1	4	4	-24	-15	0.0	-0.2	2.6	2.4
Chemicals	3%	47.6	218		6.7	3.5	BA3/B1	3	3	-27	-32	-0.1	-0.2	2.4	3.8
Metals & Mining	3%	38.3	194		6.4	3.2	BA3/B1	8	8	-30	-7	0.1	-0.2	3.0	1.8
Paper	0%	7.2	711		11.5	2.6	B1/B2	2	2	-1	42	-0.2	-0.4	1.5	-2.9
Capital Goods	10%	149.8	214		6.6	3.1	BA3/B1	3	3	-17	1	0.1	-0.1	1.8	-0.1
Aerospace & Defense	2%	30.7	108		5.7	2.9	BA3/B1	4	4	-22	8	-0.1	-0.2	1.3	1.1
Building Materials	2%	36.3	345		8.0	3.4	BA3/B1	-2	-2	-17	21	0.6	0.4	2.4	-1.1
Diversified Manufacturing	2%	27.8	158		6.0	2.6	BA2/BA3	-1	-1	-20	50	0.0	-0.1	1.1	0.9
Construction Machinery	1%	21.5	122		5.7	3.0	BA3/B1	40	40	-33	-7	-0.1	-0.3	1.9	1.6
Packaging	2%	25.2	319		7.6	3.2	B1/B2	1	1	-6	-23	-0.1	-0.3	2.3	-2.6
Environmental	1%	8.3	147		6.0	3.2	BA3/B1	0	0	-14	-1	-0.1	-0.2	1.1	1.2
Communications	15%	216.6	409		8.5	3.7	B1/B2	2	2	-6	18	-0.2	-0.3	1.3	1.2
Cable & Satellite	5%	79.9	557		9.8	2.9	B1/B2	3	3	3	39	-0.5	-0.4	-1.0	-0.5
Media & Entertainment	5%	80.7	312		7.6	4.2	BA3/B1	1	1	-12	-7	0.1	-0.1	3.1	1.8
Wireless	1%	19.7	266		7.2	4.1	BA3/B1	2	2	-17	6	0.0	-0.1	2.1	0.7
Wirelines	2%	36.3	376		8.3	4.2	B1/B2	4	4	-9	20	-0.3	-0.4	2.3	3.4
Consumer Cyclical	18%	271.8	236		6.8	3.2	BA3/B1	2	2	-21	-6	-0.1	-0.2	2.2	1.4
Automotive	3%	47.3	259		6.9	3.1	BA3/B1	2	2	-25	-6	-0.2	-0.3	2.5	1.4
Leisure	2%	29.2	197		6.4	3.4	BA3/B1	16	16	-15	10	-0.3	-0.4	1.4	0.9
Gaming	3%	45.3	261		7.1	3.0	BA3/B1	1	1	-15	7	-0.1	-0.2	1.9	1.1
Home Construction	1%	18.4	186		6.3	2.9	BA3/B1	-2	-2	-27	-8	0.2	0.1	2.3	1.7
Lodging	1%	16.5	119		5.7	3.8	BA2/BA3	2	2	-28	-10	-0.1	-0.2	1.8	1.1
Retailers	4%	61.1	235		6.7	3.3	BA3/B1	1	1	-30	-28	0.0	-0.1	2.7	1.8
Restaurants	1%	11.7	161		6.0	3.9	BA3/B1	4	4	-14	-1	-0.2	-0.2	1.6	1.3
Consumer Cyc Services	3%	42.4	297		7.4	2.6	B1/B2	0	0	-16	7	0.0	-0.2	2.2	1.2
Consumer Non-Cyclical	11%	160.2	250		7.0	3.2	B1/B2	4	4	-11	2	-0.2	-0.3	1.6	1.6
Consumer Products	2%	24.6	336		7.7	4.0	BA3/B1	1	1	-5	3	-0.5	-0.6	0.5	-0.6
Food & Beverage	2%	28.9	174		6.2	3.5	B1/B2	3	3	-5	2	-0.2	-0.3	1.1	1.1
Healthcare	5%	71.5	215		6.6	2.8	B1/B2	7	7	-14	3	-0.1	-0.2	1.8	2.1
Pharmaceuticals	2%	25.2	388		8.4	3.0	B2/B3	1	1	-17	-11	-0.2	-0.3	3.3	3.3
Supermarkets	1%	9.8	158		6.0	3.7	BA2/BA3	2	2	-5	129	-0.2	-0.3	0.4	0.3
Tobacco	0%	0.3	186		6.5	3.3	BA3/BA3	-4	-4	-25	8	0.0	0.0	2.3	0.4
Energy	11%	167.3	156		6.1	3.1	BA2/BA3	3	3	-21	-42	0.0	-0.1	1.6	4.2
Independent Energy	3%	45.0	168		6.3	2.8	BA3/B1	2	2	-17	-32	0.0	0.0	1.6	4.3
Oil Field Services	2%	27.4	172		6.4	2.9	BA3/B1	3	3	-17	-43	-0.1	-0.1	1.3	4.0
Refining	0%	6.1	220		6.8	2.3	BA3/B1	1	1	-12	-39	0.1	0.0	1.3	5.7
Midstream	6%	88.8	141		5.9	3.4	BA2/BA3	5	5	-25	-48	0.0	-0.1	1.7	4.0
Technology	9%	139.2	322		7.7	3.1	BA3/B1	-1	-1	-21	2	0.2	0.0	3.0	1.2
Transportation	2%	31.6	447		8.9	2.7	BA3/B1	2	2	-12	23	-0.4	-0.5	1.6	-0.8
Airlines	1%	13.2	383		8.1	2.6	BA3/B1	5	5	-15	38	-0.8	-0.9	1.7	-0.7
Railroads	0%	1.9	1270		17.4	2.6	B2/B3	0	0	15	32	0.0	0.1	-6.8	-7.1
Transportation Services	1%	16.5	405		8.5	2.8	B1/B2	0	0	-15	15	-0.1	-0.2	2.6	-0.1
Other Industrial	1%	15.9	211		6.7	3.1	BA3/B1	5	5	-19	-19	-0.1	-0.1	1.8	0.9
Utility	3%	44.6	159		6.1	4.0	BA2/BA3	-3	-3	-22	8	-0.1	-0.2	1.8	2.0
Financial Institutions	13%	196.6	242		6.9	3.2	BA3/B1	2	2	-25	8	-0.1	-0.2	2.5	0.7
Banking	0%	3.1	186		6.2	4.3	BA1/BA2	-1	-1	-32	-14	-0.1	-0.2	2.8	0.0
Brokerage Asset Managers Exchanges	1%	18.1	184		6.4	3.1	BA3/B1	6	6	-31	-2	-0.3	-0.4	2.7	1.5
Finance Companies	4%	63.7	262		7.1	3.1	BA2/BA3	3	3	-24	26	-0.1	-0.3	2.8	-0.1
Insurance	4%	63.0	226		6.7	3.5	BA3/B1	1	1	-23	18	0.0	-0.2	1.9	0.3
REITS	2%	29.5	243		6.8	2.8	B1/B2	1	1	-27	-10	-0.1	-0.2	3.0	2.2
Other Financial	1%	19.3	286		7.5	3.0	BA3/B1	1	1	-23	-10	-0.1	-0.2	2.8	1.9
US Corp High Yield		1486.7	263		7.1	3.3	BA3/B1	2	2	-17	-1	-0.1	-0.2	2.0	1.5

Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 24: Sector performance: EUR IG

Sector	Metrics							Spread Change				Total Return			
	%Mkt	MktVal (€bn)	OAS	Percentile	YTM	Dur	Rating	1-Wk	MTD	QTD	YTD	1-Wk	MTD	QTD	YTD
Consumer Goods	12.6	€ 397.2	82.1		3.7	4.7	A	-3	-3	-21	-2	-0.06	-0.28	1.63	0.52
Automobiles & Parts	5.2	€ 164.8	96.6		3.8	3.8	A	-5	-5	-30	-5	-0.02	-0.22	1.72	0.62
Beverages	2.9	€ 92.2	79.8		3.8	5.9	A	-1	-2	-15	-1	-0.10	-0.36	1.70	0.54
Food Producers	2.0	€ 62.1	68.0		3.6	5.2	A	-2	-3	-15	-2	-0.12	-0.34	1.44	0.40
Household Goods	0.2	€ 6.3	49.6		3.5	6.1	AA	-2	-3	-18	-2	-0.14	-0.35	1.36	0.36
Personal Goods	1.6	€ 51.4	63.3		3.5	4.4	A	-3	-3	-20	0	-0.10	-0.30	1.42	0.26
Tobacco	0.7	€ 20.5	104.6		3.9	4.7	BBB	-6	-6	-23	-8	0.00	-0.24	1.74	0.76
Consumer Services	4.5	€ 140.6	85.1		3.8	5.4	A	-2	-3	-15	3	-0.08	-0.32	1.44	0.38
Food & Drug Retailers	0.6	€ 19.3	82.5		3.6	4.4	BBB	-3	-3	-20	-3	-0.09	-0.29	1.50	0.52
General Retailers	0.6	€ 18.8	87.2		4.0	6.9	AA	-4	-4	-15	8	-0.07	-0.27	1.56	0.45
Media	1.0	€ 32.6	88.4		3.7	4.3	BBB	-4	-3	-22	5	-0.03	-0.24	1.55	0.38
Travel & Leisure	2.2	€ 69.9	83.7		3.8	5.8	A	-1	-1	-11	2	-0.10	-0.35	1.34	0.32
Energy	3.1	€ 97.6	91.1		3.8	5.0	A	-4	-5	-20	-7	-0.02	-0.25	1.68	0.90
Oil & Gas Producers	2.8	€ 87.4	91.3		3.8	4.9	A	-5	-5	-21	-7	-0.01	-0.24	1.70	0.90
Oil Equipment / Services & Distribution	0.2	€ 7.7	89.8		3.9	5.6	A	-2	-2	1	13	-0.14	-0.39	1.54	0.89
Renewable Energy	0.1	€ 2.5	89.1		3.7	5.0	BBB	-3	-5	-17	-12	-0.14	-0.35	1.47	1.05
Industrials	9.6	€ 303.1	78.0		3.6	4.5	A	-3	-4	-20	-3	-0.08	-0.29	1.51	0.45
Aerospace & Defense	0.5	€ 15.8	67.6		3.6	4.6	A	-2	-3	-17	3	-0.15	-0.35	1.32	0.17
Construction & Materials	1.6	€ 51.6	80.4		3.6	4.2	BBB	-3	-4	-22	-3	-0.06	-0.26	1.55	0.52
Electronic & Electrical Equipment	0.9	€ 28.9	67.7		3.5	4.9	A	-3	-3	-19	-2	-0.10	-0.31	1.54	0.43
General Industrials	1.3	€ 39.3	68.7		3.6	5.3	A	-4	-4	-23	-6	-0.07	-0.30	1.68	0.49
Industrial Engineering	1.1	€ 35.4	74.5		3.5	3.4	A	-3	-3	-23	-3	-0.08	-0.23	1.44	0.52
Industrial Transportation	3.3	€ 102.4	82.4		3.7	4.8	A	-3	-3	-18	-2	-0.09	-0.32	1.44	0.36
Support Services	0.9	€ 29.7	93.2		3.7	4.1	BBB	-2	-3	-23	-2	-0.08	-0.26	1.56	0.60
Health Care	5.5	€ 173.4	75.4		3.8	5.9	A	-3	-3	-18	-4	-0.12	-0.37	1.54	0.56
Health Care Equipment & Services	2.4	€ 74.8	80.3		3.8	5.8	A	-2	-3	-18	-3	-0.11	-0.35	1.54	0.52
Pharmaceuticals & Biotechnology	3.1	€ 98.6	71.8		3.7	6.0	A	-4	-4	-19	-5	-0.13	-0.39	1.54	0.60
Basic Materials	2.7	€ 84.2	80.4		3.7	4.9	A	-4	-4	-24	-6	-0.09	-0.33	1.69	0.62
Chemicals	2.0	€ 63.8	79.3		3.7	5.0	A	-3	-3	-23	-5	-0.09	-0.34	1.69	0.60
Forestry & Paper	0.2	€ 5.6	83.4		3.6	4.2	BBB	-7	-7	-33	-3	0.01	-0.21	1.92	0.50
Industrial Metals	0.1	€ 3.7	85.7		3.7	5.1	BBB	-4	-5	-24	-7	-0.14	-0.36	1.69	0.71
Mining	0.4	€ 11.1	83.8		3.6	4.2	BBB	-2	-3	-25	-13	-0.12	-0.30	1.59	0.78
Technology	2.7	€ 84.9	89.8		3.9	5.8	A	-5	-6	-20	3	-0.01	-0.24	1.63	0.24
Software & Computer Services	2.3	€ 72.2	92.8		4.0	6.2	A	-6	-7	-20	3	0.00	-0.24	1.65	0.24
Technology Hardware & Equipment	0.4	€ 12.7	60.8		3.4	3.7	A	-4	-5	-28	1	-0.07	-0.23	1.51	0.24
Telecommunications	5.5	€ 172.3	89.1		3.8	5.5	BBB	-3	-3	-19	-2	-0.09	-0.34	1.61	0.54
Fixed Line Telecommunications	4.4	€ 138.0	87.9		3.8	5.5	BBB	-2	-3	-18	-3	-0.10	-0.35	1.57	0.53
Mobile Telecommunications	1.1	€ 34.3	94.4		3.9	5.3	BBB	-4	-4	-20	-1	-0.06	-0.31	1.74	0.57
Financials	42.2	€ 1,325.6	92.8		3.7	4.1	A	-2	-3	-23	-2	-0.10	-0.29	1.54	0.53
Banks Sen	26.3	€ 827.1	81.6		3.6	3.9	A	-2	-3	-21	-3	-0.10	-0.28	1.42	0.51
Banks Sub	4.1	€ 129.8	124.9		4.0	3.6	BBB	-1	-2	-29	-2	-0.14	-0.30	1.68	0.69
Financial Services Sen	2.8	€ 88.7	80.0		3.7	4.6	A	-3	-4	-20	-3	-0.10	-0.30	1.55	0.62
Financial Services Sub	0.0	€ 1.0	104.4		3.7	2.1	A	-9	-9	-44	-34	0.07	-0.04	1.52	1.38
Insurance Sen	1.4	€ 45.4	75.4		3.6	4.2	A	-2	-3	-24	0	-0.12	-0.29	1.54	0.34
Insurance Sub	2.7	€ 84.3	134.6		4.2	4.9	BBB	0	-1	-21	1	-0.15	-0.36	1.73	0.54
Real Estate Sen	4.7	€ 148.7	110.1		3.9	4.8	BBB	-3	-4	-29	1	-0.08	-0.29	2.01	0.44
Real Estate Sub	0.0	€ 0.5	171.0		4.5	5.7	A	-1	-6	-36	0	0.03	0.01	3.02	0.00
Utilities	11.5	€ 362.5	93.3		3.8	5.2	BBB	-4	-4	-16	-1	-0.06	-0.30	1.51	0.59
Electricity	7.0	€ 219.3	95.9		3.9	5.2	BBB	-4	-4	-17	-1	-0.05	-0.29	1.54	0.61
Gas / Water & Multiutilities	4.6	€ 143.3	89.2		3.8	5.2	BBB	-4	-4	-15	-1	-0.07	-0.32	1.46	0.55
Index	100.0	€ 3,141.5	87.7		3.7	4.7	A	-3	-3	-20	-2	-0.09	-0.30	1.56	0.53

Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 25: Sector performance: EUR HY

Sector	Metrics							Spread Change				Total Return			
	%Mkt	MktVal (€bn)	OAS	Percentile	YTM	Dur	Rating	1-Wk	MTD	QTD	YTD	1-Wk	MTD	QTD	YTD
Consumers	27.4	€ 125.5	291		5.6	2.7	BB	-16	-16	-66	15	0.28	0.12	2.96	1.21
Consumer Goods	15.2	€ 69.5	264		5.4	2.8	BB	-12	-13	-57	28	0.21	0.09	2.85	1.25
Consumer Services	12.2	€ 56.0	325		6.0	2.7	BB	-22	-21	-68	7	0.35	0.15	3.11	1.16
Energy	10.0	€ 45.6	244		5.2	3.1	BB	-8	-7	-45	-14	0.05	-0.12	1.86	0.97
Energy	2.3	€ 10.3	242		5.1	2.7	BB	-10	-10	-69	-59	0.14	0.00	2.55	2.64
Utilities	7.7	€ 35.3	244		5.2	3.2	BB	-8	-7	-39	-2	0.03	-0.15	1.66	0.51
Industrials	31.1	€ 142.0	323		6.0	2.7	BB	-10	-10	-71	-21	0.16	0.01	3.19	1.92
Basic Materials	6.1	€ 27.9	411		6.8	2.7	B	-1	-6	-162	-152	-0.08	-0.07	6.05	5.88
Construction & Materials	1.5	€ 7.0	510		7.8	3.0	BB	-157	-159	-204	-98	0.08	0.01	3.59	0.11
Health Care	8.9	€ 40.5	306		5.8	2.6	BB	13	11	-35	40	0.03	-0.15	2.32	1.36
Industrials	14.6	€ 66.6	276		5.5	2.7	BB	-20	-19	-65	-10	0.34	0.14	2.64	1.02
TMT	20.0	€ 91.5	313		5.9	3.4	BB	-9	-5	-52	27	0.32	0.06	2.98	0.90
Technology	4.1	€ 18.9	419		7.0	3.5	B	-8	1	-67	41	0.75	0.30	3.62	-0.51
Telecommunications	15.9	€ 72.6	284		5.6	3.3	BB	-10	-8	-52	11	0.21	0.00	2.82	1.19
Financials	11.5	€ 52.6	358		6.3	3.2	BB	-10	-10	-54	31	0.03	-0.12	2.22	0.08
Core Financials	4.3	€ 19.7	191		4.6	3.1	BB	-2	-3	-48	12	-0.04	-0.16	2.05	0.56
Financial Services	1.5	€ 6.9	866		11.4	2.2	B	-31	-30	186	265	-0.34	-0.48	-1.83	-3.19
Real Estate	5.7	€ 26.0	386		6.6	3.4	BB	-13	-14	-78	8	0.17	0.00	3.54	0.68
Index	100.0	€ 457.2	308		5.8	2.9	BB	-11	-10	-61	6	0.20	0.02	2.84	1.22

0 50 100

Source: iBoxx, Goldman Sachs Global Investment Research

Performance dashboard: USD Credit ETFs

Exhibit 26: USD ETF performance

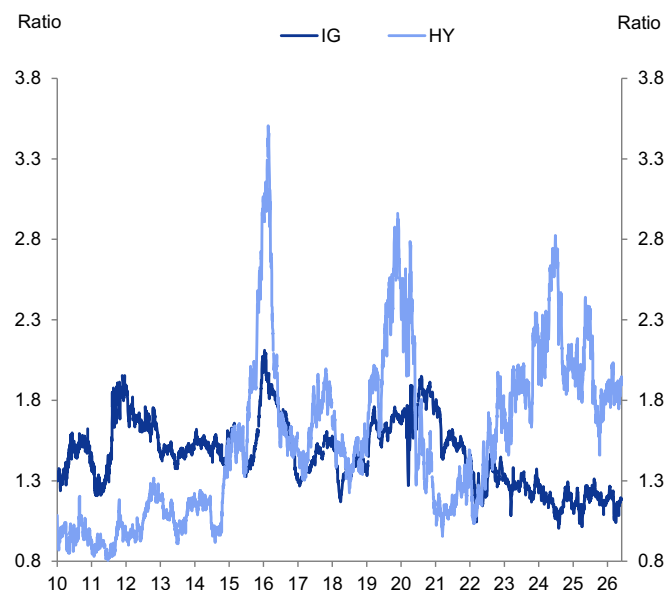
Sector	ETF Ticker	AUM	Weekly Flow (\$Mn.)	YTD cum flow (% of AUM)	Shares Outstanding (Mn.)	Short Interest (%)	Price	% NAV Premium	YTD Total Return	QTD Total Return	MTD Total Return	Benchmark Index
Investment Grade	LQD	\$29.46B	-84	0.2%	271	19.34	108.92	-0.01%	0.7	1.1	0.0	iBoxx USD Liquid Investment Grade Index
	VCSH	\$43.86B	931	25.7%	557	0.91	78.88	0.04%	0.8	0.6	0.0	Bloomberg Barclays US Corporate 1-5 Yrs Total Return Index Value Unhedged
	VCIT	\$66.42B	585	34.3%	808	2.27	82.45	0.04%	0.4	0.9	-0.1	Bloomberg Barclays US Credit Corporate 5-10 Yrs Total Return Index Value Unhedged
	FLOT	\$9.56B	76	26.2%	188	--	50.94	0.16%	1.9	1.0	0.2	Bloomberg Barclays US FRN < 5 yr Total Return Index Value Unhedged
	IGSB	\$22.36B	435	6.2%	428	0.73	52.32	0.05%	0.8	0.7	0.0	ICE BofAML 1-5 Year US Corporate Bond Index
	IGIB	\$18.17B	151	21.1%	343	1.88	53.04	0.03%	0.4	0.9	-0.1	Bloomberg Barclays U.S. Interim
	SPSB	\$10.23B	45	23.2%	342	1.70	29.96	-0.01%	1.0	0.7	0.0	Bloomberg Barclays US Corporate 1-3 Yrs Total Return Index Value Unhedged
	SPIB	\$10.96B	-655	9.7%	329	3.35	33.40	0.05%	0.6	0.7	-0.1	Bloomberg Barclays Intermediate Corporate Total Return Index Value Unhedged
	VCIT	\$8.38B	-114	-38.7%	112	23.86	75.12	0.15%	1.3	1.9	0.1	Bloomberg Barclays Long U.S. Corporate Total Return Index Value Unhedged
High Yield	HYG	\$16.18B	392	8.9%	203	42.22	79.78	0.11%	1.4	1.8	-0.2	iBoxx USD Liquid High Yield Index
	JNK	\$7.76B	116	1.4%	81	11.06	96.12	0.08%	1.6	2.1	-0.1	Bloomberg Barclays VLI: High Yield Total Return Index Value Unhedged USD
	SJNK	\$4.79B	52	3.6%	192	2.36	24.96	0.14%	1.4	1.7	-0.1	Bloomberg Barclays US High Yield 350M Cash 0-5yr 2% Cap Total Return Index
	SHYG	\$7.53B	0	22.9%	178	0.64	42.30	0.11%	1.6	1.7	-0.1	Markit iBoxx USD Liquid High Yield 0-5 Total Return Index
	HYS	\$1.68B	5	28.7%	18	0.04	93.27	0.05%	1.4	1.8	-0.1	ICE BofAML 0-5 Year US High Yield Constrained Index
	HYLS	\$1.65B	6	1.3%	40	0.22	40.83	0.09%	0.3	1.8	-0.2	ICE BofAML US High Yield Constrained Index
	ANGL	\$3.08B	3	-0.6%	106	0.94	29.07	0.00%	1.7	2.9	0.0	ICE BofAML US Fallen Angel High Yield Index
	BKLN	\$7.15B	2	-20.2%	349	24.59	20.51	-0.06%	0.2	1.5	0.2	S&P/LSTA U.S. Leveraged Loan 100 Index
	SRLN	\$5.21B	0	-30.9%	129	6.62	40.38	-0.21%	0.7	2.4	0.2	Markit iBoxx USD Liquid Leverage Loans Total Return Index
Loans	FTSL	\$2.3B	2	6.9%	51	0.16	44.97	0.09%	0.6	1.4	0.1	S&P/LSTA US Leveraged Loan 100 Market Value Index
	EMB	\$14.36B	268	-2.7%	150	11.55	96.14	0.22%	2.1	3.7	0.1	J.P. Morgan EMBI Global Core Index
	EMLC	\$4.85B	0	78.0%	191	1.34	25.45	-0.21%	1.1	3.0	-0.1	J.P. Morgan Government Bond Index Emerging Markets Global Core Index
EM	JAAA	\$27.76B	528	67.9%	549	17.34	50.56	0.00%	1.9	1.2	0.0	
	PAAA	\$10.23B	153	503.5%	199	--	51.33	0.01%	2.0	1.0	0.0	
	JBBB	\$1.2B	0	-11.7%	25	--	47.37	0.36%	1.8	2.6	0.1	
	CLOI	\$1.35B	11	70.2%	26	11.22	52.79	0.05%	2.1	1.4	0.1	
	CLOA	\$2.18B	36	234.7%	42	--	51.83	-0.02%	2.1	1.1	0.1	
CLOs												

Source: Bloomberg, data compiled by Goldman Sachs Global Investment Research

Performance dashboard: Bond-level dispersion

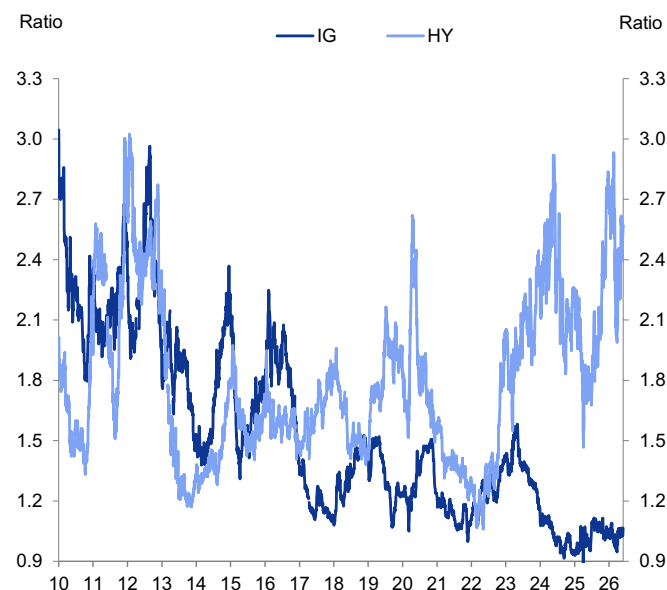
Note: Spread dispersion is the inter-decile range divided by the median across the bond index constituents and return dispersion is the inter-decile range of 10-day returns divided by the median across the bond index constituents.

Exhibit 27: USD spread dispersion



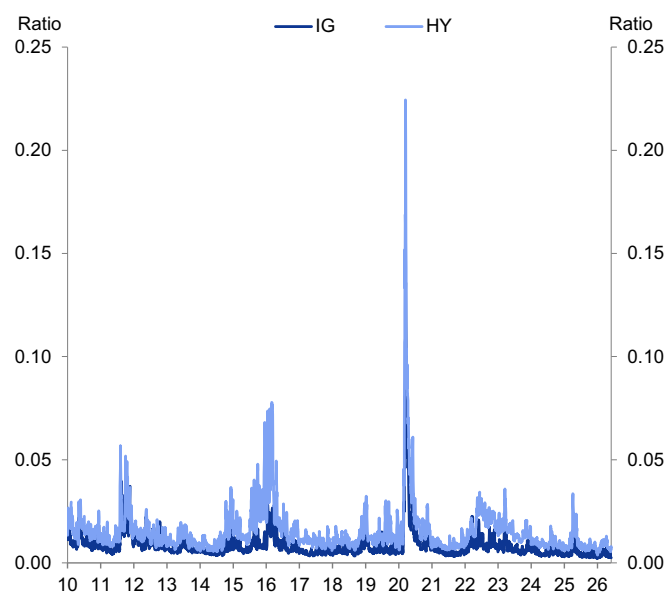
Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 28: EUR spread dispersion



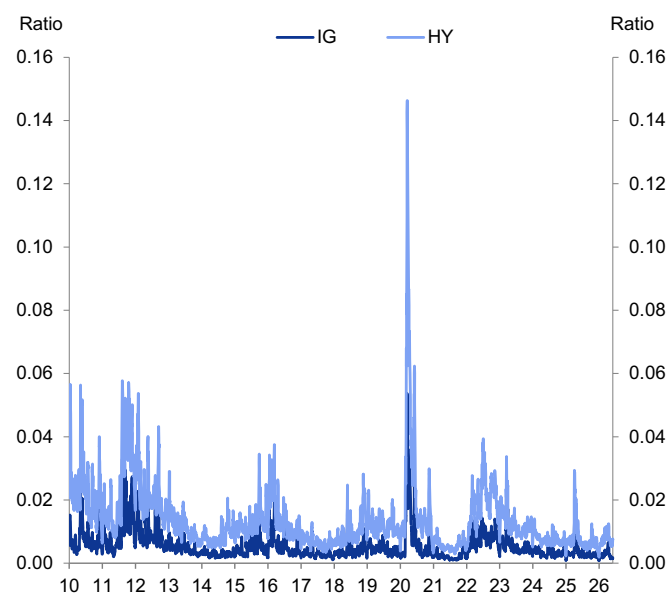
Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 29: USD return dispersion



Source: iBoxx, Goldman Sachs Global Investment Research

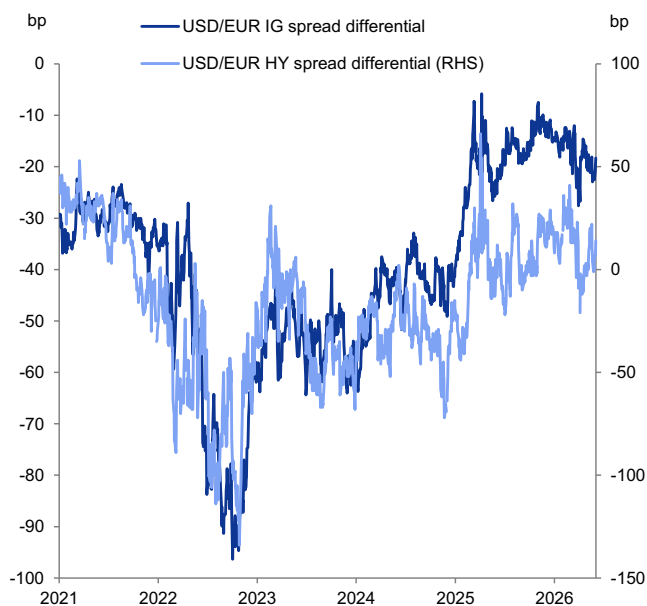
Exhibit 30: EUR return dispersion



Source: iBoxx, Goldman Sachs Global Investment Research

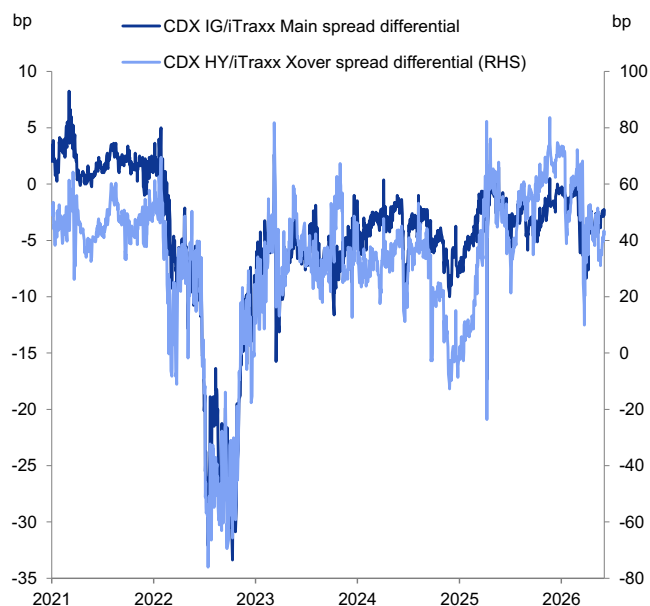
Relative value dashboard: Excess spreads in the USD and EUR markets

Exhibit 31: USD/EUR IG and HY spread differential



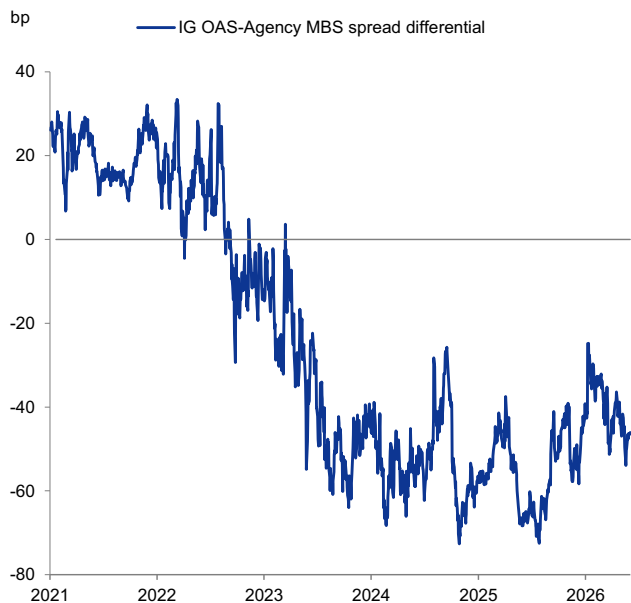
Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 32: CDX IG/iTraxx Main and CDX HY/iTraxx Xover spread differential



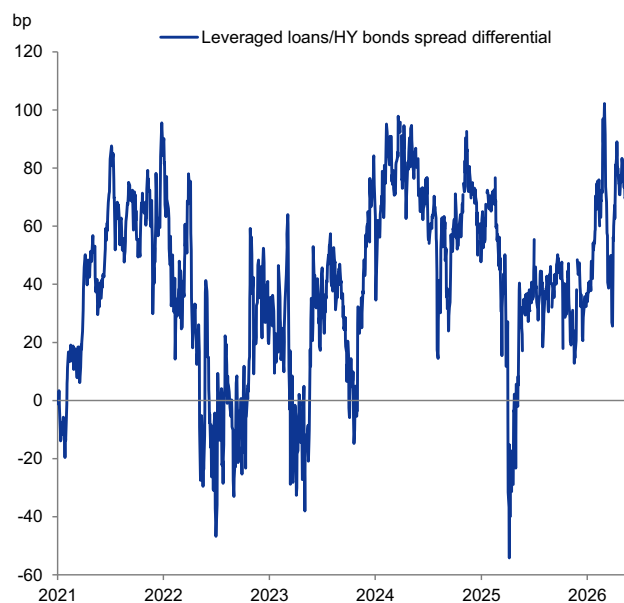
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 33: IG OAS/Agency MBS spread differential



Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 34: Leveraged loans/HY bonds spread differential



Source: Bloomberg, Morningstar LCD, Goldman Sachs Global Investment Research

Performance dashboard: Biggest movers in the USD bond market

Exhibit 35: Top and bottom 20 performers in the USD IG market

Issuer	Sector	Security	Amount outstanding (\$mn)	Ratings (Moody's/S&P/Fitch)	1-week price return (%)	1-week price change (pts)	Current price
Top 20 performers							
Flowers Foods Inc	Food	FLO 6.2 03/15/55	300	Baa3/BBB-/BBB-	2.6%	2.2	85.08
Concentrix Corp	Software	CNXC 6.85 08/02/33	550	Baa3/BBB-/BBB	1.9%	1.8	91.59
Santander Uk Group Hldgs	Banks	SANUK 5.136 09/22/36	1,000	Baa1/BBB/A	1.7%	1.6	97.15
Phillips Edison Grocery	REITS	PECO 2 5/8 11/15/31	350	Baa2/BBB/--	1.6%	1.4	87.36
Salesforce Inc	Software	CRM 6.7 03/15/66	1,000	A2/A+/-	1.5%	1.5	103.18
American Honda Finance	Auto Manufacturers	HNDA 5.1 01/08/36	500	A3/BBB+/A-	1.4%	1.4	98.59
Nvidia Corp	Semiconductors	NVDA 2.85 04/01/30	1,500	Aa1/AA/-	1.4%	1.3	94.92
Ford Motor Company	Auto Manufacturers	F 6.1 08/19/32	1,750	Ba1/BBB-/BBB-	1.3%	1.4	102.57
Vale Overseas Limited	Iron/Steel	VALEBZ 3 3/4 07/08/30	1,500	--/--	1.3%	1.3	95.69
Humana Inc	Healthcare-Services	HUM 4.8 03/15/47	400	Baa2/BBB/BBB-	1.3%	1.1	82.96
Paypal Holdings Inc	Commercial Services	PYPL 5.15 06/01/34	850	A3/A-/A-	1.2%	1.2	99.64
Dick'S Sporting Goods	Retail	DKS 4.1 01/15/52	750	Baa2/BBB/--	1.2%	0.9	72.37
Hp Enterprise Co	Computers	HPE 5.6 10/15/54	1,500	Baa2/BBB/BBB+	1.1%	1.1	92.60
Helmerich & Payne Inc	Oil&Gas	HP 5 1/2 12/01/34	549	Baa2/BBB/--	1.1%	1.1	98.86
Dentsply Sirona Inc	Healthcare-Products	XRAY 3 1/4 06/01/30	750	Baa2/BBB/--	1.1%	1.0	92.60
Jpmorgan Chase & Co	Banks	JPM 4 1/4 10/01/27	1,500	A2/A-/A	1.1%	1.1	99.75
Charter Comm Opt Llc/Cap	Media	CHTR 5 1/8 07/01/49	1,250	Ba1/BBB-/BBB-	1.1%	0.8	76.89
Johnson & Johnson	Pharmaceuticals	JNJ 3 5/8 03/03/37	1,500	Aaa/AAA/-	1.0%	0.9	89.58
Block Financial Llc	Commercial Services	HRB 5 3/8 09/15/32	350	Baa3/BBB/--	1.0%	1.0	98.52
Comcast Corp	Media	CMCSA 4.049 11/01/52	1,006	A3/A-/A-	1.0%	0.7	72.25
Bottom 20 performers							
Ibm Corp	Computers	IBM 5.1 02/06/53	650	A3/A-/A-	-3.5%	-3.2	92.40
Kilroy Realty Lp	REITS	KRC 2.65 11/15/33	450	Baa3/BBB-/BBB-	-3.1%	-2.5	83.06
Ford Motor Company	Auto Manufacturers	F 5.291 12/08/46	1,300	Ba1/BBB-/BBB-	-3.1%	-2.6	84.03
Booz Allen Hamilton Inc	Computers	BAH 5.95 08/04/33	650	Baa3/BBB/--	-2.7%	-2.8	105.35
Fiserv Inc	Software	FISV 5 5/8 08/21/33	1,300	Baa2/BBB/--	-2.5%	-2.6	103.60
Flowers Foods Inc	Food	FLO 2.4 03/15/31	500	Baa3/BBB-/BBB-	-2.5%	-2.2	88.45
Reinsurance Grp Of Amer	Insurance	RGA 5 3/4 09/15/34	650	Baa1/A/BBB+	-2.4%	-2.5	104.82
At&T Inc	Telecommunications	T 4.85 03/01/39	956	Baa2/BBB/BBB+	-2.4%	-2.3	95.21
Gartner Inc	Computers	IT 4.95 03/20/31	350	Baa3/BBB-/BBB	-2.4%	-2.4	99.76
Brookfield Finance Inc	Private Equity	BNCN 5.675 01/15/35	450	A3/A-/A-	-2.3%	-2.4	103.78
Johnson & Johnson	Pharmaceuticals	JNJ 4.95 06/01/34	850	Aaa/AAA/-	-2.3%	-2.4	105.17
Kemper Corp	Insurance	KMPR 2.4 09/30/30	400	Baa3/BBB-/BBB-	-2.3%	-2.0	89.41
Altria Group Inc	Agriculture	MO 5 1/4 08/06/35	500	A3/BBB+/BBB+	-2.1%	-2.2	101.97
Nextera Energy Capital	Electric	NEE 5 1/4 02/28/53	1,150	Baa1/BBB+/A-	-2.0%	-1.8	91.08
Raymond James Financial	Diversified Finan Serv	RJF 4.9 09/11/35	650	A3/A-/A-	-2.0%	-2.0	99.14
Cigna Group/The	Healthcare-Services	CI 4.8 08/15/38	2,195	Baa1/A-/BBB+	-2.0%	-1.9	96.36
Sempra	Electric	SRE 6 3/8 04/01/56	800	Baa3/BBB-/BBB-	-2.0%	-2.0	102.68
Eli Lilly & Co	Pharmaceuticals	LLY 4.9 10/15/35	1,250	Aa3/AA/-	-1.9%	-2.0	101.55
At&T Inc	Telecommunications	T 5.55 08/15/41	541	Baa2/BBB/BBB+	-1.9%	-1.9	98.05
General Motors Co	Auto Manufacturers	GM 5.2 04/01/45	1,250	Baa2/BBB/BBB	-1.9%	-1.7	88.98

Note: Only Bloomberg index-eligible bonds.

Source: Bloomberg, TRACE, data compiled by Goldman Sachs Global Investment Research

Exhibit 36: Top and bottom 20 performers in the USD HY market

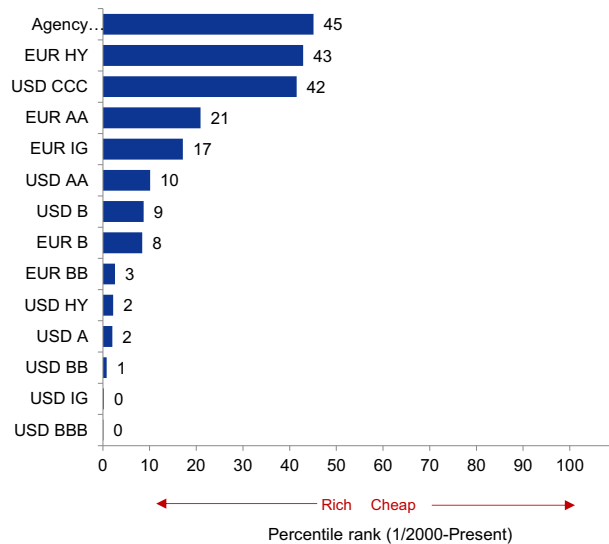
Issuer	Sector	Security	Amount outstanding (\$mn)	Ratings (Moody's/S&P/Fitch)	1-week price return (%)	1-week price change (pts)	Current price
Top 20 performers							
Csc Holdings LLC	Media	CSCHLD 5 11/15/31	500	Ca/CCC-/	12.8%	3.0	23.50
Csc Holdings LLC	Media	CSCHLD 3 3/8 02/15/31	1,000	Caa2/CCC+/	5.0%	2.8	55.50
Xerox Corporation	Office/Business Equip	XRX 13 1/2 04/15/31	500	Caa1/CCC-/	4.4%	2.9	67.00
Xerox Corporation	Office/Business Equip	XRX 4.8 03/01/35	250	Ca/CCC-/	4.1%	1.5	36.03
Csc Holdings LLC	Media	CSCHLD 4 1/2 11/15/31	1,500	Caa2/CCC+/	4.0%	2.3	56.75
Csc Holdings LLC	Media	CSCHLD 4 1/8 12/01/30	1,100	Caa2/CCC+/	3.9%	2.3	57.00
Ziprecruiter Inc	Internet	ZIP 5 01/15/30	550	B3/B/B	3.7%	2.5	68.72
Taylor Morrison Comm	Home Builders	TMHC 5 3/4 11/15/32	525	Ba1/BB+/	3.1%	3.1	100.48
Csc Holdings LLC	Media	CSCHLD 6 1/2 02/01/29	1,750	Caa2/CCC+/	2.5%	1.5	59.00
Cnt Prnt/Cdk Glo li/Fin	Software	CDK 8 06/15/29	755	B3/B-/	2.3%	1.3	55.00
Cleveland-Cliffs Inc	Iron/Steel	CLF 6 1/4 10/01/40	235	B3/B-/B-	1.8%	1.5	81.90
Iheartcommunications Inc	Media	IHRT 10 7/8 05/01/30	675	Caa3/CCC-/	1.8%	1.5	85.00
Paramount Global	Media	PARA 5.85 09/01/43	1,250	Baa3/BB+/BB+	1.7%	1.3	73.75
Taylor Morrison Comm	Home Builders	TMHC 5 1/8 08/01/30	500	Ba1/BB+/	1.6%	1.6	99.13
Sabre Gbl Inc	Leisure Time	SABHLD 10 3/4 03/15/30	470	B3/B-/	1.3%	1.2	89.05
Virtusa Corp	Computers	VRTU 7 1/8 12/15/28	350	Caa1/B-/	1.2%	1.0	83.27
Paramount Global	Media	PARA 4 3/8 03/15/43	1,446	Baa3/BB+/BB+	1.2%	0.8	63.75
Amc Entertainment Hldgs	Entertainment	AMC 7 1/2 02/15/29	360	Caa3/CCC+/	1.1%	1.0	87.25
Long Ridge Energy LLC	Electric	LONRID 8 3/4 02/15/32	600	B1/B-/	1.1%	1.1	103.75
Innophos Holdings Inc	Chemicals	IPHS 11 1/2 06/15/29	461	Caa1/B-/	1.0%	1.0	94.00
Bottom 20 performers							
Cntrl Parent/Cdk Gbl Inc	Software	CDK 7 1/4 06/15/29	750	B3/B-/	-27.0%	-18.8	69.50
Csc Holdings LLC	Media	CSCHLD 7 1/2 04/01/28	1,046	Ca/CCC-/	-10.3%	-3.5	34.00
Jetblue Airways/Loyalty	Airlines	JBLU 9 7/8 09/20/31	2,000	Caa1/B+/B	-5.4%	-5.0	92.13
Whirlpool Corp	Home Furnishings	WHR 4 1/2 06/01/46	500	B2/B+/BB-	-5.0%	-3.2	63.69
Whirlpool Corp	Home Furnishings	WHR 4.6 05/15/50	500	B2/B+/BB-	-4.6%	-2.9	63.19
Lcpr Sr Secured Fin Dac	Media	LILAPR 6 3/4 10/15/27	1,161	Caa3/CCC+/CCC+	-3.9%	-2.5	64.49
Embecka Corp	Healthcare-Products	EMBC 5 02/15/30	500	B3/B-/	-3.1%	-2.4	77.50
Embarq LLC	Telecommunications	EMBARQ 7.995 06/01/36	1,215	Ca/CCC-/	-3.0%	-0.8	27.00
Goodyear Tire & Rubber	Auto Parts&Equipment	GT 5 5/8 04/30/33	450	B2/B+/BB-	-3.0%	-2.6	87.13
Csc Holdings LLC	Media	CSCHLD 11 3/4 01/31/29	2,050	Caa2/CCC+/	-2.9%	-1.9	64.00
Csc Holdings LLC	Media	CSCHLD 11 1/4 05/15/28	1,000	Caa2/CCC+/	-2.6%	-1.8	66.63
Cobra Acquisitionco LLC	Diversified Finan Serv	EXETFI 6 3/8 11/01/29	400	B3/CCC/B-	-2.6%	-2.2	86.50
Whirlpool Corp	Home Furnishings	WHR 5 3/4 03/01/34	300	B2/B+/BB-	-2.5%	-2.1	83.61
Whirlpool Corp	Home Furnishings	WHR 6 1/2 06/15/33	600	B2/B+/BB-	-2.5%	-2.3	90.75
Whirlpool Corp	Home Furnishings	WHR 4.7 05/14/32	300	B2/B+/BB-	-2.4%	-2.0	82.50
Chs/Community Health Sys	Healthcare-Services	CYH 6 1/8 04/01/30	1,227	Caa3/CCC-/CCC-	-2.3%	-2.1	91.13
Navient Corp	Diversified Finan Serv	NAV 9 3/8 10/15/31	500	Ba3/BB-/BB-	-2.2%	-2.2	101.55
Kingpin Intermediate Ho	Leisure Time	LUCK 7 1/4 10/15/32	500	B3/B-/	-2.1%	-1.9	87.25
Owens-Brockway	Packaging&Containers	OI 7 1/4 05/15/31	690	B3/B+/	-2.1%	-2.1	98.43
Whirlpool Corp	Home Furnishings	WHR 5 1/2 03/01/33	300	B2/B+/BB-	-2.1%	-1.8	83.75

Note: Only Bloomberg index-eligible bonds.

Source: Bloomberg, TRACE, data compiled by Goldman Sachs Global Investment Research

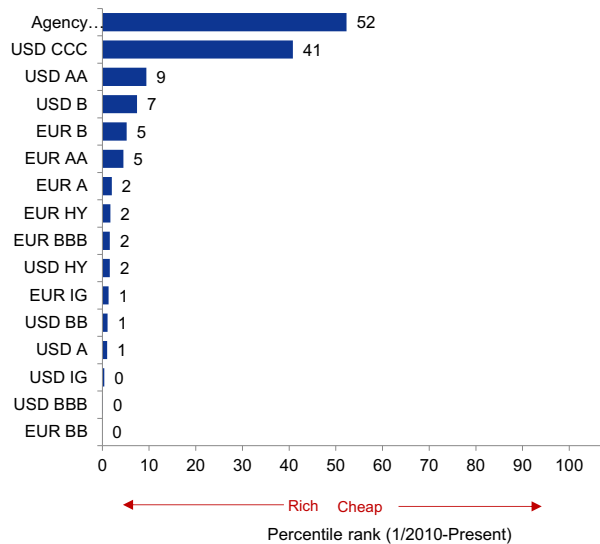
Valuation dashboard: Percentile ranks for spreads and yields

Exhibit 37: Percentile ranks for spreads across markets (2000-current)



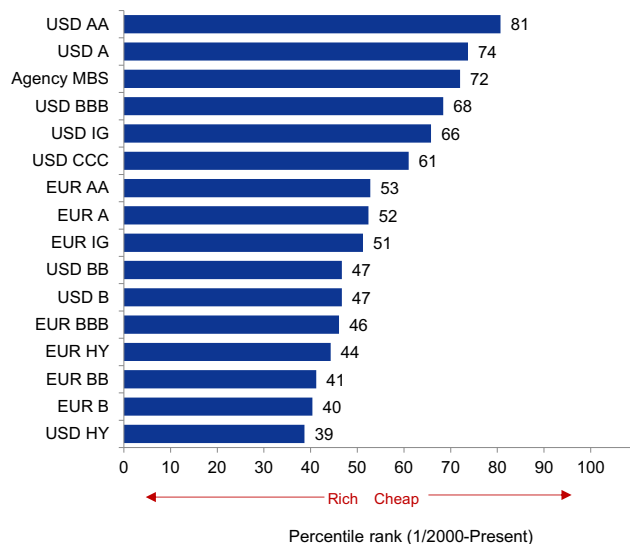
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 38: Percentile ranks for spreads across markets (2010-current)



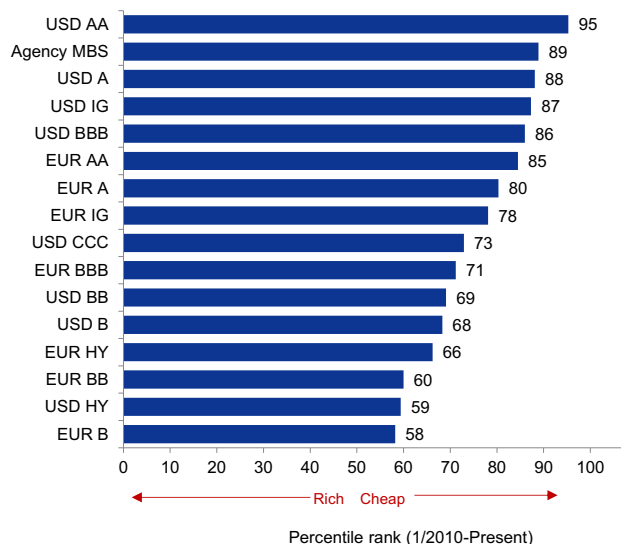
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 39: Percentile ranks for yields across markets (2000-current)



Source: Bloomberg, Goldman Sachs Global Investment Research

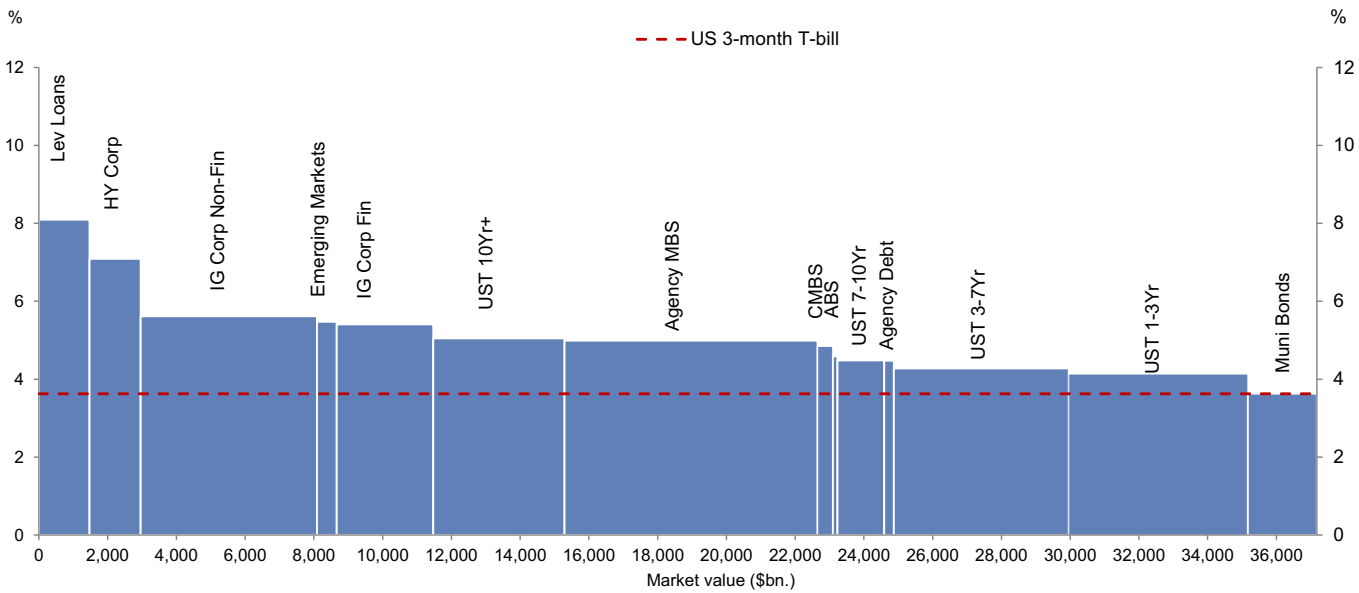
Exhibit 40: Percentile ranks for yields across markets (2010-current)



Source: Bloomberg, Goldman Sachs Global Investment Research

Valuation dashboard: Sizing the carry in USD and EUR fixed income markets

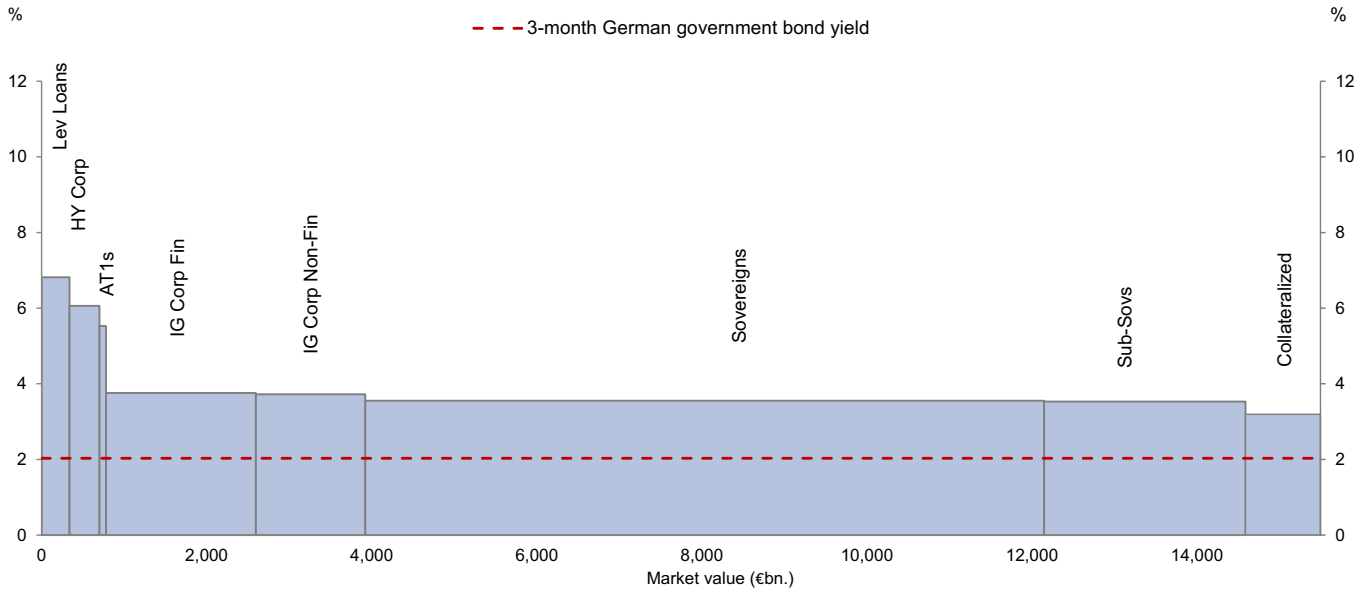
Exhibit 41: Average yield vs. total market value outstanding across the USD fixed income complex



Note: We use the iBoxx indices for the IG corporate credit and US Treasury markets, the Bloomberg indices for the HY, MBS, CMBS and ABS markets, and the Morningstar LSTA index for the leveraged loan market.

Source: iBoxx, Bloomberg, Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 42: Average yield vs. total market value outstanding for the EUR fixed income markets

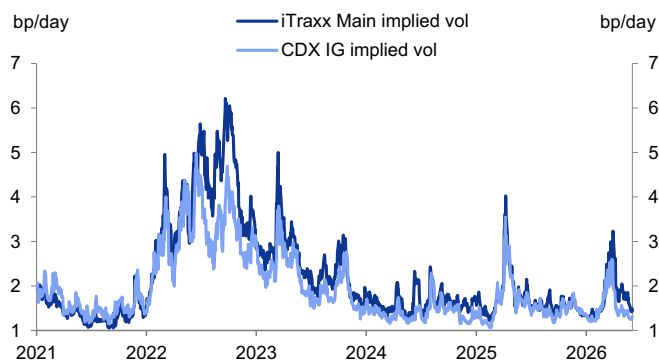


Source: iBoxx, Goldman Sachs Global Investment Research

Valuation dashboard: Macro index volatility

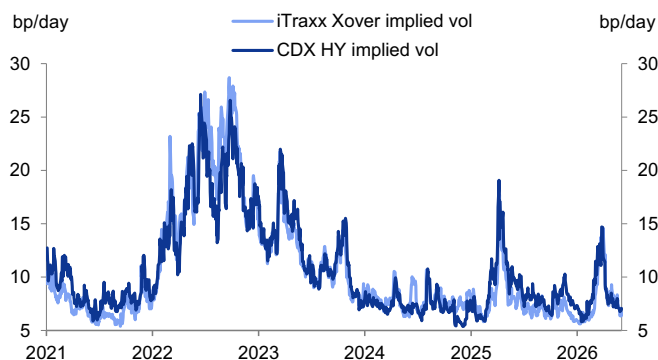
Note: Skew is calculated using difference in 3m implied volatility between 25-delta payers and receivers, divided by 3m ATM implied volatility.

Exhibit 43: CDX IG and iTraxx Main vol 3m ATM implied volatility



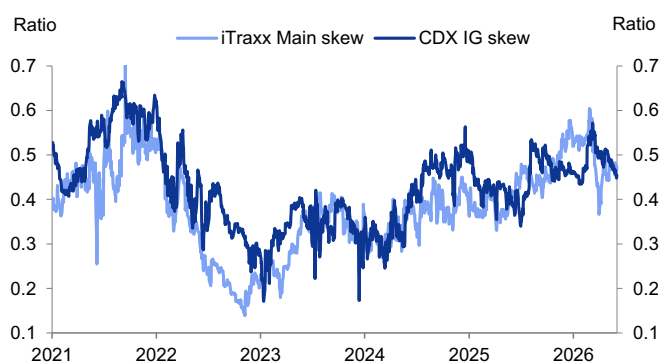
Source: Goldman Sachs, Goldman Sachs Global Investment Research

Exhibit 44: CDX HY and iTraxx Xover vol 3m ATM implied volatility



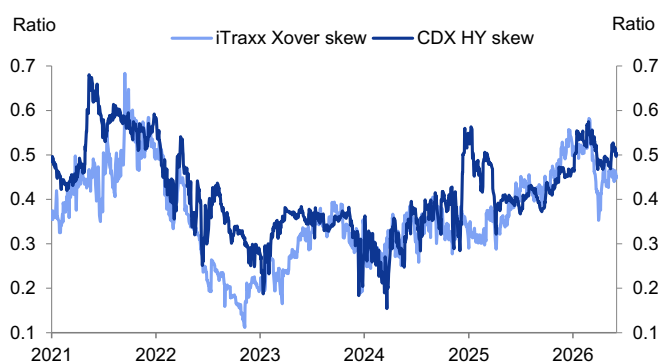
Source: Goldman Sachs, Goldman Sachs Global Investment Research

Exhibit 45: CDX IG and iTraxx Main skew



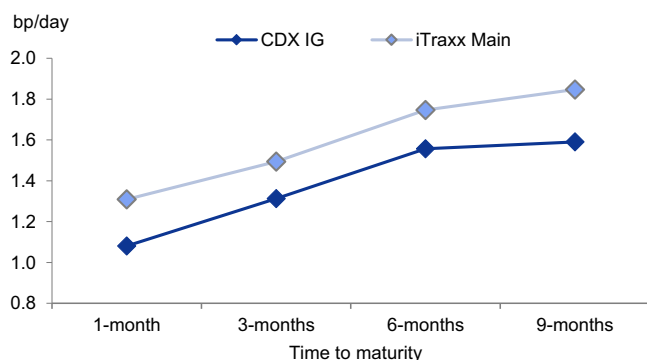
Source: Goldman Sachs, Goldman Sachs Global Investment Research

Exhibit 46: CDX HY and iTraxx Xover skew



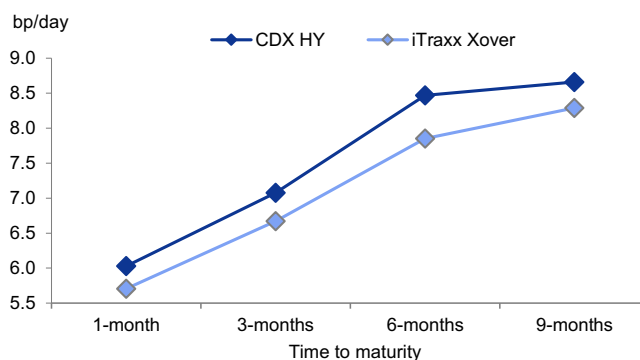
Source: Goldman Sachs, Goldman Sachs Global Investment Research

Exhibit 47: CDX IG and iTraxx Main ATM implied volatility term structure



Source: Goldman Sachs, Goldman Sachs Global Investment Research

Exhibit 48: CDX HY and iTraxx Xover ATM implied volatility term structure



Source: Goldman Sachs, Goldman Sachs Global Investment Research

Valuation dashboard: CLOs

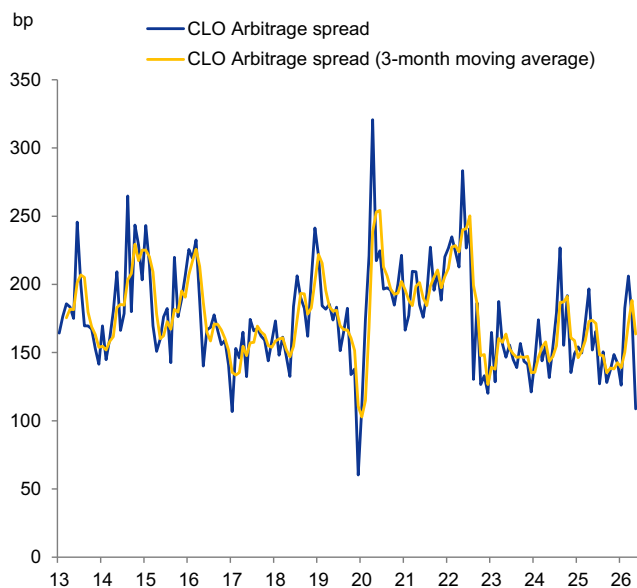
Exhibit 49: CLO spread percentile ranks

USD	Current spread (bp)	Percentile rank		
		3-year	5-year	10-year
AAA	106	14%	8%	4%
AA	143	3%	2%	1%
A	163	0%	0%	0%
BBB	283	32%	20%	14%
BB	758	62%	57%	64%

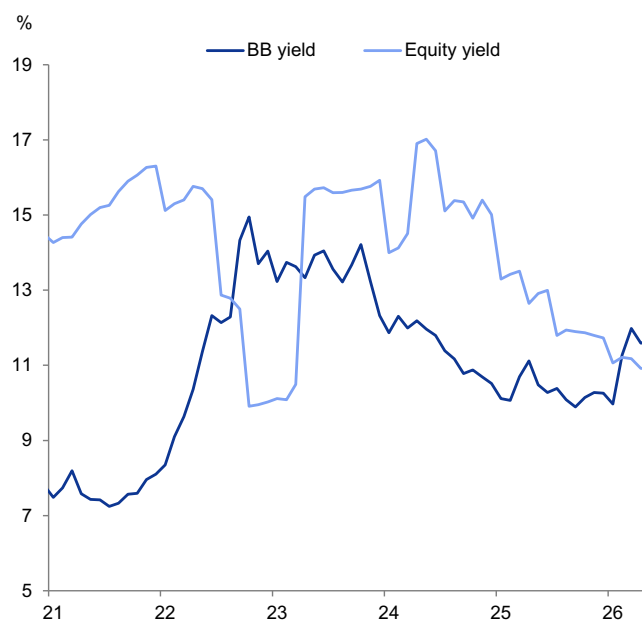
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 51: CLO new issue arbitrage spread

New issue CLO arbitrage spread, i.e., the difference between new issue loan spreads and new issue CLO spreads (across the capital stack)



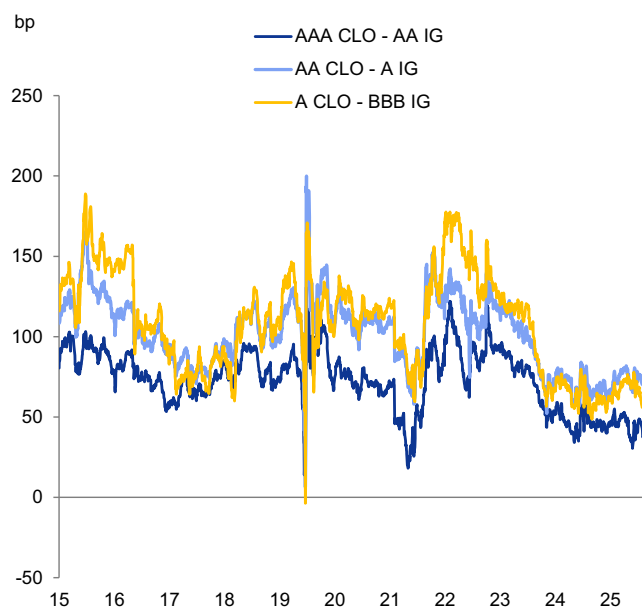
Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 50: CLO equity vs. CLO BB yield


Source: Intex, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 52: CLO vs. USD IG spread differential

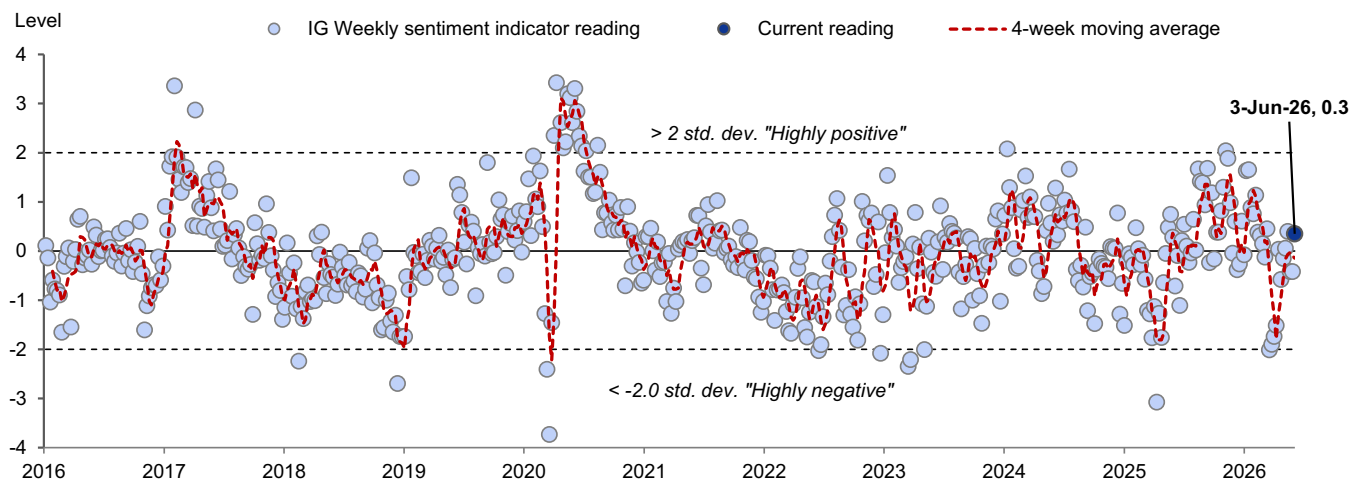
CLO by rating and USD IG bond by rating spread differential



Source: Bloomberg, Goldman Sachs Global Investment Research

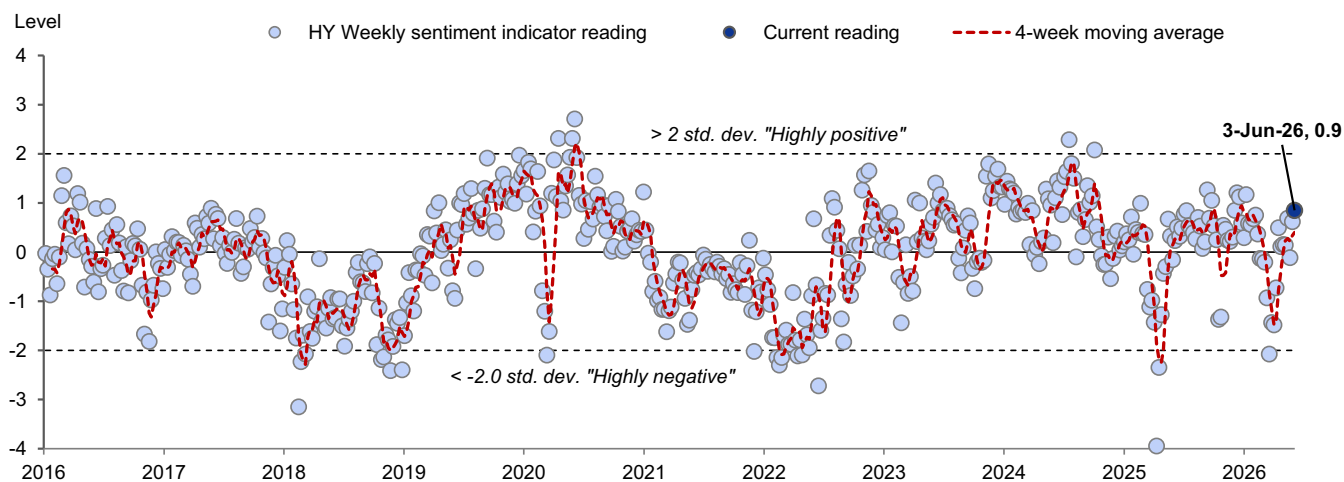
USD IG and HY Sentiment indicators

Exhibit 53: IG Weekly sentiment indicator



Source: See table below.

Exhibit 54: HY Weekly sentiment indicator



Source: See table below.

Exhibit 55: Variables included in the USD IG and HY weekly sentiment indicators

Measure	Frequency	Data Source
Synthetic index Positioning	Weekly	DTCC
Net Dealer Positioning	Weekly	FRBNY
Net Dealer Lift	Weekly	TRACE
Active US Credit Fund flows	Weekly	EPFR
Passive US Credit Fund flows	Weekly	EPFR
US Money Market Fund flows	Weekly	EPFR
Short interest on the largest credit ETFs	Semimonthly	GS FICC & Equities
Credit Mutual Fund cash positions	Monthly	ICI

Source: Goldman Sachs Global Investment Research

Microstructure dashboard: Top 20 most traded capital structures in the USD IG and HY markets

Exhibit 56: Top 20 most liquid issuers by weekly turnover in the USD IG market

Ticker	Issuer	Notional bonds outstanding (\$bn)	Notional volume traded on the week (\$mn)	Weekly turnover ratio (Wed. to Wed. %)	Median rating	Average age of bonds outstanding (Years)	Average duration of bonds outstanding (Years)
NOW	ServiceNow Inc	5.5	187.8	3.4	A	1.6	6.2
CPB	The Campbell's Company	5.7	152.0	2.7	BBB-	4.5	6.3
CRM	Salesforce Inc	33.5	847.1	2.5	A	1.6	7.3
ECL	Ecolab Inc	11.0	215.4	2.0	A-	2.5	6.8
META	Meta Platforms Inc	84.0	1603.6	1.9	AA-	1.2	9.9
BCRED	Blackstone Private Credit Fund	6.6	125.3	1.9	BBB-	1.3	4.0
EXPE	Expedia Group Inc	5.5	103.4	1.9	BBB	4.3	4.3
CAG	Conagra Brands Inc	5.3	94.2	1.8	BBB-	6.0	5.9
GOOGL	Alphabet Inc	51.0	895.8	1.8	AA	1.4	10.0
ORCL	Oracle Corp	117.2	1950.0	1.7	BBB	4.2	8.9
BHI	Baker Hughes Holdings LLC / Baker Hughes Co-Obligor Inc	11.4	177.9	1.6	A-	4.3	8.0
HPE	Hewlett Packard Enterprise Co	14.9	228.6	1.5	BBB	2.6	5.6
PYPL	PayPal Holdings Inc	10.0	148.6	1.5	A-	3.7	7.0
UBER	Uber Technologies Inc	7.8	112.9	1.5	BBB+	2.0	6.5
HUM	Humana Inc	12.4	176.3	1.4	BBB	3.5	7.6
AMZN	Amazon.com Inc	97.5	1387.1	1.4	AA-	3.1	8.9
DVN	Devon Energy Corp	6.2	86.4	1.4	BBB+	9.6	9.1
SUZANO	Suzano Austria GmbH	7.8	102.5	1.3	BBB-	5.9	5.2
FANG	Diamondback Energy Inc	9.3	119.9	1.3	BBB	3.0	8.3
ARCC	Ares Capital Corp	8.3	106.4	1.3	BBB	2.3	3.3

Source: TRACE, iBoxx, data compiled by Goldman Sachs Global Investment Research

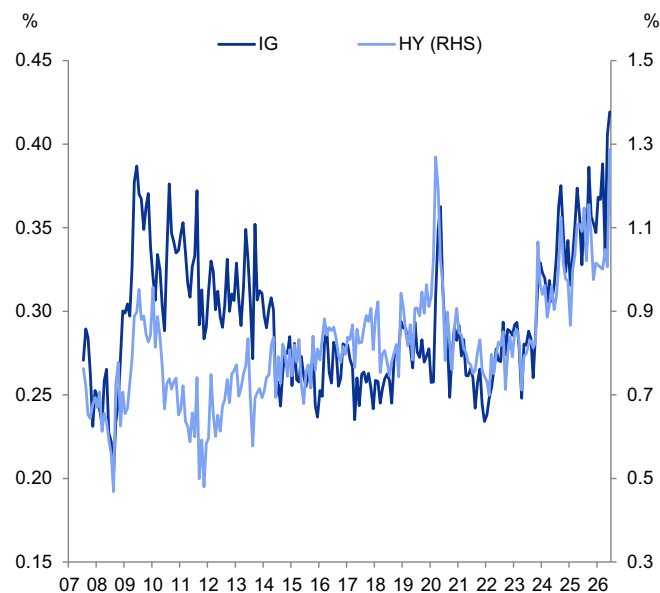
Exhibit 57: Top 20 most liquid issuers by weekly turnover in the USD HY market

Ticker	Issuer	Notional bonds outstanding (\$bn)	Notional volume traded on the week (\$mn)	Weekly turnover ratio (Wed. to Wed. %)	Median rating	Average age of bonds outstanding (Years)	Average duration of bonds outstanding (Years)
JBLU	JetBlue Airways Corp / JetBlue Loyalty LP	2.0	86.3	4.3	B	1.8	4.0
IHRT	iHeartCommunications Inc	2.1	62.9	3.1	CCC+	1.4	3.1
GT	The Goodyear Tire & Rubber Company	3.0	88.0	3.0	B+	3.9	4.0
FSK	FS KKR Capital Corp	2.9	68.0	2.4	BB+	2.5	2.9
MGM	MGM Resorts International	2.4	54.5	2.3	BB-	3.1	2.3
CRWV	CoreWeave Inc	6.5	147.9	2.3	B+	0.6	3.2
PFORGE	APLD ComputeCo 2 LLC	2.2	48.1	2.2	BB-	0.2	2.9
CZR	Caesars Entertainment Inc	5.8	109.1	1.9	BB-	3.0	3.4
XRX	Xerox Holdings Corp	2.0	36.2	1.8	CCC-	2.9	2.8
GTN	Gray Media Inc	4.8	84.4	1.8	CCC	2.9	3.6
RRD	RR Donnelley & Sons Co	2.4	40.4	1.7	B-	1.1	2.7
FRTITA	Fertitta Entertainment LLC / Fertitta Entertainment Finance Co	2.3	34.8	1.5	CCC	4.3	2.9
MIK	Michaels Cos Inc/The	2.8	41.3	1.5	B-	0.2	5.2
CAR	Avis Budget Car Rental LLC / Avis Budget Finance Inc	3.2	47.9	1.5	B	3.0	2.4
NCX	NOVA Chemicals Corp	2.7	40.0	1.5	BB-	5.4	1.4
SABHLD	Sabre GLBL Inc	3.2	47.5	1.5	B-	0.8	2.7
VSTJET	VistaJet Malta Finance PLC / Vista Management Holding Inc	2.0	29.4	1.5	B-	2.9	3.0
SNAP	Snap Inc	2.1	29.2	1.4	BB-	1.2	5.6
UAL	United Airlines Holdings Inc	2.0	28.5	1.4	BB+	0.3	3.4
PARA	Paramount Global	3.7	52.8	1.4	BB+	6.8	3.6

Source: TRACE, iBoxx, data compiled by Goldman Sachs Global Investment Research

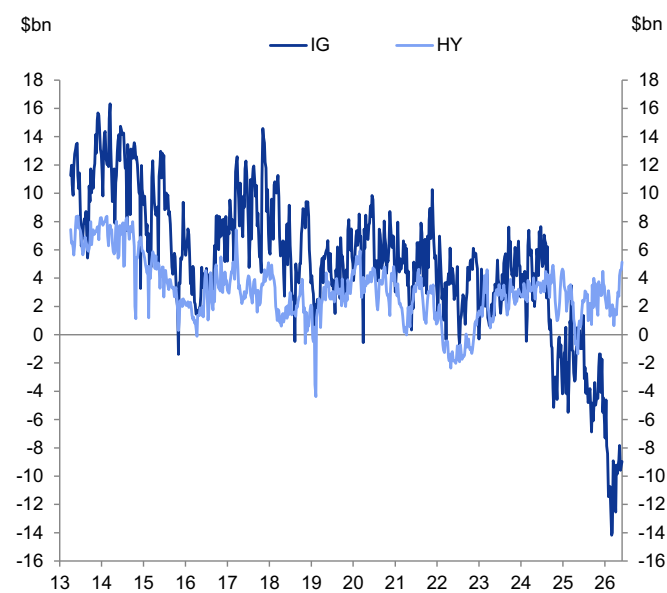
Microstructure dashboard: Liquidity conditions in the USD secondary bond market

Exhibit 58: Seasonally adjusted daily average turnover (trading volume/amount outstanding)



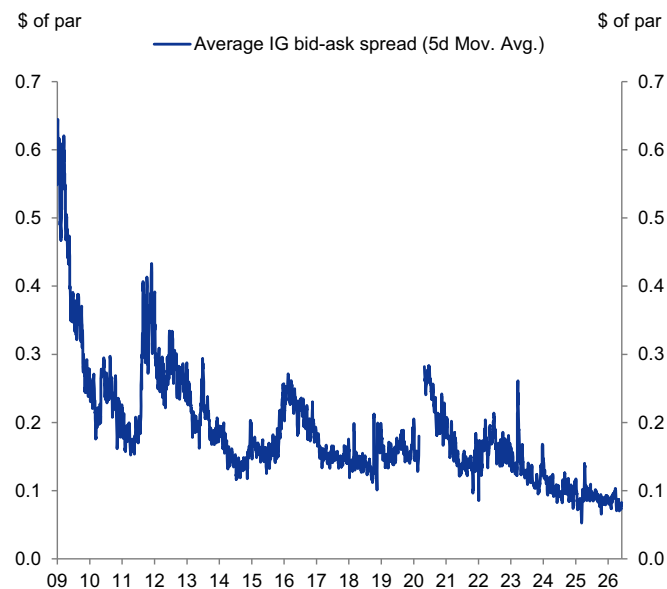
Source: TRACE, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 59: Primary dealer net positioning in USD corporate bonds



Source: FRBNY, Goldman Sachs Global Investment Research

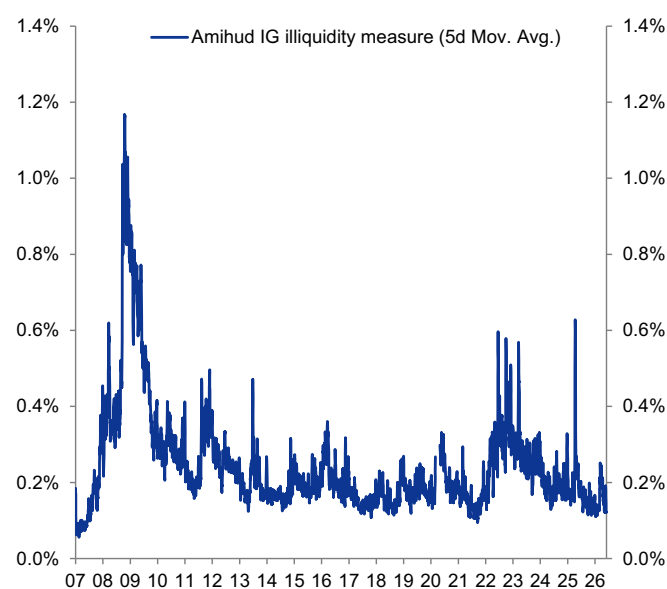
Exhibit 60: Average IG bid-ask spread 5-day moving average



Note: March and April 2020 are omitted.

Source: Trace, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 61: Amihud IG illiquidity measure 5-day moving average

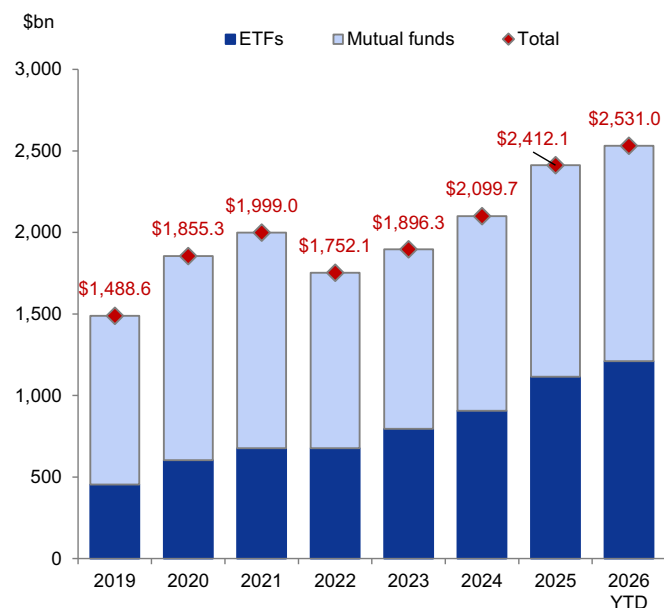


Note: March and April 2020 are omitted.

Source: TRACE, Bloomberg, Goldman Sachs Global Investment Research

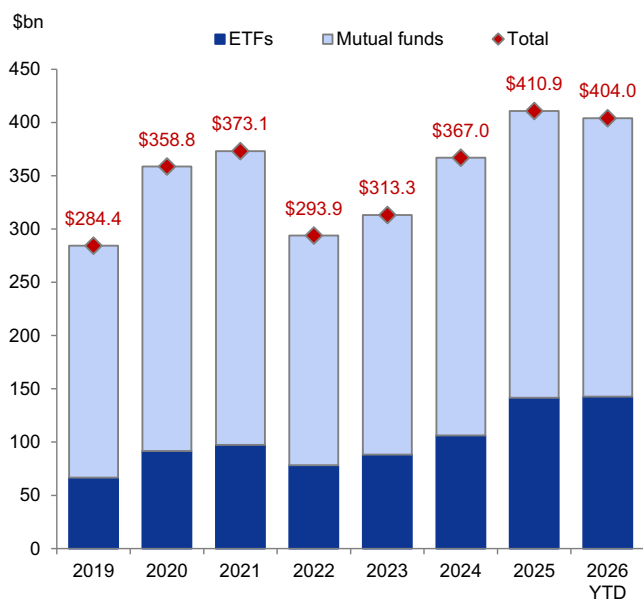
Flow dashboard: Mutual fund and ETF AUM

Exhibit 62: AUM in USD IG-focused Mutual Funds and ETFs



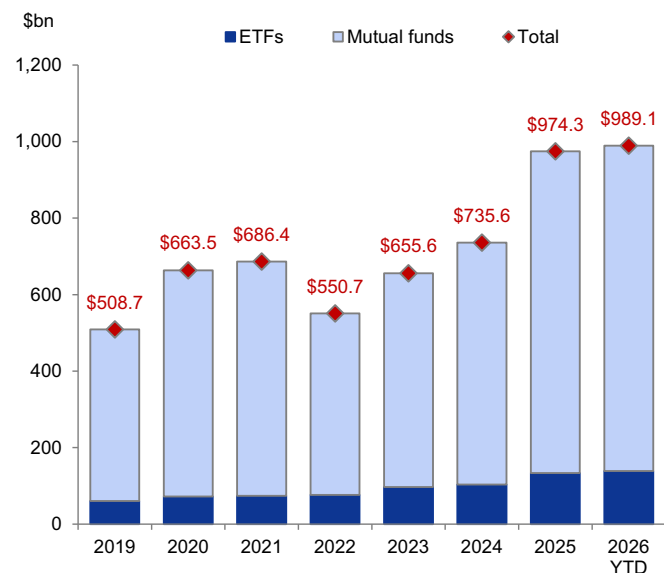
Source: EPFR, Goldman Sachs Global Investment Research

Exhibit 63: AUM in USD HY-focused Mutual Funds and ETFs



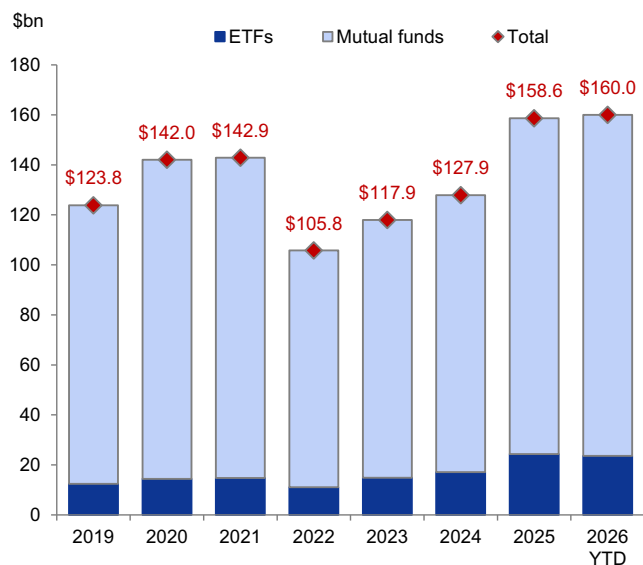
Source: EPFR, Goldman Sachs Global Investment Research

Exhibit 64: AUM in EUR IG-focused Mutual Funds and ETFs



Source: EPFR, Goldman Sachs Global Investment Research

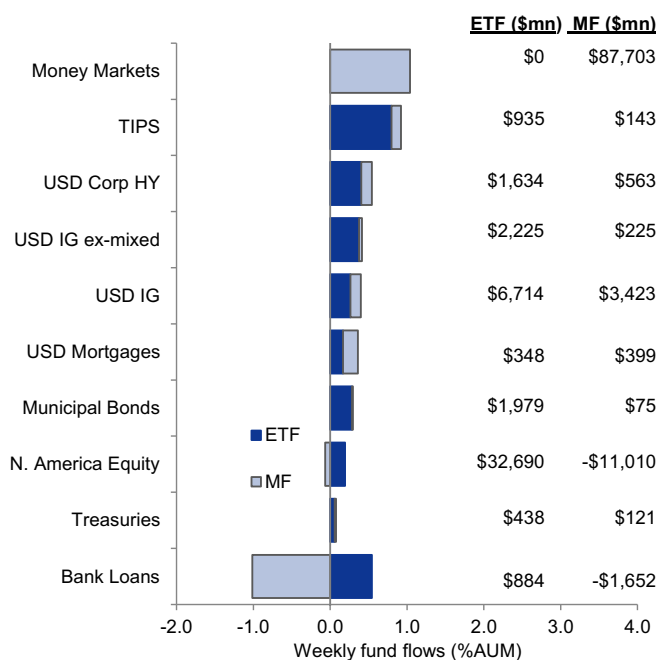
Exhibit 65: AUM in EUR HY-focused Mutual Funds and ETFs



Source: EPFR, Goldman Sachs Global Investment Research

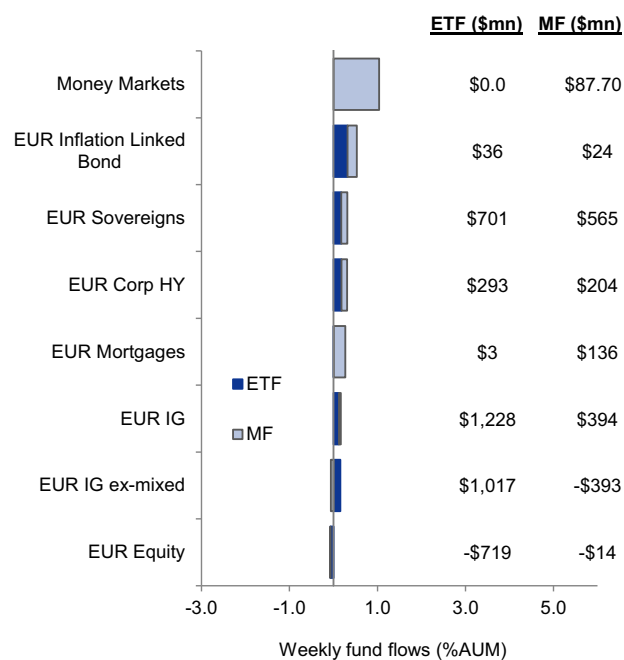
Flow dashboard: Mutual fund and ETF flows

Exhibit 66: Weekly mutual fund flows into US fixed income and equity markets



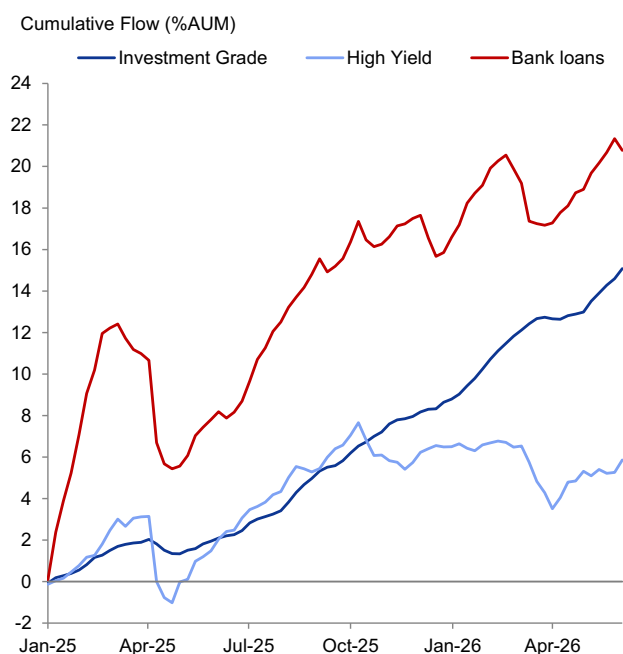
Source: EPFR, Goldman Sachs Global Investment Research

Exhibit 67: Weekly mutual fund flows into European fixed income and equity markets



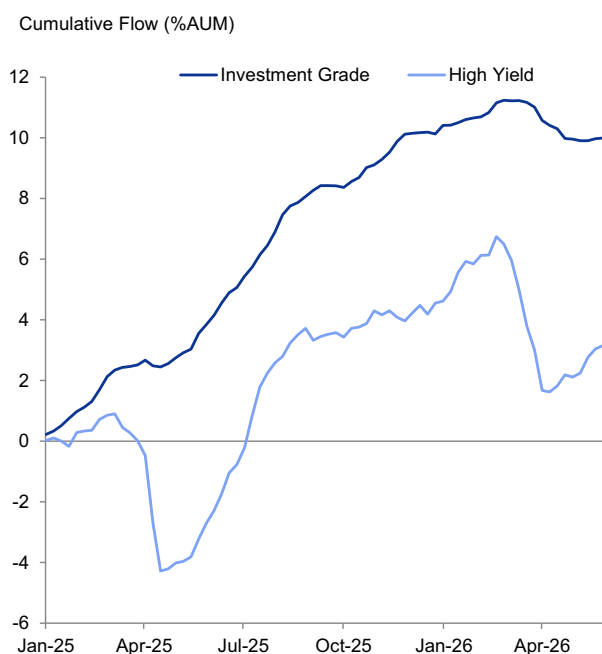
Source: EPFR, Goldman Sachs Global Investment Research

Exhibit 68: Year-to-date cumulative flows as a percentage of beginning of the year AUM for US credit funds



Source: EPFR, Goldman Sachs Global Investment Research

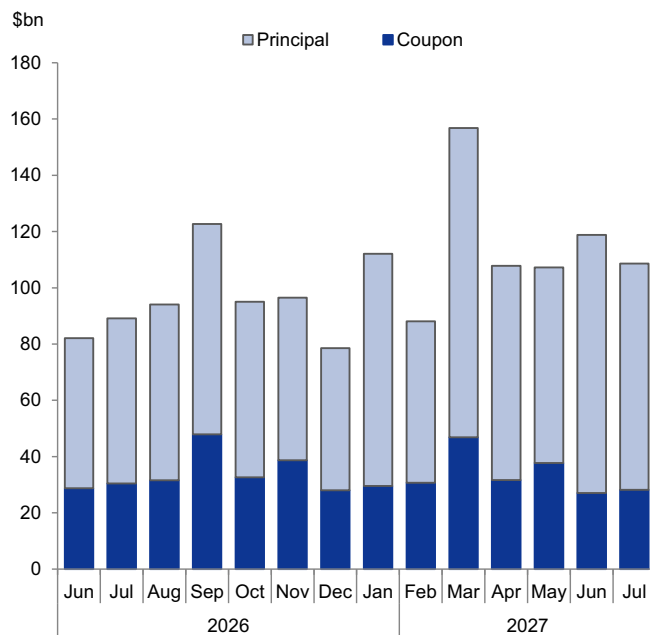
Exhibit 69: Year-to-date cumulative flows as a percentage of beginning of the year AUM for European credit funds



Source: EPFR, Goldman Sachs Global Investment Research

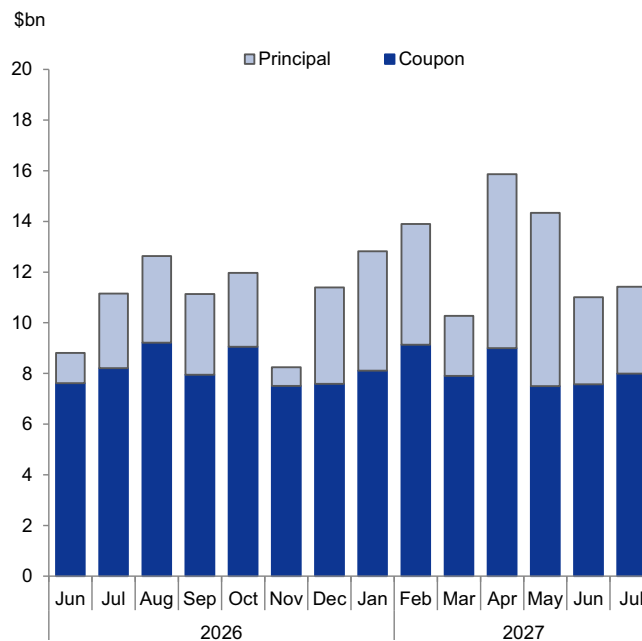
Flow dashboard: Upcoming coupon and principal payments

Exhibit 70: Our estimates of upcoming coupon and principal payments in the USD IG market



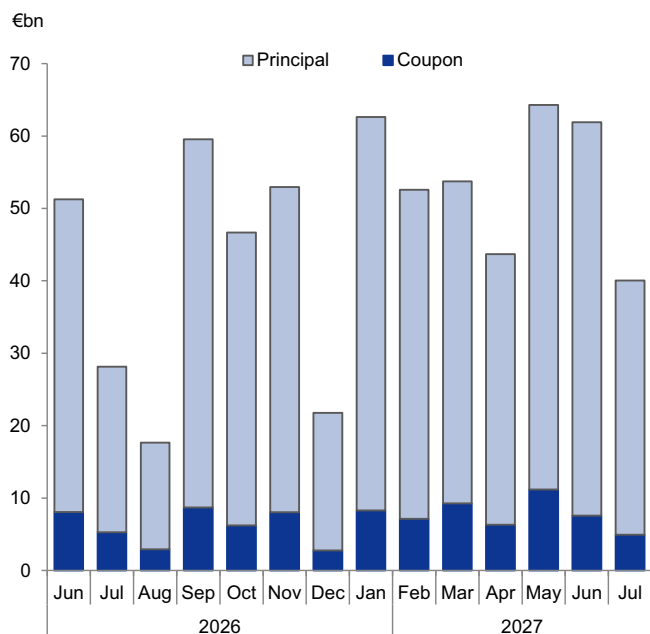
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 71: Our estimates of upcoming coupon and principal payments in the USD HY market



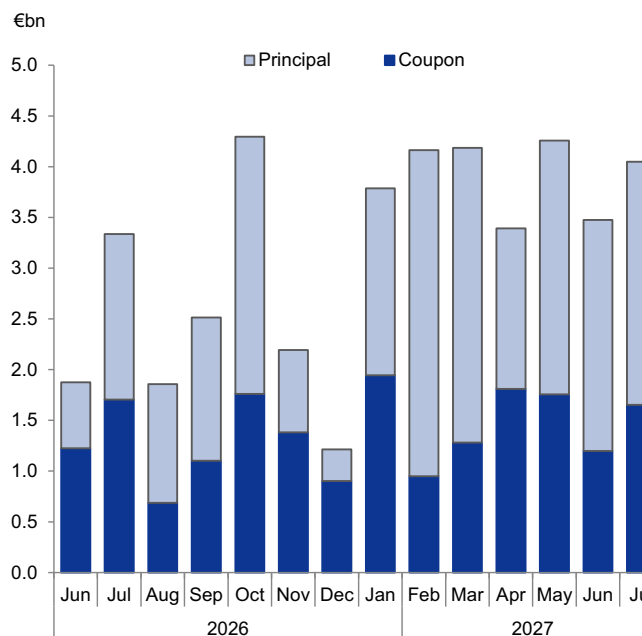
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 72: Our estimates of upcoming coupon and principal payments in the EUR IG market



Source: Bloomberg, Goldman Sachs Global Investment Research

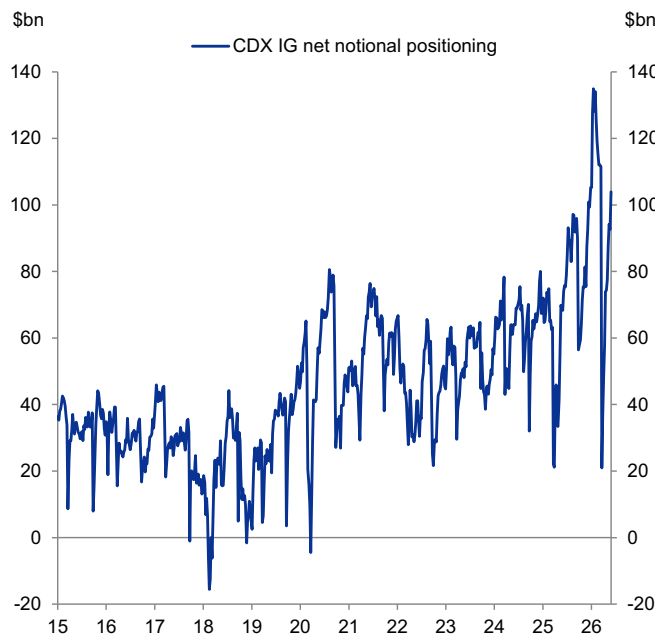
Exhibit 73: Our estimates of upcoming coupon and principal payments in the EUR HY market



Source: Bloomberg, Goldman Sachs Global Investment Research

Flow dashboard: Positioning in macro indices

Exhibit 74: Net notional positioning in the CDX IG 5-year on-the-run contract



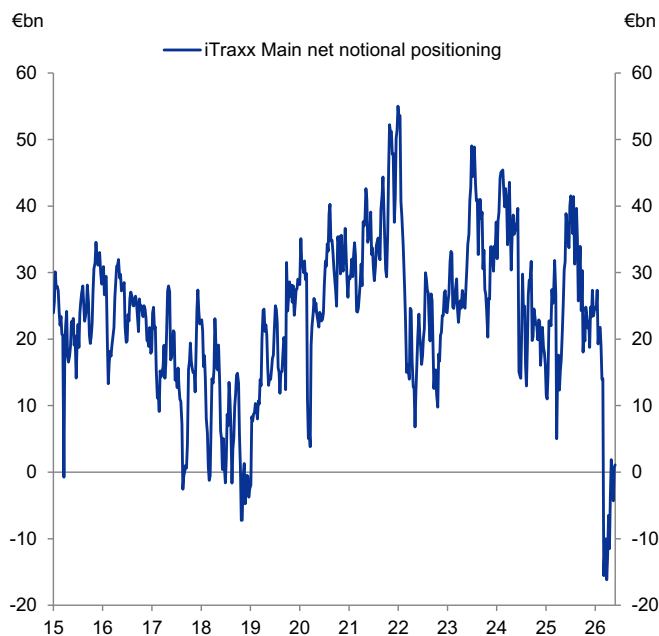
Source: DTCC, Goldman Sachs Global Investment Research

Exhibit 75: Net notional positioning in the CDX HY 5-year on-the-run contract



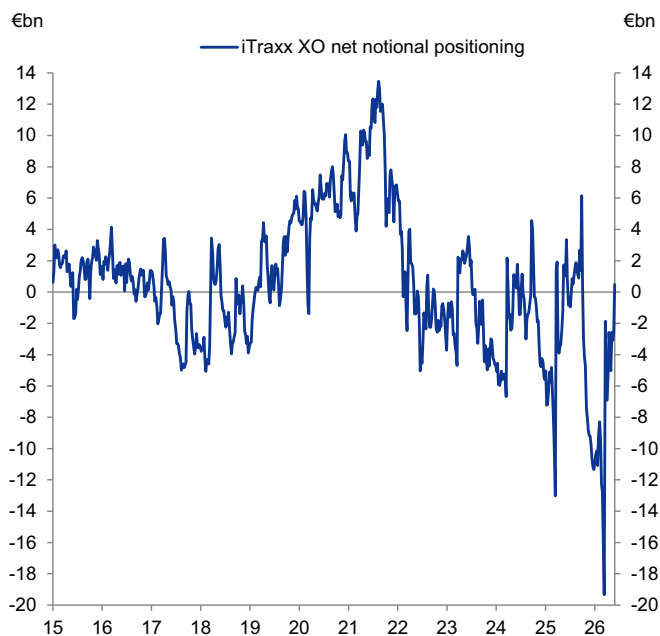
Source: DTCC, Goldman Sachs Global Investment Research

Exhibit 76: Net notional positioning in the iTraxx Main 5-year on-the-run contract



Source: DTCC, Goldman Sachs Global Investment Research

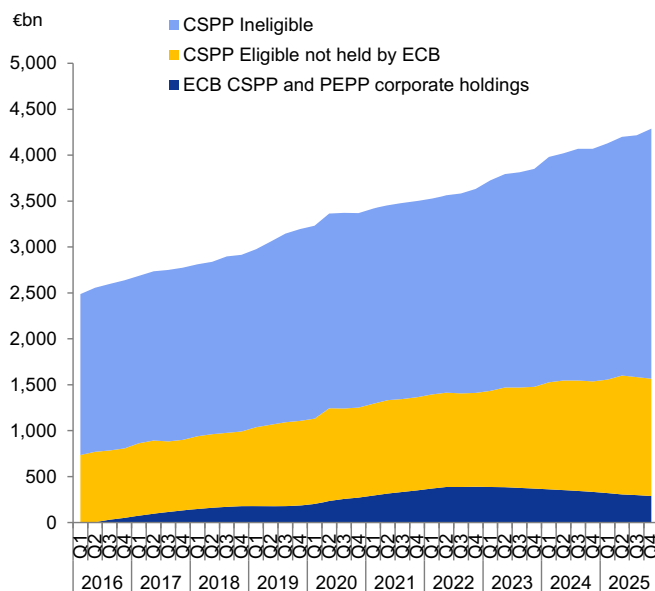
Exhibit 77: Net notional positioning in iTraxx over 5-year on-the-run contract



Source: DTCC, Goldman Sachs Global Investment Research

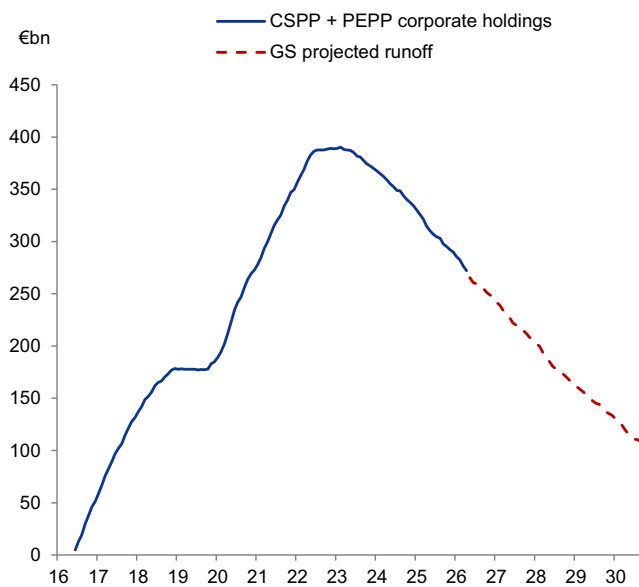
Flow dashboard: ECB QT

Exhibit 78: Ownership share of the ECB in the EUR IG market



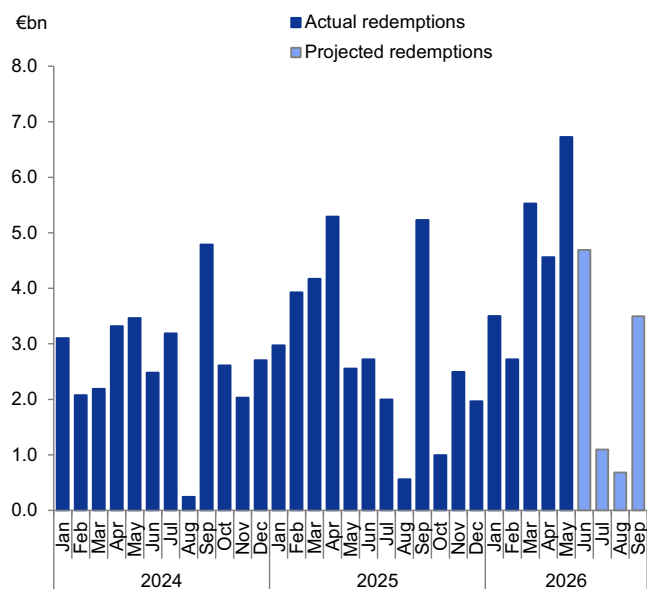
Source: ECB, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 79: Run-off projections for the ECB corporate bond holdings



Source: ECB, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 80: CSPP monthly run-off pace



Source: ECB, Goldman Sachs Global Investment Research

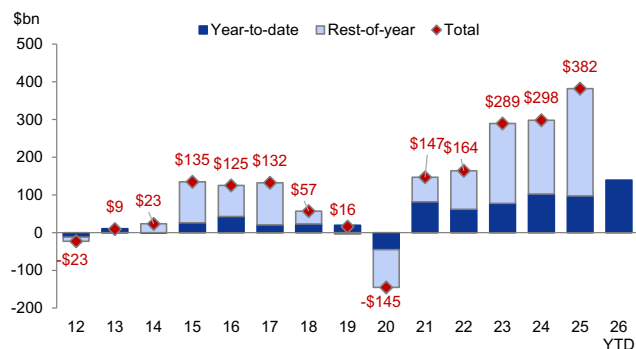
Exhibit 81: Performance of CSPP-eligible bonds vs. non-eligible bonds



Source: iBoxx, Goldman Sachs Global Investment Research

Foreign demand and hedging cost

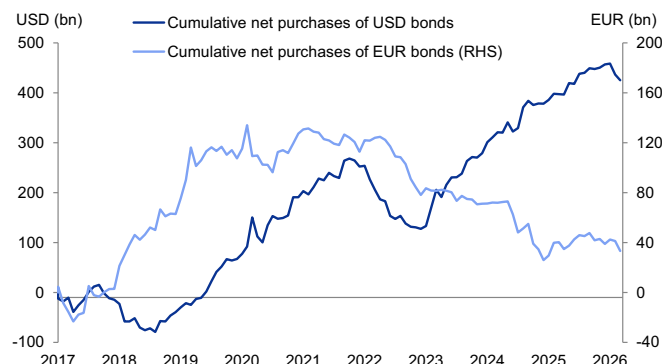
Exhibit 82: Net purchases of USD corporate bonds by foreign investors



Note: As of March 2026.

Source: Department of Treasury, Goldman Sachs Global Investment Research

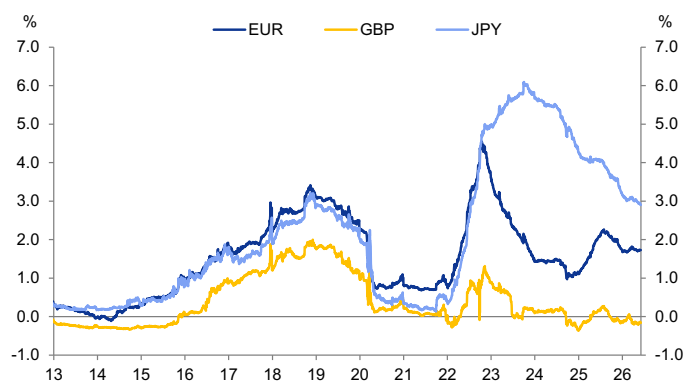
Exhibit 83: Cumulative net foreign purchases of USD and EUR bonds by Japanese investors



Note: As of March 2026.

Source: Bank of Japan, Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 84: Annualized 3-month rolling cost of USD hedging against EUR, GBP and JPY

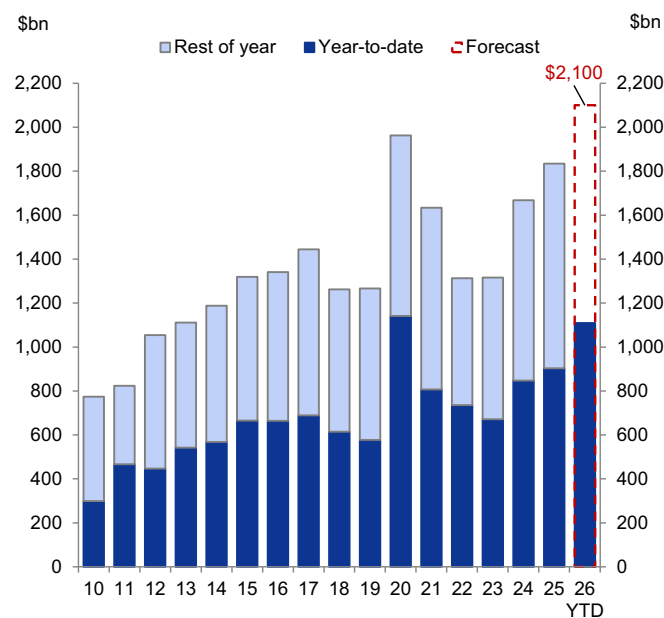


Source: Goldman Sachs FICC and Equities, Goldman Sachs Global Investment Research

Primary market dashboard: USD gross issuance

Exhibit 85: USD IG gross issuance

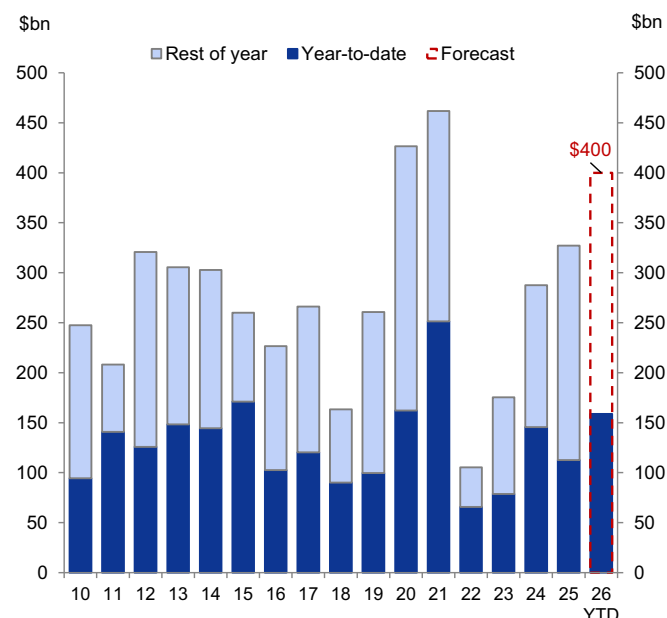
Historical investment grade issuance: Same period year to date vs. rest of the year



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 86: USD HY gross issuance

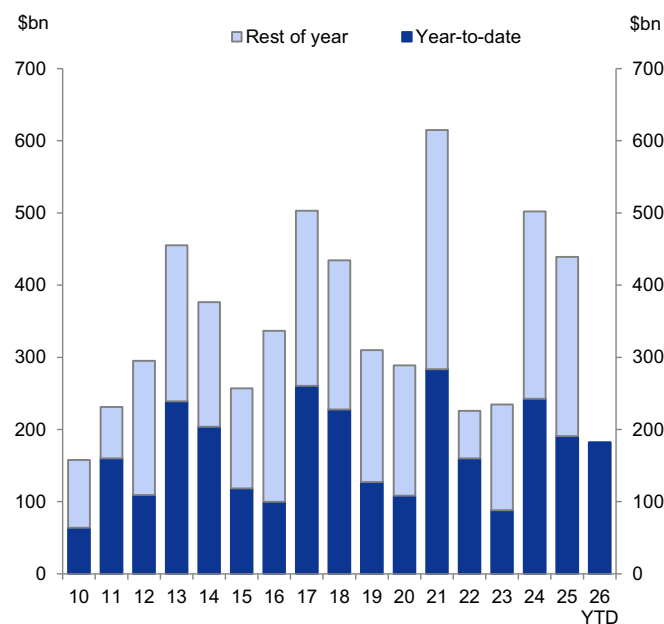
Historical USD high yield issuance: Same period year to date vs. rest of the year



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 87: USD institutional leveraged loan gross issuance

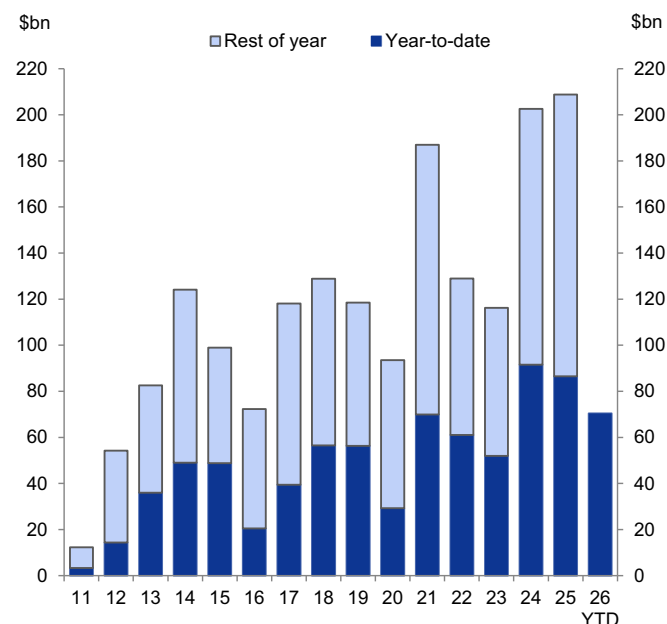
Historical leveraged loan issuance: Same period year to date vs. rest of the year



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 88: USD CLO creation

Historical USD CLO creation: Same period year to date vs. rest of the year

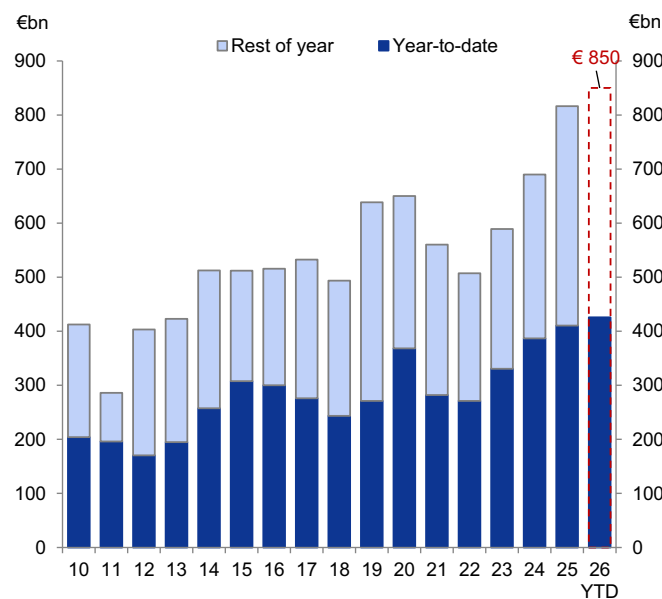


Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Primary market dashboard: EUR gross issuance

Exhibit 89: EUR IG gross issuance

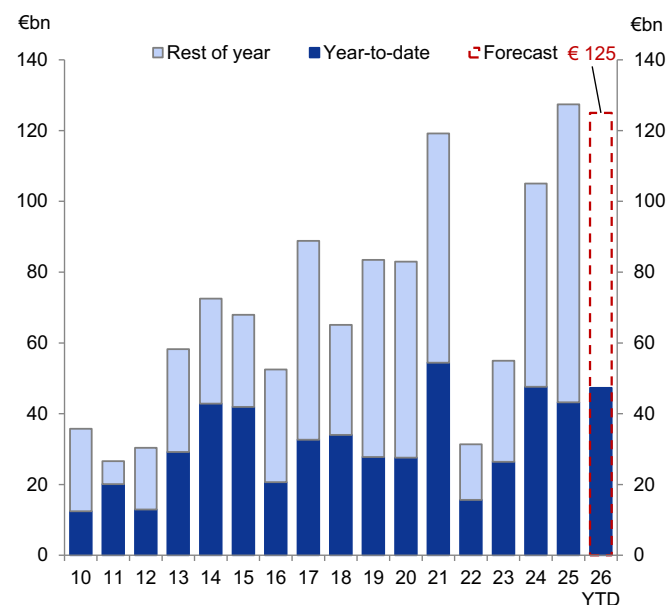
Historical EUR investment grade issuance: Same period year to date vs. rest of the year



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 90: EUR HY gross issuance

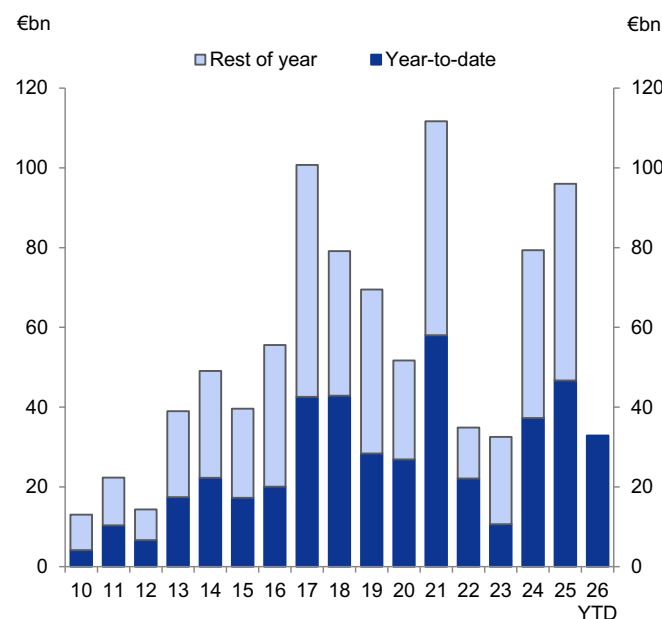
Historical EUR high yield issuance: Same period year to date vs. rest of the year



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 91: EUR Leveraged loan gross issuance

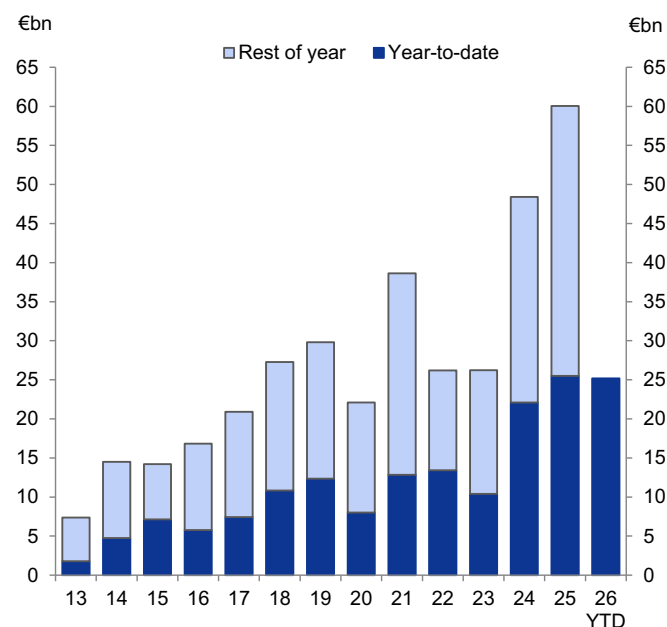
Historical EUR leveraged loan issuance: Same period year to date vs. rest of the year



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 92: EUR CLO creation

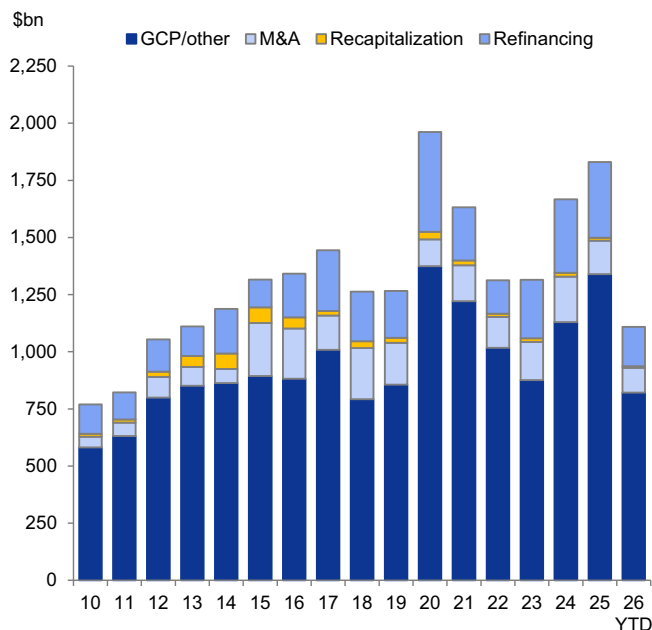
Historical EUR CLO creation broken: Same period year to date vs. rest of the year



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

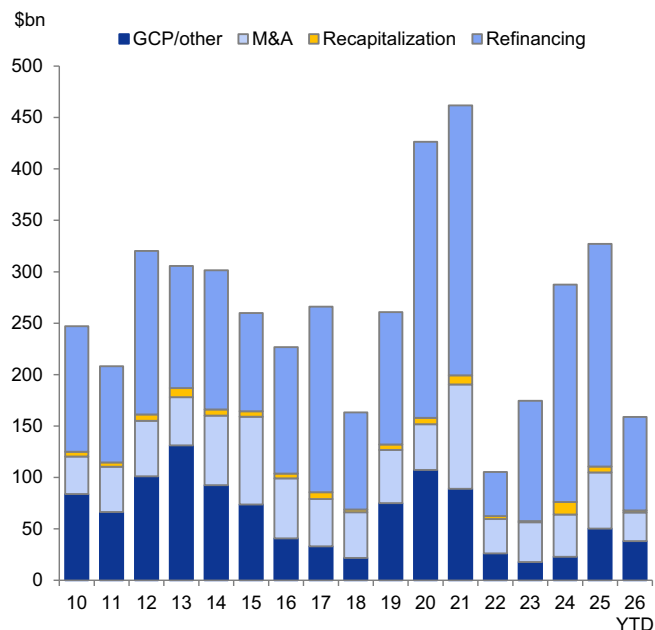
Primary market dashboard: Use of proceeds

Exhibit 93: Annual USD IG issuance volumes by use of proceeds



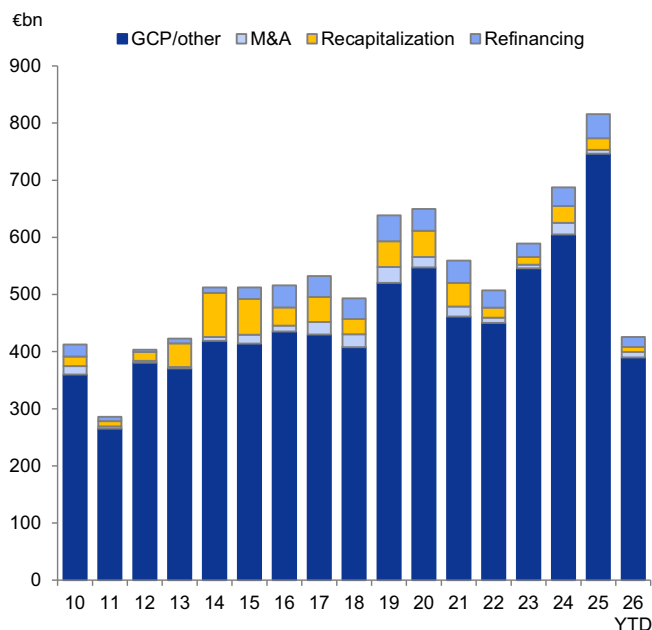
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 94: USD HY issuance volumes by use of proceeds



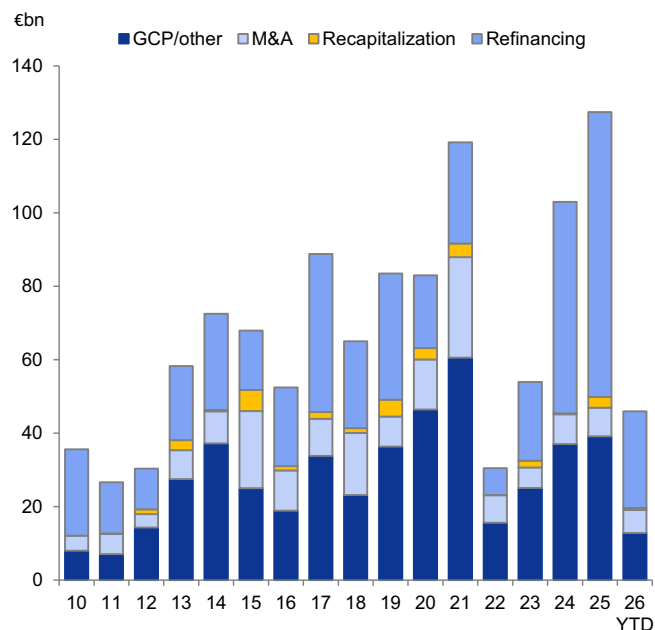
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 95: EUR IG issuance volumes by use of proceeds



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 96: EUR HY issuance volumes by use of proceeds

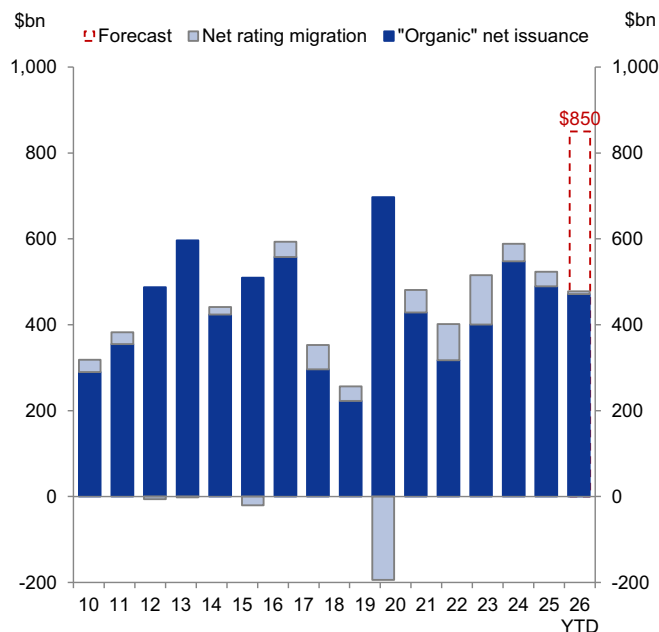


Source: Dealogic, Goldman Sachs Global Investment Research

Primary market dashboard: Net issuance

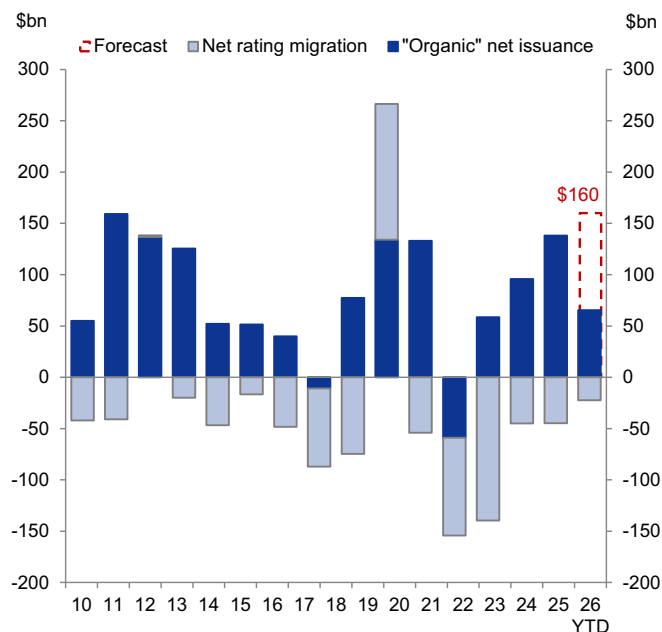
Note: Decomposition of the change in par value outstanding into net ratings migration vs. "organic" net issuance (gross issuance minus redemptions/matured bonds)

Exhibit 97: Annual USD IG net issuance



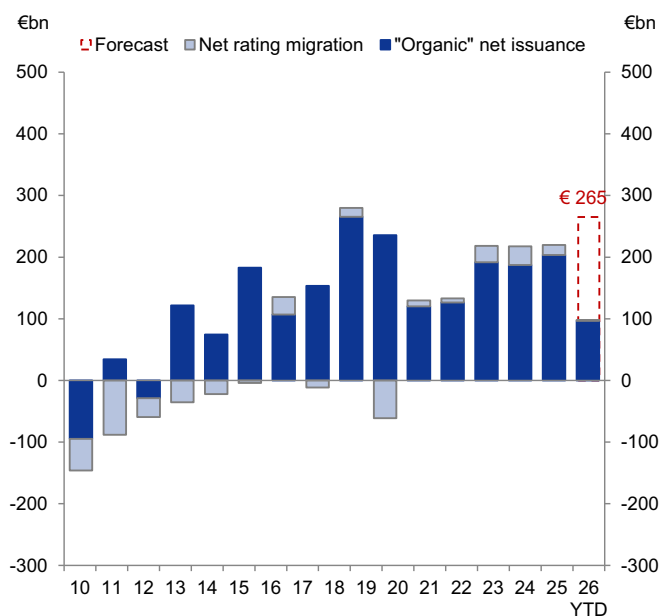
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 98: Annual USD HY net issuance



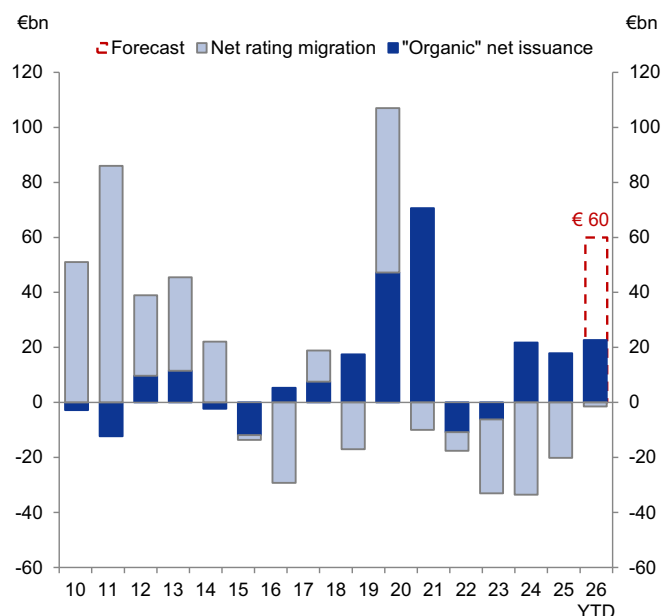
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 99: Annual EUR IG net issuance



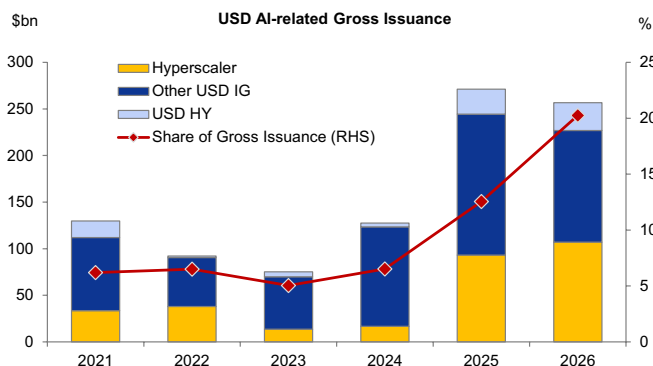
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 100: Annual EUR HY net issuance

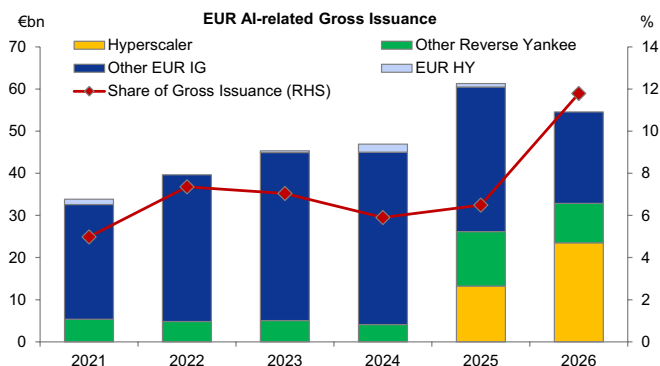


Source: Bloomberg, Goldman Sachs Global Investment Research

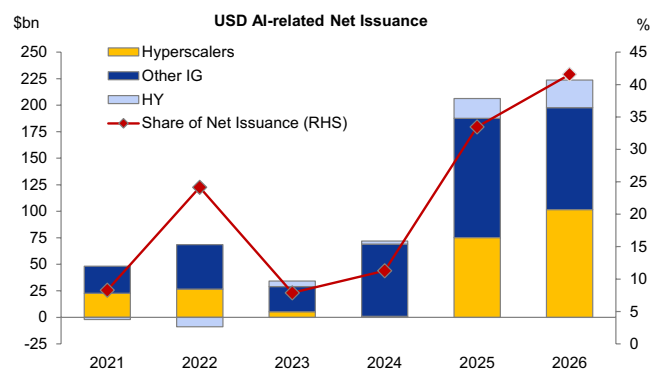
Primary market dashboard: AI-related issuance for the USD and EUR markets

Exhibit 101: USD AI-related Gross Issuance


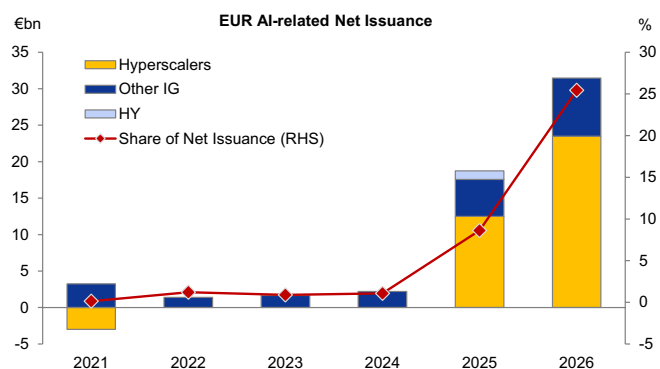
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 102: EUR AI-related Gross Issuance


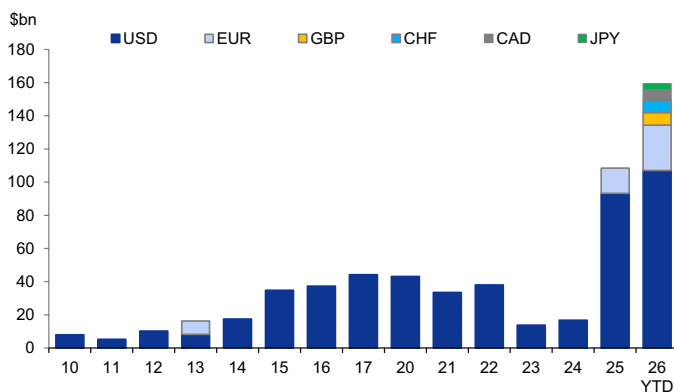
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 103: USD AI-related net issuance


Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 104: EUR AI-related net issuance


Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 105: Hyperscaler debt issuance across currencies (in USD-equivalent)


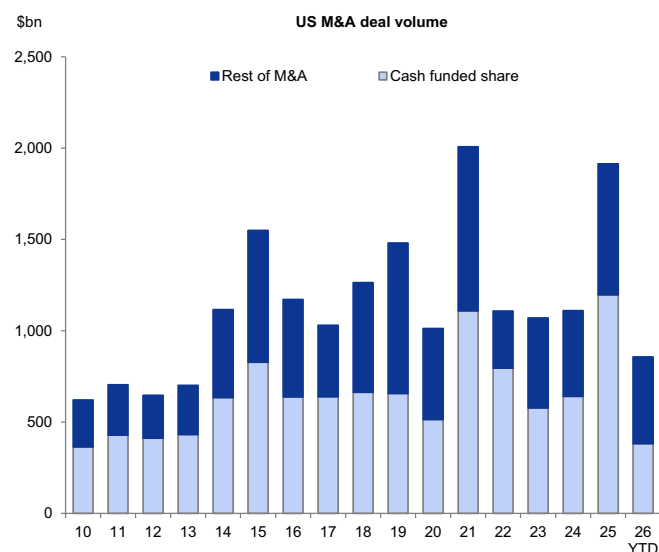
Note: Captures debt issued by the following companies: Alphabet (Google), Amazon, Microsoft, Meta, and Oracle.

Source: Dealogic, Goldman Sachs Global Investment Research

M&A volume

Exhibit 106: Annual US M&A deal volume

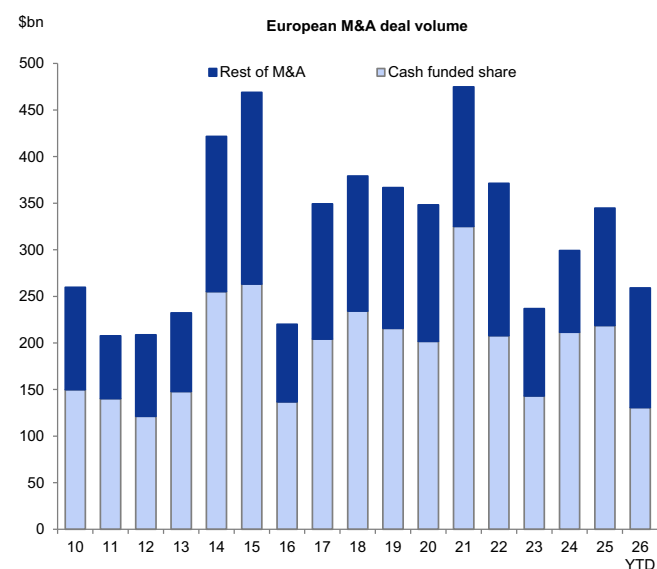
Includes strategic and sponsored deals



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 107: Annual European M&A deal volume

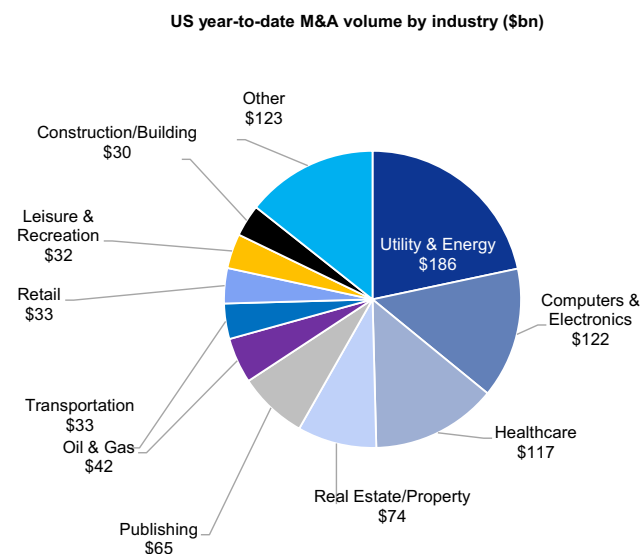
Includes strategic and sponsored deals



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 108: Year-to-date US M&A volume by sector

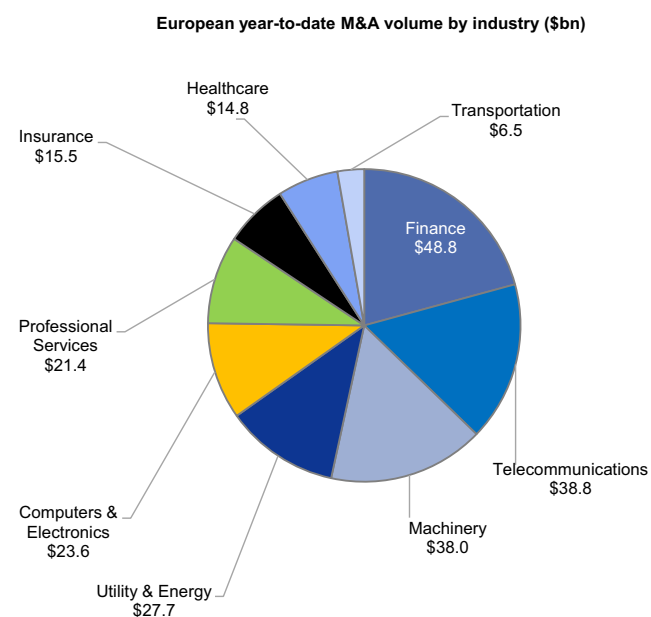
Includes strategic and sponsored deals



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 109: Year-to-date European M&A volume by sector

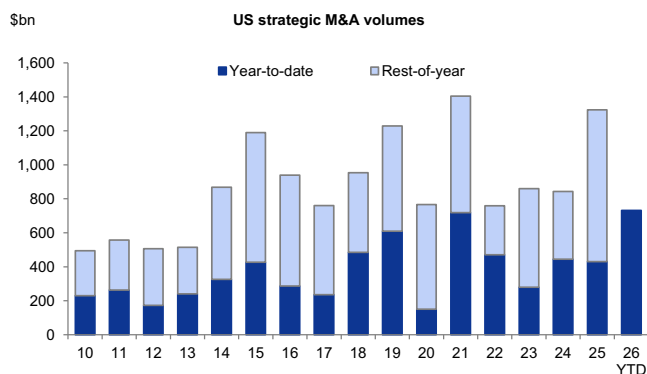
Includes strategic and sponsored deals



Source: Dealogic, Goldman Sachs Global Investment Research

M&A volume (Strategic deals only)

Exhibit 110: Annual US strategic M&A volumes



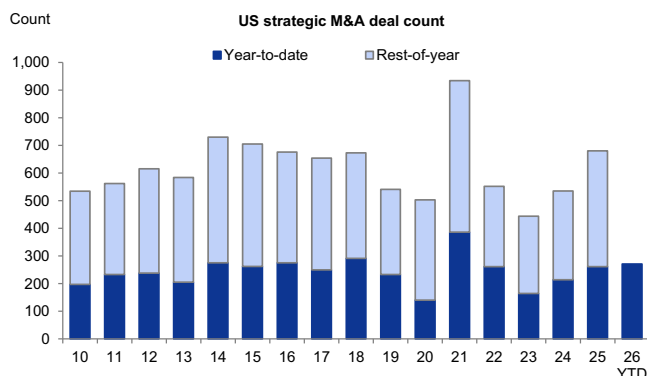
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 111: Annual European strategic M&A volumes



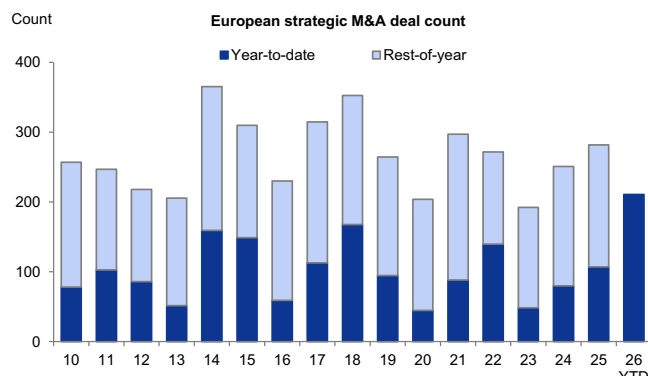
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 112: Annual US strategic M&A deal count



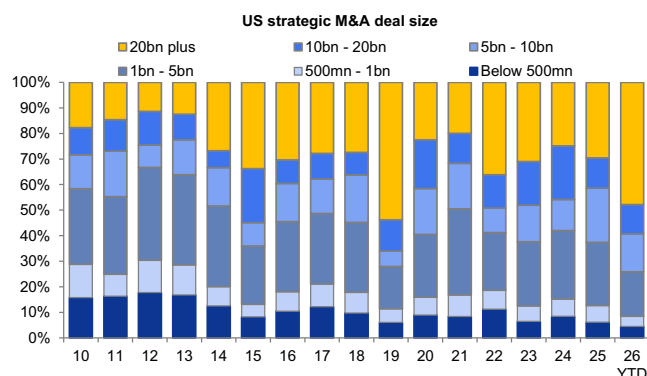
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 113: Annual European strategic M&A deal count



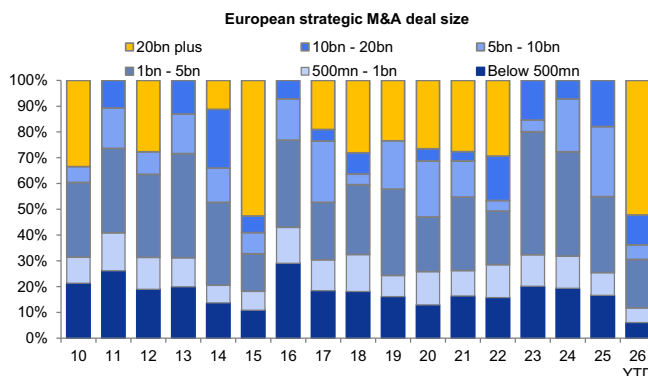
Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 114: Annual US strategic M&A deal size



Source: Dealogic, Goldman Sachs Global Investment Research

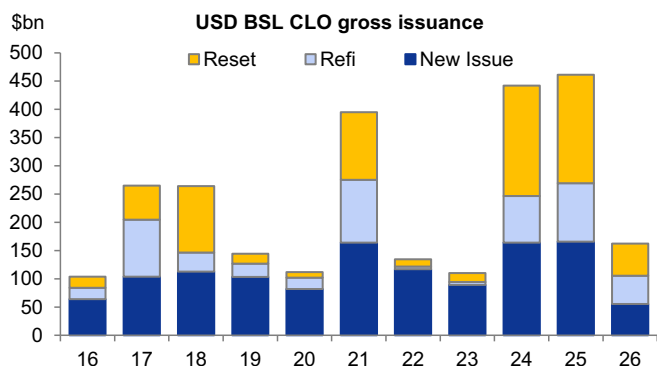
Exhibit 115: Annual European strategic M&A deal size



Source: Dealogic, Goldman Sachs Global Investment Research

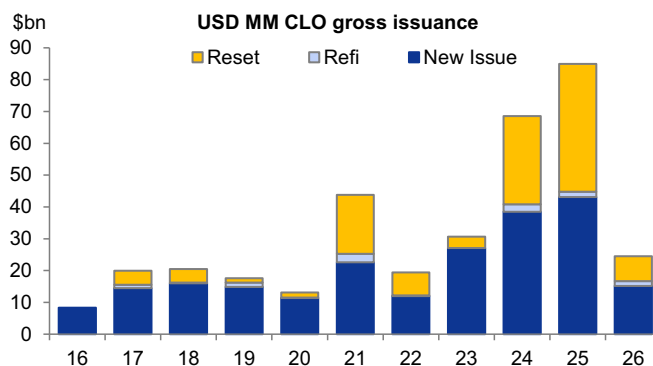
Primary market dashboard: CLO creation and refinancing

Exhibit 116: Annual USD Broadly syndicated loan CLO creation



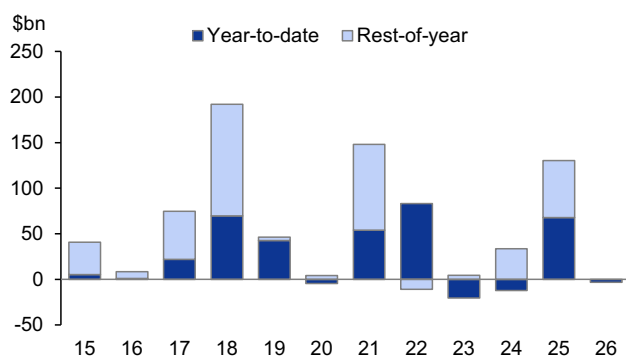
Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 117: Annual USD Middle market CLO creation



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

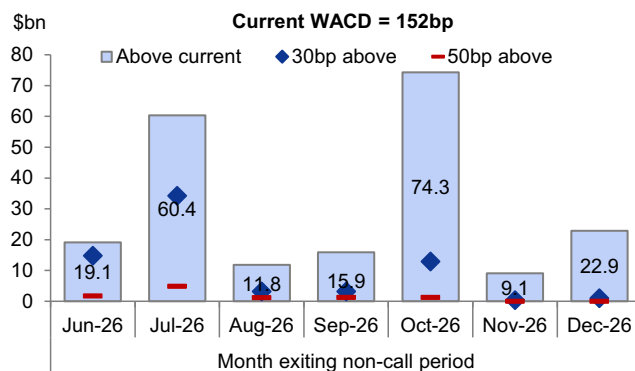
Exhibit 118: Annual USD leveraged loan net issuance



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 119: USD CLO refinance monitor

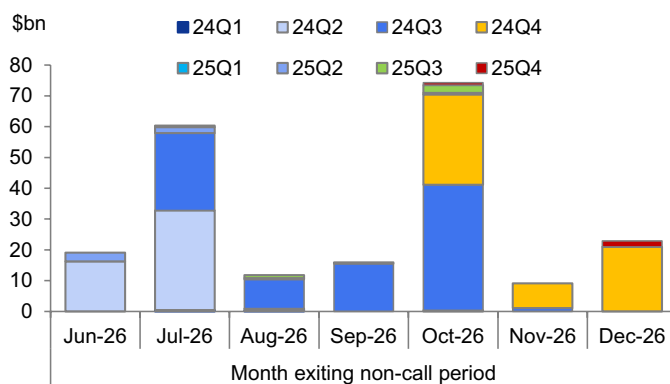
Upcoming notional above prevailing WACD



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 120: USD CLO refinance monitor

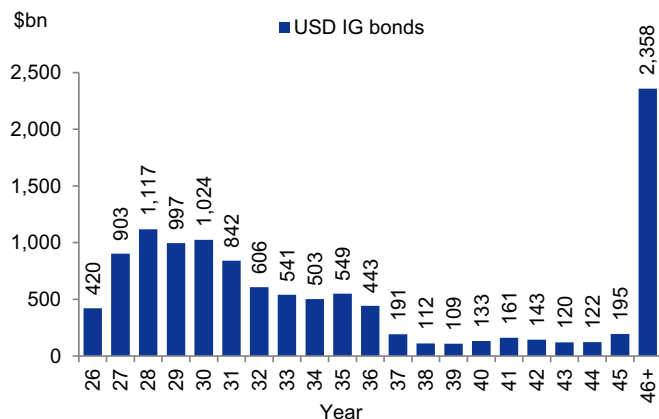
Notional above prevailing WACD by origination quarter



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

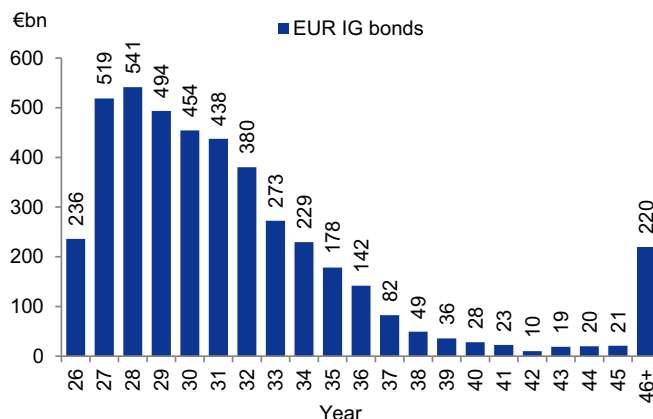
Primary market dashboard: Maturity walls

Exhibit 121: USD IG bond market maturity wall



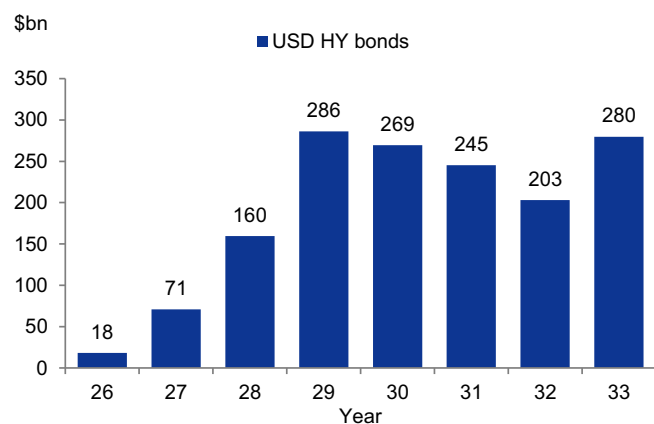
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 122: EUR IG bond market maturity wall



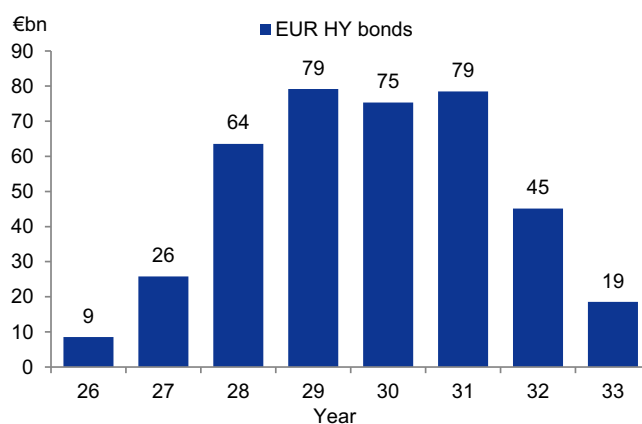
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 123: USD HY bond market maturity wall



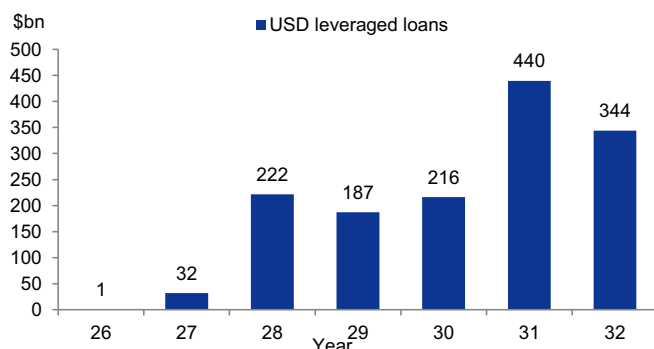
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 124: EUR HY bond market maturity wall



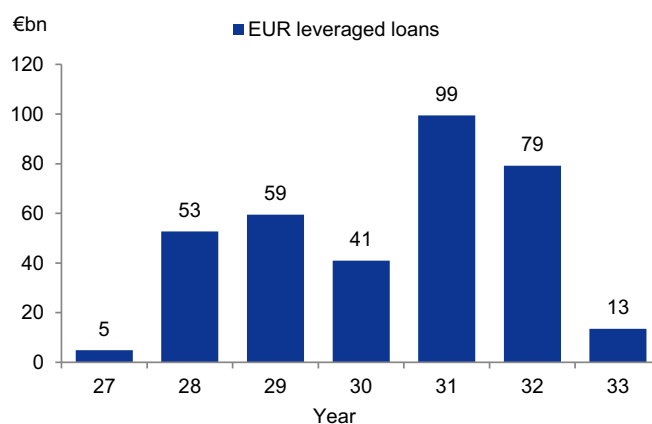
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 125: USD leveraged loan market maturity wall



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

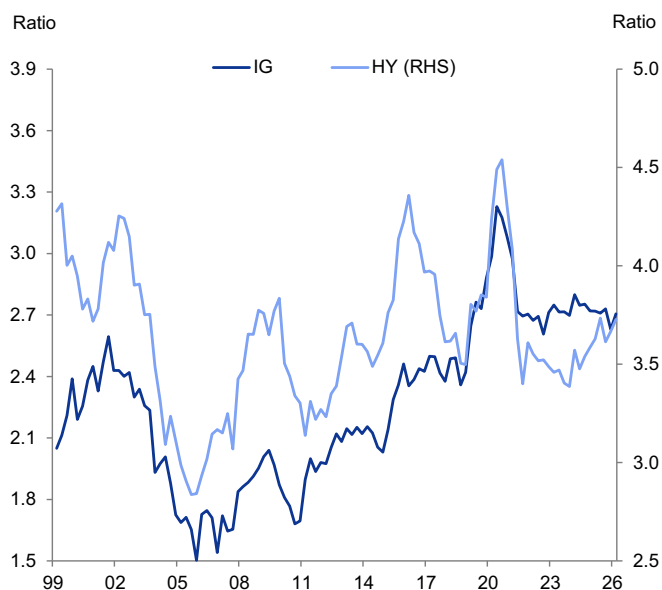
Exhibit 126: EUR leveraged loan market maturity wall



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

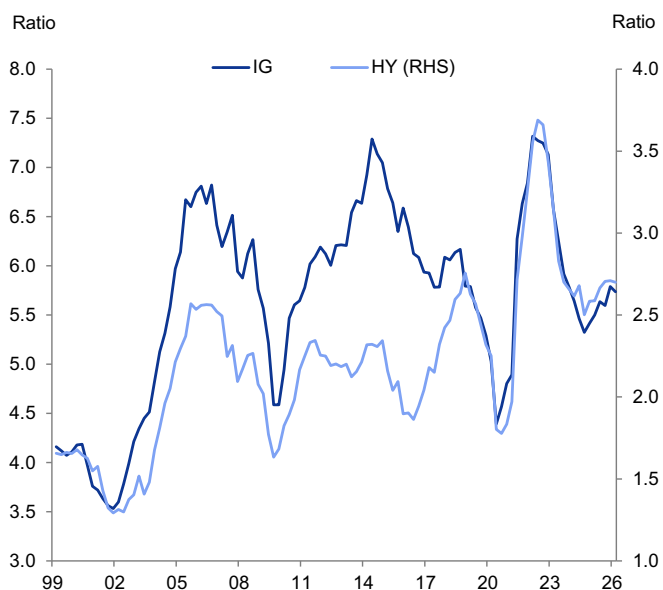
Fundamental dashboard: Credit metrics for US non-financial corporations

Exhibit 127: Net debt to EBITDA ratios for the median US IG and HY-rated non-financial corporations



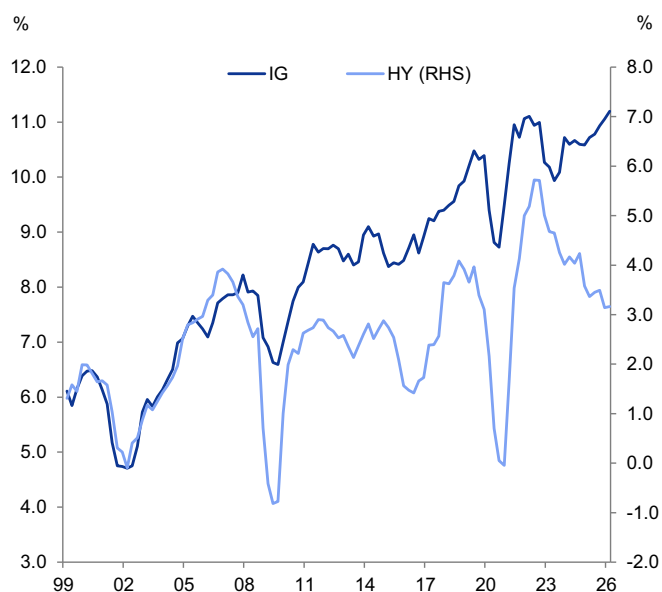
Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 128: Interest coverage ratios for the median US IG and HY-rated non-financial corporations



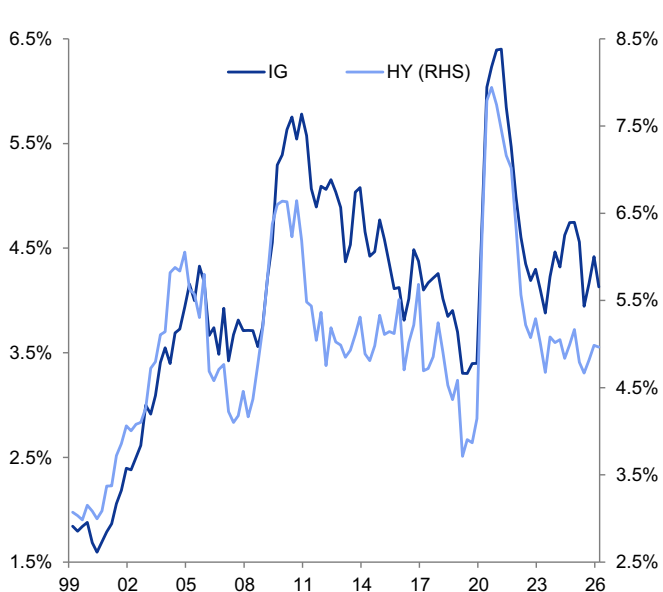
Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 129: Net margins for the median US IG and HY-rated non-financial corporations



Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 130: Cash to total assets ratios for the median US IG and HY-rated corporations

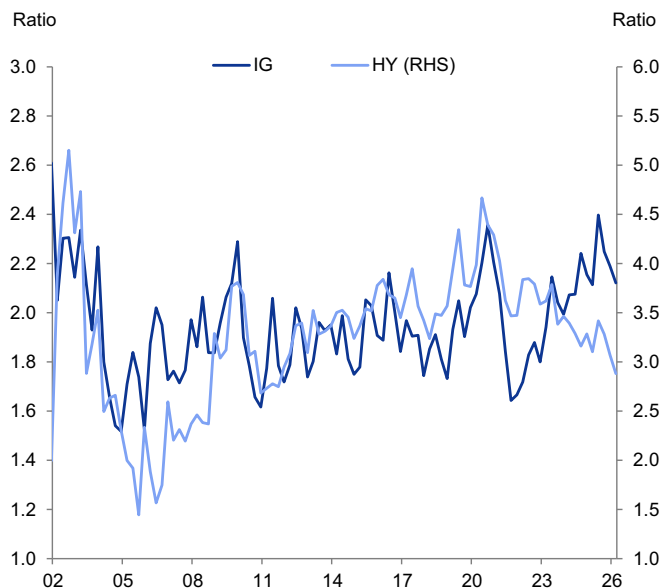


Source: FactSet, Goldman Sachs Global Investment Research

Fundamental dashboard: Credit metrics for European non-financial corporations

Exhibit 131: Net debt to EBITDA ratios for the median European IG and HY-rated non-financial corporations

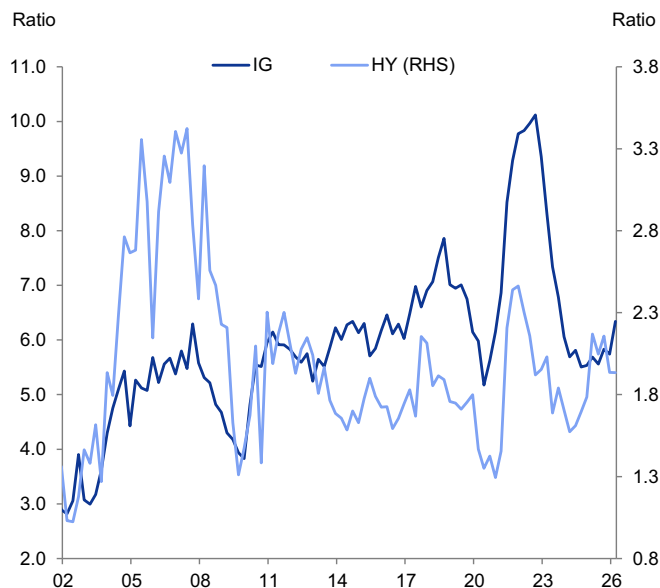
We include firms domiciled in the Euro area, UK, Switzerland, Sweden, Denmark and Norway



Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 132: Interest coverage ratios for the median European IG and HY-rated non-financial corporations

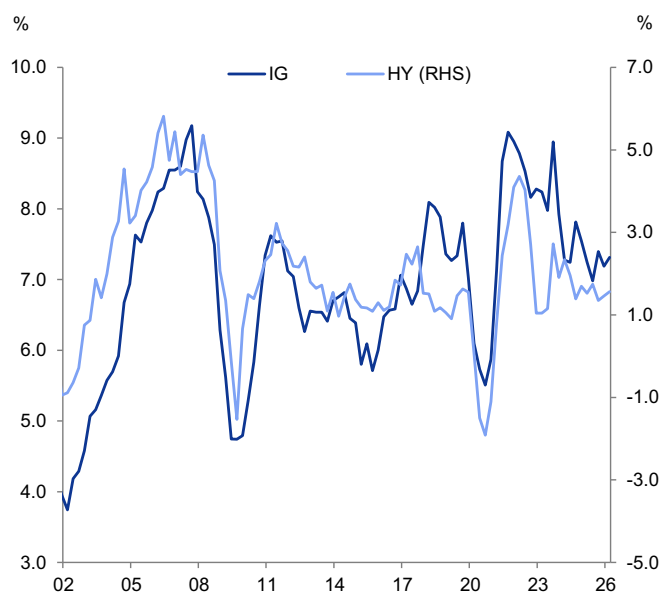
We include firms domiciled in the Euro area, UK, Switzerland, Sweden, Denmark and Norway



Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 133: Net margins for the median European IG and HY-rated non-financial corporations

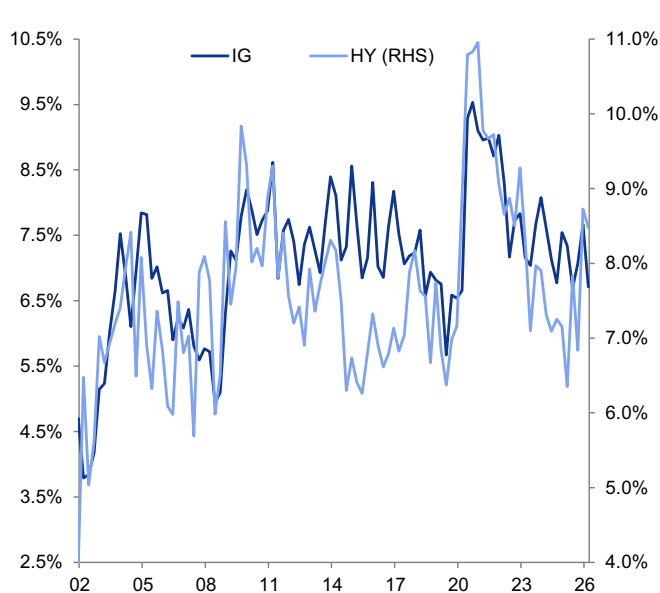
We include firms domiciled in the Euro area, UK, Switzerland, Sweden, Denmark and Norway



Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 134: Cash to total assets ratios for the median European IG and HY-rated non-financial corporations

We include firms domiciled in the Euro area, UK, Switzerland, Sweden, Denmark and Norway

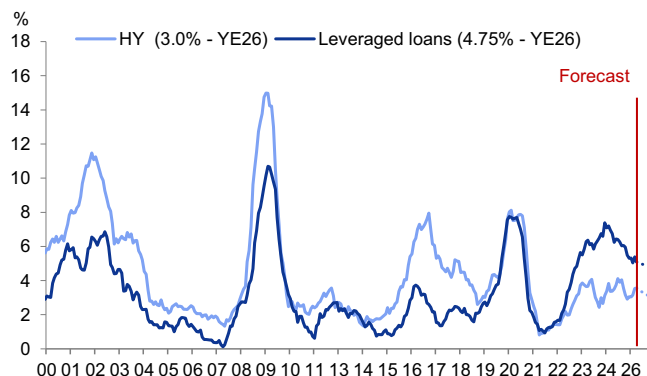


Source: FactSet, Goldman Sachs Global Investment Research

Default and ratings migration dashboard: Default trends

Exhibit 135: Default rates for US HY bond and leveraged loan issuers

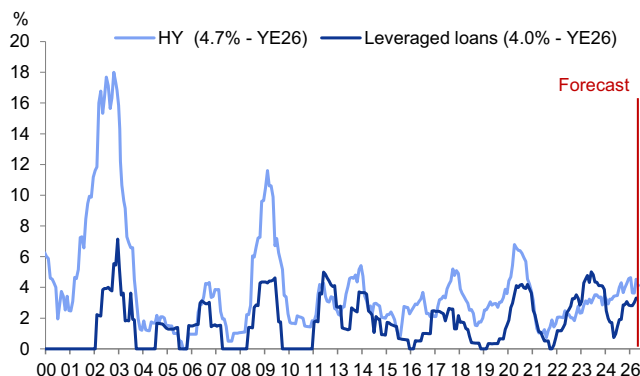
12-month trailing issuer-weighted default rate in the US HY bond and leveraged loan markets



Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 136: Default rates for European bond and leveraged loan issuers

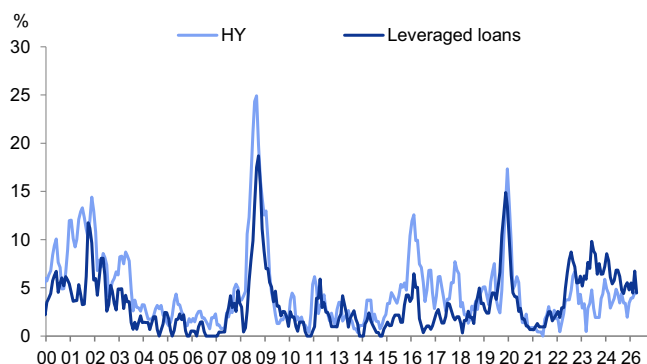
12-month trailing issuer-weighted default rate in the European HY bond and leveraged loan markets



Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 137: High frequency default rates for US HY bond and leveraged loan issuers

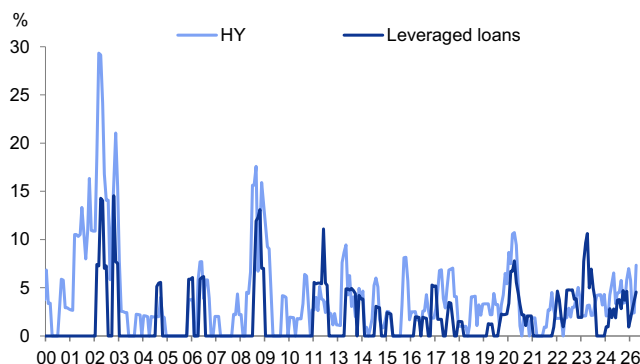
3-month trailing issuer-weighted default rate in the US HY bond and leveraged loan markets



Source: Moody's, Goldman Sachs Global Investment Research

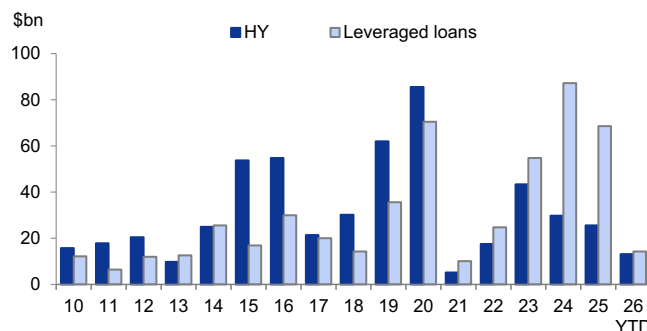
Exhibit 138: High frequency default rates for European HY bond and leveraged loan issuers

3-month trailing issuer-weighted default rate in the European HY bond and leveraged loan markets



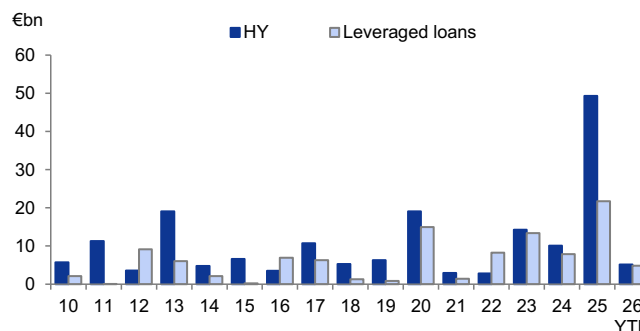
Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 139: Annual par value amounts of defaulted USD HY bonds and leveraged loans



Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 140: Par value amounts of defaulted EUR HY bonds and leveraged loans



Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 141: Top 5 largest defaults in the USD HY bond and leveraged loan markets in the past 12 months

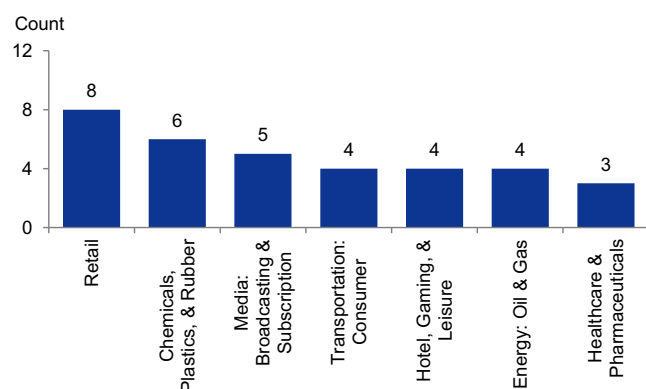
	Default date	Issuer	Industry	Default type	Notional amount (\$bn)
HY	1/29/2026	LABL, Inc.	Containers, Packaging, & Glass	Prepackaged Chapter 11	2.9
	11/20/2025	New Fortress Energy Inc.	Energy: Oil & Gas	Missed interest payment	2.7
	1/14/2026	Saks Global Enterprises LLC	Retail	Chapter 11	2.7
	4/16/2026	QVC, Inc.	Retail	Prepackaged Chapter 11	2.7
	8/8/2025	Saks Global Enterprises LLC	Retail	Distressed exchange	2.1
Loans	9/28/2025	First Brands Group, LLC	Automotive	Chapter 11	4.7
	6/30/2025	Radiate HoldCo, LLC	Telecommunications	Distressed exchange	3.8
	5/30/2025	Quest Identity Intermediate Limited	High Tech Industries	Distressed exchange	3.2
	5/2/2025	Ivanti Software, Inc.	High Tech Industries	Distressed exchange	2.8
	2/17/2026	Vibrantz Technologies Inc.	Chemicals, Plastics, & Rubber	Distressed exchange	2.6

Source: Moody's, data compiled by Goldman Sachs Global Investment Research

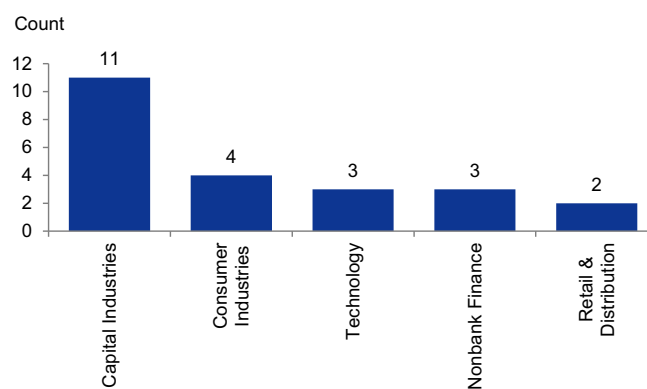
Exhibit 142: Top 5 largest defaults in the EUR HY bond and leveraged loan markets in the past 12 months

	Default date	Issuer	Industry	Default type	Notional amount (€bn)
HY	6/10/2025	Altice France SAS	Telecommunications	Suspension of payments	6.0
	6/16/2025	Altice France Holding S.A.	Telecommunications	Missed interest payment	2.1
	2/19/2026	Garfunkelux Holdco 3 S.A.	Finance	Distressed exchange	1.7
	6/25/2025	Garfunkelux Holdco 3 S.A.	Finance	Distressed exchange	1.7
	11/12/2025	Ardagh Group S.A.	Containers, Packaging, & Glass	Distressed exchange	1.4
Loans	6/10/2025	Altice France SAS	Telecommunications	Suspension of payments	3.5
	6/3/2025	Colisee Group	Healthcare & Pharmaceuticals	Missed interest payment	1.3
	4/30/2026	Poseidon BidCo S.A.S	High Tech Industries	Missed interest payment	1.3
	1/29/2026	Eos Finco S.a.r.l.	Wholesale	Distressed exchange	1.3
	5/9/2025	Eos Finco S.a.r.l.	Wholesale	Distressed exchange	1.0

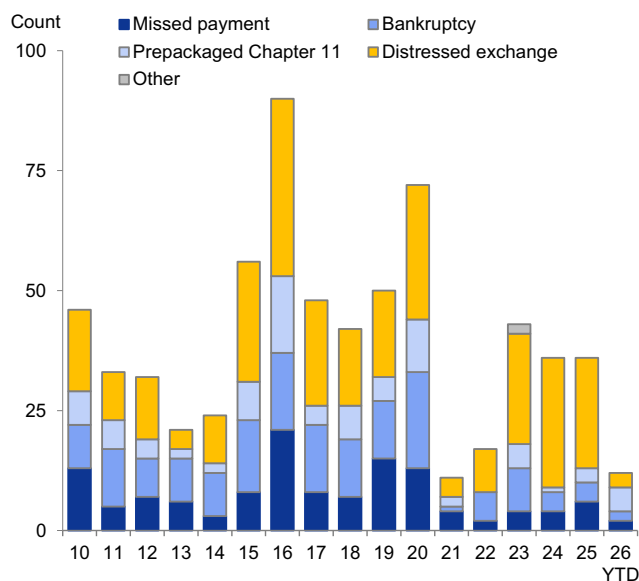
Source: Moody's, data compiled by Goldman Sachs Global Investment Research

Exhibit 143: Industry default count in the US

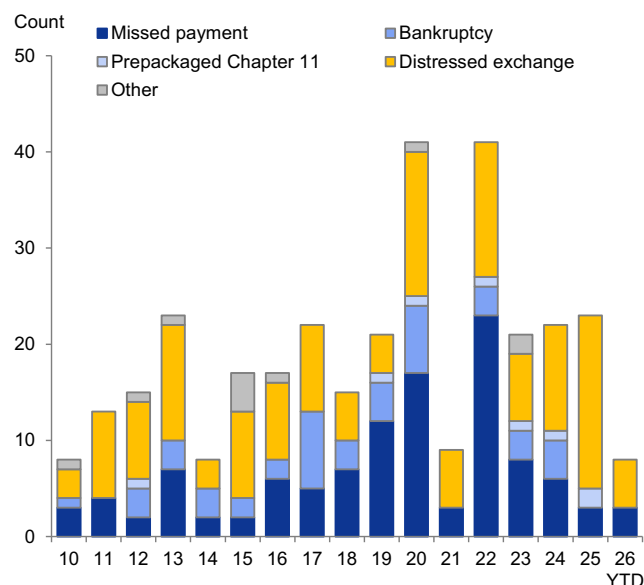
Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 144: Industry default count in Europe

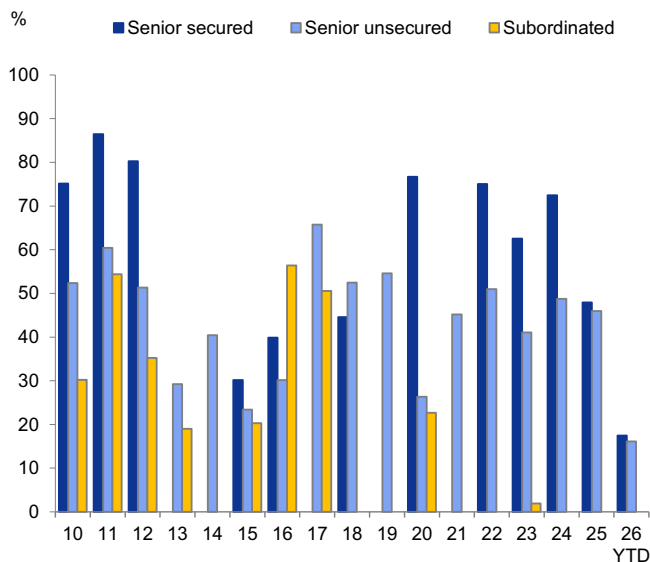
Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 145: Composition of defaults in the USD HY bond market by type

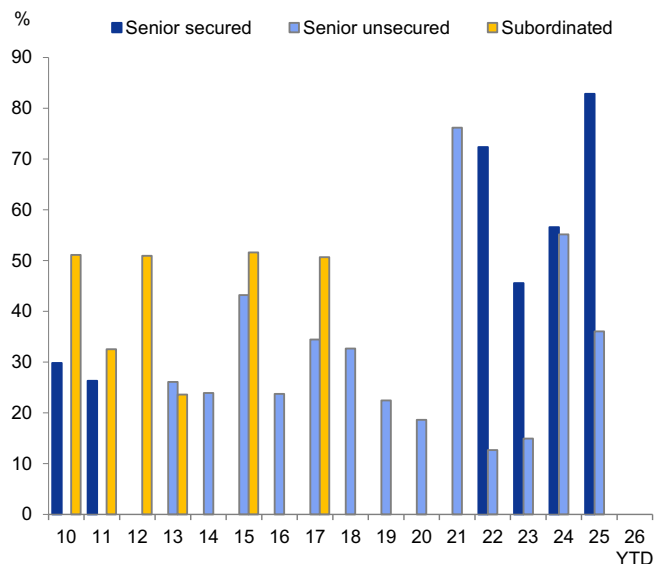
Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 146: Composition of defaults in the EUR HY bond market by type

Source: Moody's, Goldman Sachs Global Investment Research

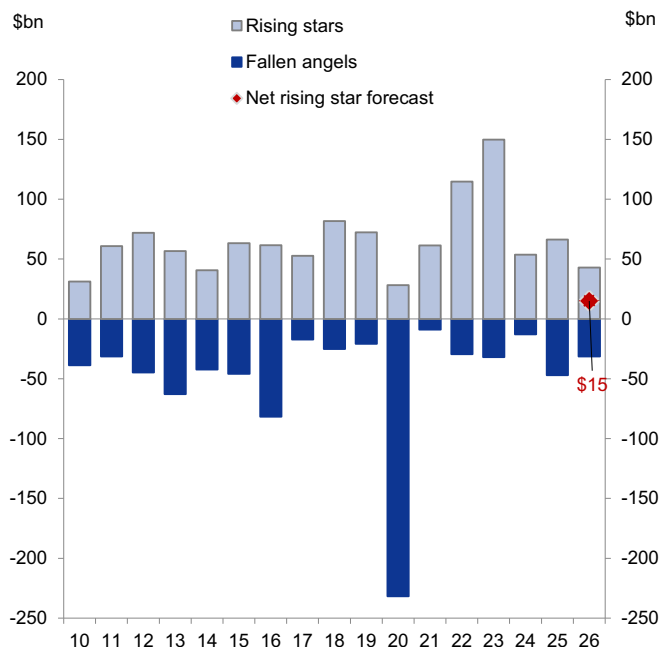
Exhibit 147: Annual average recovery rates by seniority for US issuers

Source: Moody's, Goldman Sachs Global Investment Research

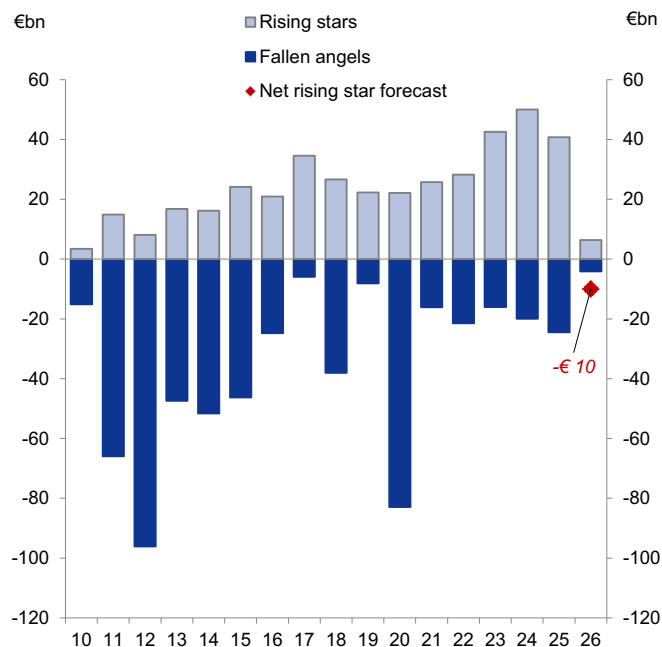
Exhibit 148: Annual average recovery rates by seniority for European issuers

Source: Moody's, Goldman Sachs Global Investment Research

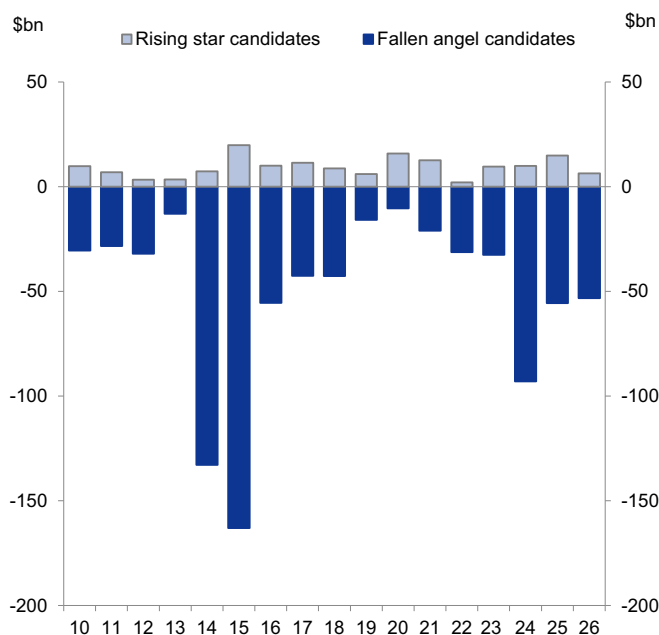
Default and ratings migration dashboard: Upgrades and downgrades

Exhibit 149: Fallen angels vs. rising stars in the USD market


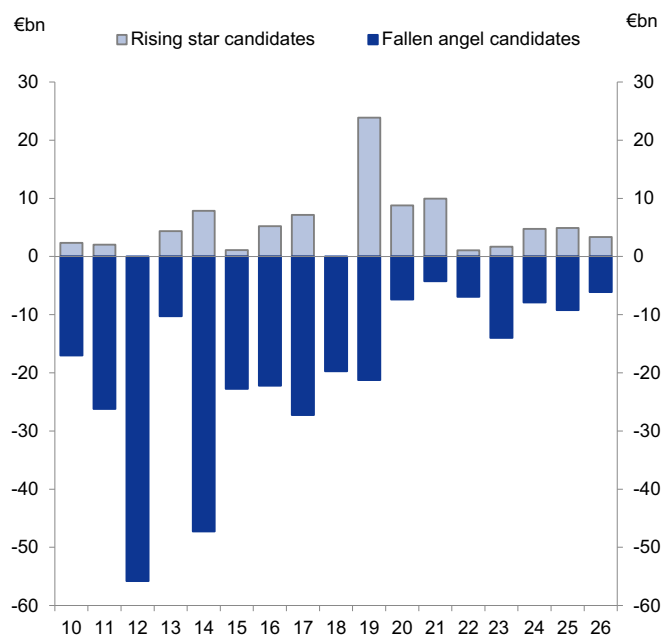
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 150: Fallen angels vs. rising stars in the EUR market


Source: Bloomberg, Goldman Sachs Global Investment Research

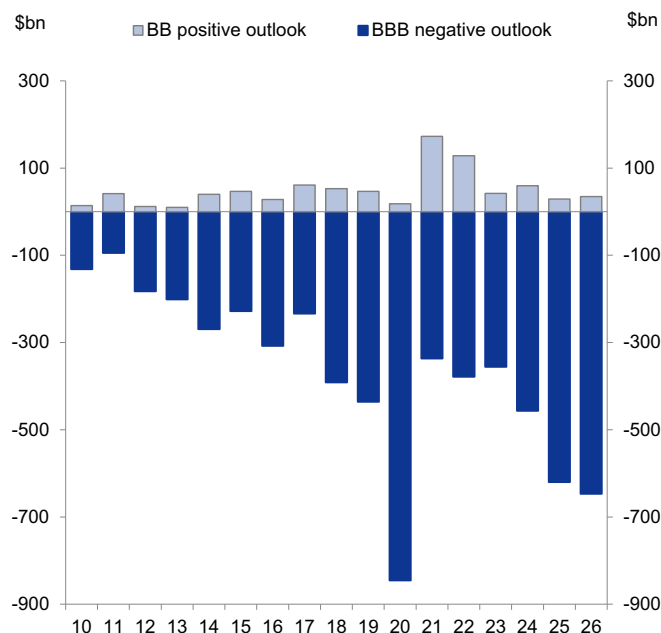
Exhibit 151: BB-rated bonds on upgrade watch (upgrade candidates) vs. BBB-rated bonds on downgrade watch (downgrade candidates) in the USD market


Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 152: BB-rated bonds on upgrade watch (upgrade candidates) vs. BBB-rated bonds on downgrade watch (downgrade candidates) in the EUR market


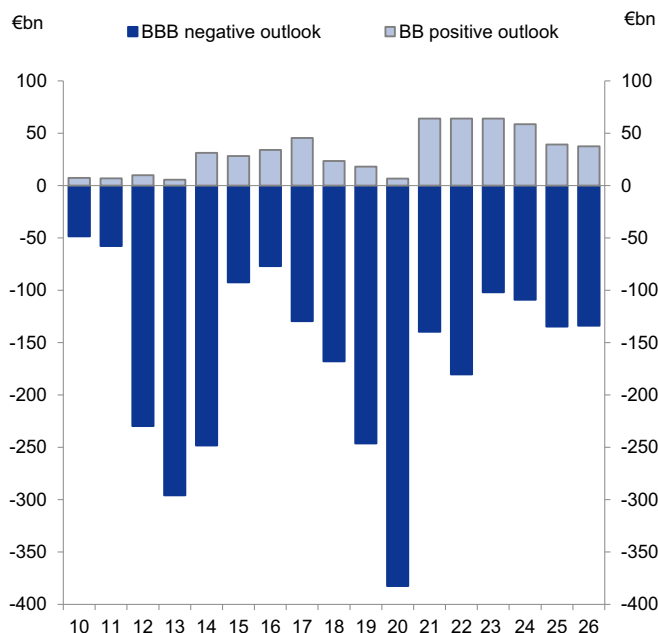
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 153: BB-rated bonds on outlook positive vs. BBB-rated bonds on outlook negative in the USD market



Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 154: BB-rated bonds on outlook positive vs. BBB-rated bonds on outlook negative in the EUR market



Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 155: Top 10 USD fallen angel issuers in the last 12 months

Company	Notional downgraded (\$bn)
Centene Corp	14.7
Paramount Global	13.0
Fs Kkr Capital Corp	3.8
Fmc Corp	2.5
Huntsman International L	1.5
Banco Santander Mexico	1.3
Pacificorp	0.9
Blackrock Tcp Cap Corp	0.7
Horizon Mutual Holdings	0.6
Ses Americom Inc	0.5

Source: Bloomberg, data compiled by Goldman Sachs Global Investment Research

Exhibit 156: Top 10 USD rising star issuers in the last 12 months

Company	Notional upgraded (\$bn)
Vistra Operations Co Llc	6.3
Medline Borrower Lp	4.5
Office Cherifien Des Pho	4.1
1011778 Bc / New Red Fin	4.0
Permian Resourc Optg Llc	3.6
Unicredit Spa	3.5
Frontier Communications	2.8
Frontier Communications	2.8
Constellation En Gen Llc	2.3
Kioxia Holdings Corp	2.2

Source: Bloomberg, data compiled by Goldman Sachs Global Investment Research

Exhibit 157: Top 10 EUR Fallen angel issues

Company	Notional downgraded (€bn)
Ses Sa	3.4
Lanxess Ag	2.2
Worldline Sa/France	2.2
Citycon Treasury Bv	1.7
Verallia Sa	1.6
Mediobanca Di Cred Fin	1.5
Unibail-Rodamco-Westfld	1.1
Adecco Int Financial Svs	0.5
Illimity Bank Spa	0.3
Solaben Luxembourg	0.1

Source: Bloomberg, data compiled by Goldman Sachs Global Investment Research

Exhibit 158: Top 10 EUR rising star issuers in the last 12 months

Company	Notional upgraded (€bn)
Banca Monte Dei Paschi S	3.0
Terna Rete Elettrica	1.9
Aib Group Plc	1.7
Asr Nederland Nv	1.5
Unipol Assicurazioni Spa	1.3
Intesa Sanpaolo Assicura	1.1
Bper Banca	1.0
Piraeus Bank Sa	1.0
Czechoslovak Group	1.0
Abanca Corp Bancaria Sa	1.0

Source: Bloomberg, data compiled by Goldman Sachs Global Investment Research

Disclosure Appendix

Reg AC

We, Amanda Lynam, CPA, Spencer Rogers, CFA, Sara Grut and Shamshad Ali, hereby certify that all of the views expressed in this report accurately reflect our personal views, which have not been influenced by considerations of the firm's business or client relationships.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Amanda Lynam, CPA Goldman Sachs & Co. LLC, Spencer Rogers, CFA Goldman Sachs & Co. LLC, Sara Grut Goldman Sachs International, Shamshad Ali Goldman Sachs & Co. LLC.

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

Disclosures

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; for equity securities, market making and/or specialist role. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at <https://www.gs.com/worldwide/brazil/area/gir/index.html>. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details can be found at: <https://www.goldmansachs.com/worldwide/india/documents/Grievance-Redressal-and-Escalation-Matrix.pdf>, and a copy of the annual audit compliance report can be found at this link: <https://publishing.gs.com/content/site/india-annual-compliance-report.html>. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is

defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at <https://www.gs.com/disclosures/europeanpolicy.html> which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C.; Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (<https://www.sipc.org>).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research, unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is focused on investment themes across markets, industries and sectors. It does not attempt to distinguish between the prospects or performance of, or provide analysis of, individual companies within any industry or sector we describe.

Any trading recommendation in this research relating to an equity or credit security or securities within an industry or sector is reflective of the investment theme being discussed and is not a recommendation of any such security in isolation.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at <https://www.theocc.com/about/publications/character-risks.jsp> and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your

individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g., marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to <https://research.gs.com>.

Disclosure information is also available at <https://www.gs.com/research/hedge.html> or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. This foregoing restriction includes, without limitation, using, extracting, downloading or retrieving this information, in whole or in part, to train or finetune a machine learning or artificial intelligence system, or to provide or reproduce this information, in whole or in part, as a prompt or input to any such system.